Power System Control Technology
&
Operational Training

Power Electronics Module



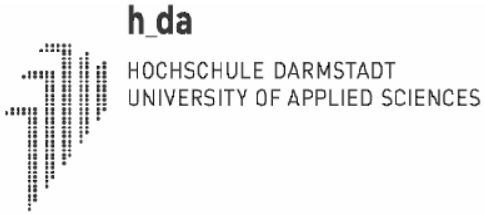
h_da
HOCHSCHULE DARMSTADT
UNIVERSITY OF APPLIED SCIENCES

Master of Sciences
in Electrical Engineering

2014

University of Applied Sciences - Hochschule Darmstadt
FB EIT
Prof. Dr.-Ing. Dieter Metz
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Power System Control Technology & Operational Training



FB Elektrotechnik und Informationstechnik
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Master of Sciences in Electrical Engineering
System Design and Technology

2014

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Outline of this course

This course is designed for the "Power Electronics Module", a module of the h_da Master of Science Program "Electrical Engineering - System Design and Technology" presented by the University of Applied Sciences - Hochschule Darmstadt. The module wants to give the students an extended understanding of electrical power systems, power system control technology, power system operation and management. To follow the course the participants need a basic understanding of electrical power systems and information technology on a bachelor level.

What is the program? First the participants have an introduction in the power grids structure and in the technology of central computer based control systems, remote lines and information transfer (RTUs). Then participants were put in the position of control centre engineers. They learn how to operate the power system by a control centre SCADA system. The main technical questions of operational control are addressed: Normal operations like changing transformers or bus bars and how to influence load, load flow, voltages, etc. with special emphasis to the needs of medium voltage level will be treaded including communication to local staff. The impact of reactive power to the voltage and active power to the frequency, compensators, the power system regulations and other aspects on the dispatcher's levels are addressed. The effects of network failures witch depend from the treatment of the transformer' neutral point (including Peterson coils for failure compensation) will be discussed and exercised. Some network failures and emergencies are triggered by the trainer and the course participants have to react by locating and clearing the failures using SCADA and relays information and doing e.g. switching operations.

Besides the technical discussions a commercial thinking in operating the grid is forced. Commercial indicators and balances have to be followed indicating e.g. costs of losses and outages with respect to the grid companies' position and contracts in a deregulated energy market.

The context will be discussed theoretically and will be practically exercised by using a dynamic real time training simulator RESY-NES*. The training facilities include a standard SCADA control system, RESY-EVU*. SCADA standard functions will be learned, checked, tuned and adapted to certain operators' applications.

The combination of theoretical discussions and practical exercises during the training sessions very well supports understanding of power system behaviour and operational control. The dynamic training simulator provides a realistic environment for electrical power system experiments and exploration that can not be exercised in real grids because of damages and customer cut offs.

We wish you a good success.

Your trainer team
Prof. Dr.-Ing Dieter Metz / Dr.-Ing. Klaus-Martin Graf / Dipl.-Ing. Martin Anthes

*** RESY-PMC and RESY-NES are products of Hereschwerke Automation Company**

Please note two important things:

1.) It is extremely important to prepare each training meeting by reading the text and the next exercises because training time is short.

2.) During the lab hours do your snapshots and write down the results in short words as draft version. The final lab report and formatting is done later (at home / in the student's training rooms).

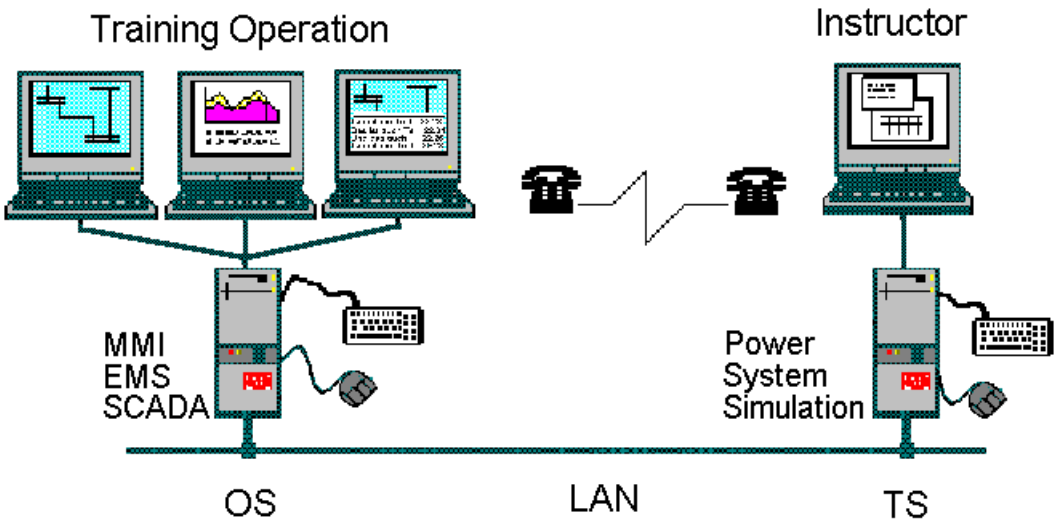
Your Remarks:

Outline of this course book

The outline of this course book is to inform how to start the training system, to perform, to exercise and to stop the training. It wants to give needed background information and to guide the participants through the lectures and exercises.

The system uses LINUX (for control and simulation) and Windows XP (for MMI) in parallel (VM-ware). Please ask the trainers how to start the sequence of building up the training workstation. Please note:

1. Switch on both computers (OS and TS)
2. Wait some seconds and type in "trainer1" only TS, OS has to wait for the process.
3. After "training" in green letters is shown type in "trainer1" at OS.



After starting the system the training system is ready to run. **Please wait for an introduction in the SCADA handling by the trainer.** There is a full declaration of a 400/110/20-kV grid including lines, transformers, generators and loads including the switchgear and all important regulation components. Before acting as a power system control engineer it is very important to get familiar first with

- the power system structure
- the components and the limitations
- the SCADA system functions
- the operational displays and the handling

Participants have to study the implemented power grid intensively. Only after having this knowledge the participants are allowed to execute normal operations first. During the exercises, the power system status can be followed and influenced by the SCADA tools and displays. Such are station pictures, load flow displays, line diagrams and a lot of other displays.

After being familiar with the grid and the SCADA tools the participants are able to be confronted with power system emergencies which have to be located and cleared…

Abbreviations and Idioms

	Meaning
akt	actual
Al	Aluminium
BV	Screen variable
DMD	Data model
DMS	Distribution Management System
VDN	German UHV Utility Association
EVU	Utility Company
h_da	University of Applied Sciences - Hochschule Darmstadt
HV	High Voltage (110-kV)
fun	System functions for data maintenance
FW	Remote lines
KW	Power plant
MMI	Men-Machine-Interface
MMK	Men-Machine-Communication
MTU	Master Terminal Unit
MV	Medium Voltage (20-kV ... 6-kV)
NTS, NES	Network Training Simulator
PC	Personal-Computer
PV	Process Variable
pwe	Enter password
QNX	PC-operating system of QNX Software Ltd.
RESY-PMCS	CADA System (Product of Herschwerke IT-Company)
RESY-NES	Network Simulator (Product Hereschwerke IT-Company)
RTU	Remote terminal unit
St	Steel
ste	Stop and end of control system operation
UHV	Ultra High Voltage (400 kV ... 220 kV)
HV	High voltage (110 kV)
MV	Medium voltage (30kv…6kV)
LV	Low voltage (0.4 kV)
EIN	switch on
AUS	switch off

Arbeitsmarkierung	Notes linked to a process element
Verfügungsberechtigung	Allowance to operate a component

Preface

The electric power supply has to be on a high quality level because it is a main basis to develop economy and culture. Unfortunately the level of power supply quality differs very much in the world. Electricity supply for the society has been started around 1900. It rapidly increased in the number of customers, electricity consumption and transfer. Not all countries could provide an excellent access to electricity is all regions because of the costs.

In the developing countries strong efforts are made to increase reliability and stability of the power system. Main goals are: Installing new plants, increasing the efficiency and saving electrical losses. It is very surprising that in Europe and in the US, once being the starting points of electricity, the same goals appear again due to the philosophy of restructuring the energy market. After a long period of a rather high quality supply in a monopoly type of market the liberalization was implemented to reduce the costs of electricity by inducting market competition. What are the effects? Due to the competition the utilities were forced to increase productivity. A strong cost pressure came up resulting in centralization, remote acting, automation and computer based grid supervision. The effects: Reduction of staff, doing less grid maintenance and reduction of investments for replacing power system components. As automation and centralization positively affected the power quality the lack of maintenance and investments obviously did the opposite.

Controlling a power grid through a control centre using a SCADA system and remote line operation is a quite different philosophy as controlling from a desk using a distributed team of local staff, local panels and phone call orders. The speed of reactions and the efficiency of operation will increase by SCADA and automation. But to correctly operate the power system using new technology and even a new philosophy a new type of control centre engineer is needed. What is important? First the new type of staff needs a full training for efficient use of the new tools. Second, as the control area is much wider, a more general view to the grid and its components forming an interacting system is necessary. Third, as before, the knowledge of the stations' equipment and details is needed. The new type of staff is responsible for much larger systems. The decisions will affect the whole power system, e.g. components in a large distance, wrong decisions will affect assets of billions of Dollars/Euros. Finally wrong decisions strongly affect the customers' quality of electrical power supply.

The big advantage of a centralized computer based operational control by SCADA is to have access to all information, to permanently view the updates of measurements, to follow data and trends and to have a very fast access to the switchgear and regulation equipment. If failures or emergencies occur, the clearing operations can directly be executed. Because of the strong impact in the large power system modern SCADA control software allows simulations of planned operation to see the effects before execution. That philosophy obviously increases the quality of supply because wrong decisions will be avoided.

Additionally, the quality of the staff should be mentioned. Only well educated engineers guarantee safe operations and high quality level of power supply. Due to the market competition and the trend for implementing product responsibility for electricity by law it is important to demonstrate all efforts providing a high quality for electricity supply to the public. How to achieve that? Important tools are computer based control systems (SCADA systems), RTUs, simulation tools and frequent training to develop the skills of the staff.

Frequent staff training results in professional and effective operations. The computer based operational training is a rather young development. It has been first used off line in universities' courses to support students' understanding and in continuing education courses for utility engineers. Now state of the art SCADA systems provide an integrated training as a standard function in the control centres.

RESY-SCADA© systems appeared to be the first control systems providing dynamic training functions. The basic concept is to link the SCADA software to an additional software package which dynamically calculates the reactions of a power system in real-time.

The course training system has been designed and developed during the last fifteen years in a partnership between former repas-AEG Automation, now named Hereschwerke Automation and the University of Applied Sciences, Hochschule Darmstadt, with advisory support form local utilities, mainly HSE-AG Darmstadt.

I would like to acknowledge my gratitude to all industry engineers and institution as well as the students doing final thesis for their great support. All have done great for the realization of this training system. Finally, I have appreciated the former cooperation with repas-AEG Automation and currently I do with Hereschwerke/ AutomationX very much.

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1 Introduction

The increasing complexity of electrical power systems changed the requirements for operation, monitoring and controlling the power system. The change from a rather simple panel to control a power substation or a single transformer station to a large central control system in the last three decades was made to increase the security, efficiency and reliability and to decrease the costs of labour.

Remote lines allow controlling all the important power grid stations and components from a computer based operation station in the control centre. Depending from the size of the utility company the control centre personnel is responsible for supervising the customers' supply by supervising the power grid, its components and states, the load flow, the frequency, the voltage levels, the losses and much more.

Efficient grid operation is based on e.g. sensors and actors in the grid, on safe bidirectional information transfer and on available control centre hardware and software. The control centre software system provides a toolset for the control engineers to supervise and operate the power system. Components such as computers, interfaces and software have to interact correctly. A basic set of functions to monitor, to log and to control the power grid which are implemented in the control centre computers are called SCADA functions. SCADA means supervisory control and data acquisition.

Electricity is important for the society. Strong demands on security and quality justify rather expensive equipment like remote lines, computer-based control systems, SCADA-software and additional software functions as energy management (EMS), load forecast etc.

The background of expensive equipment and investment on the one hand and the peoples' demand on low energy prises on the other hand is still causing a permanent optimization processes. Finally the main business driver is a reasonable economic profit. Economic indicators calculating and qualifying the grid efficiency is a new set of SCADA functions. This has become important because of the restructured energy market. That has caused a strong view to the economic side by investors AND regulator supervision. By operating the power system the effects to the costs and assets have to be taken into account always. Asses management has become an key role.

Power system operation, technical stabilisation and economic optimisation are complex tasks. Obviously only well educated and experienced personnel using helpful control centre software tools can guarantee an effective, economic and professional operation and supervision of the power system. Men and computer system have to interact positively. Who decides? Who controls? The basic philosophy of men-machine-interaction in control centres is:

- Let the machine analyse and present the network state
- Let men decide about operations
- Let machine calculate the effects of operation and give warnings
- Let the machine execute standard operations when men decides to trigger
- Let both control each other

What philosophy is used? Computers are very fast in analysing a large amount of data. Recognition and warning about critical values or trends is done fast. Men decide about operations. The operations were checked due to the consequences by automatic simulation

before. Warnings are given like “Warning: Action switches off a load”. Men finally decide. Extreme dangerous operations may be blocked. Men decides about machines’ automatic activities which can be switched on or off, e.g. automatic voltage control by transformer tap changing. Men decide about starting any automatic control system action. The system may execute e.g. a busbar change automatically after men’s request.

During the last two decades the following software tools have shown effective support: Interlocking system to avoid mistakes in operations, checking the planned topology and load flow consequences of planned operations, study modes for outage effects of high loaded components which possibly may fail, indicating losses and its costs, indicating costs of outages and which is a rather new approach: Providing a dynamic training environment for a risk free operational training.

Please see: There is an important difference between a load flow simulation to check planned switching orders and dynamic power system training. Load flow simulation is a static calculation of just one step to the future. Dynamic power system training is working continuously on a simulated power grid taking all dynamics and real time effects into account. This kind of training requires also the mathematical models of the protection relays’ behaviour during overload and short circuits, the simulation of voltage and frequency regulations and other dynamic effects of power system components. A dynamic training mode provides a real-time feeling during the training. It needs an original workstation running all SCADA, EMS and MMI software functions which are linked a dynamic and real-time power system simulation.

An increasing number of utilities support the staff by training mode facilities in the control centres. Well trained staff increases security, quality and even the companies’ profit.

1.1 Power systems in Germany and in Europe

The total consumption of electricity in Germany has been rather stable during the last 5 years slightly increasing from 620 to 650 TWh (1 TWh = 10^6 MWh). The German peak load has increased from 71 to 80 GW (1 GW = 10^3 MW).

The electricity production in Germany is a mix of different types of primary energy and plants including a significant and increasing number of renewable. See the production sources (year 2008):

Nuclear	**27 %**	**Brown coal**	**24 %**
Pit coal	**24 %**	**Gas**	**11 %**
Water	**4 %**	**Wind**	**8 %**
Else	**2 %**		

By (still valid) Act of Parliament in Germany the nuclear power plants will be stopped completely during the next 15 years. The wind energy is the most increasing figure by 30% a year. In 2009 the installed wind power was equal to 22 GW which is close to 30% of peak load. As wind power is fluctuating the contribution to energy is just around 8%. To match the customers' needs, it is important to use physical storage as well as doing demand side management. A significant amount of backup plants have to be installed. Newly installed conventional plants are mainly gas turbines using gas and steam technology leading to efficiency up to 60%. Combined heating (CH) is available only in larger cities. Besides the hydro storage plants which are mainly in the southern parts of Germany there are some storage plants using air buffers in the northern parts on base of former salt mining installations. Some more figures concerning the grid:

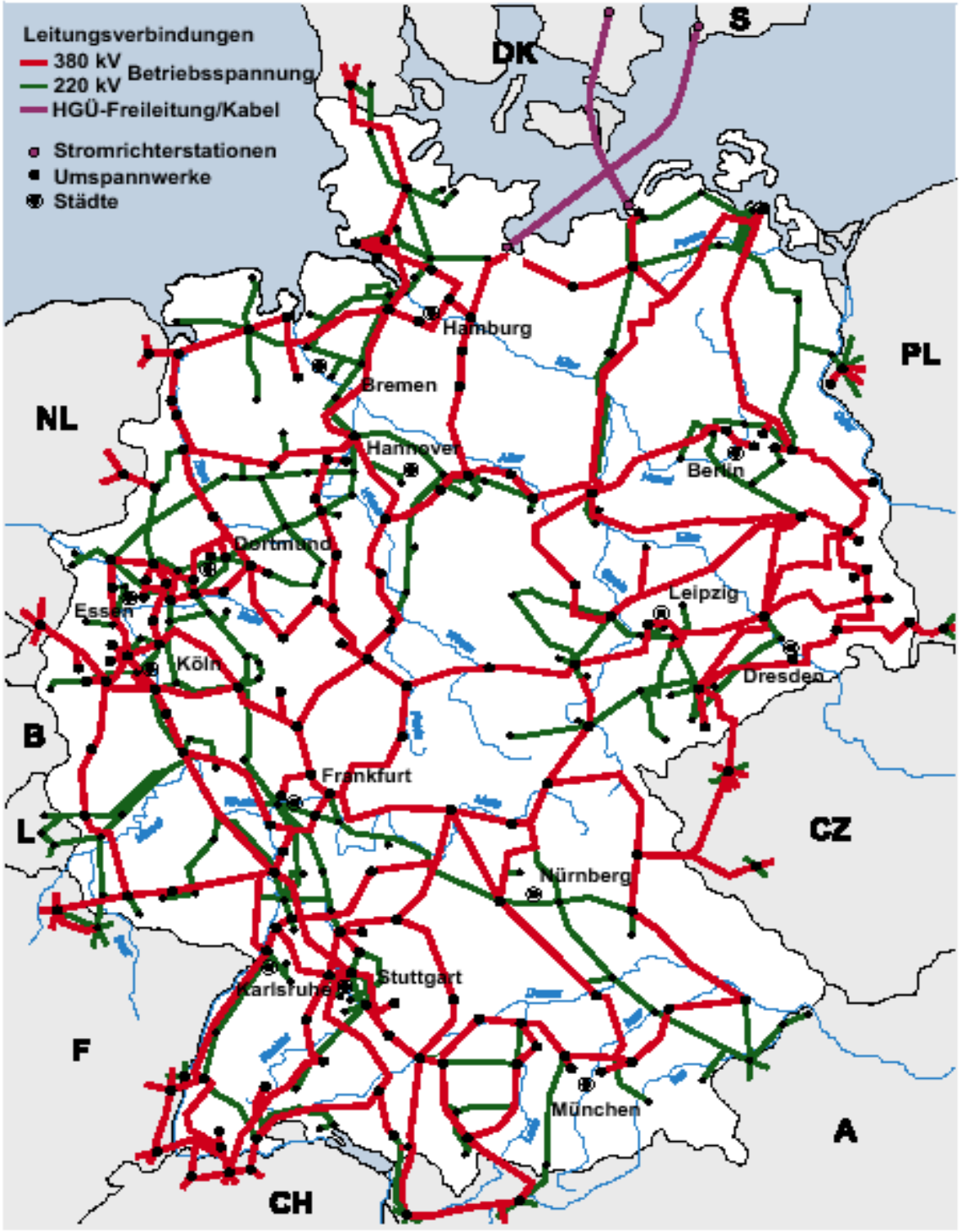
Length of lines and cables in Germany
(figures from year 2002, did not change significantly)**:**

UHV	(400 ... 220 kV)	38.500 km
HV	(110 ... 36 kV)	74.400 km
MV	(36 ... 6 kV)	472.300 km
LV	(1 ... 0,4 kV)	945.600 km

Transformers in Germany:

UHV/HV	1.100
HV/MV	7.300
MV/LV	557.300

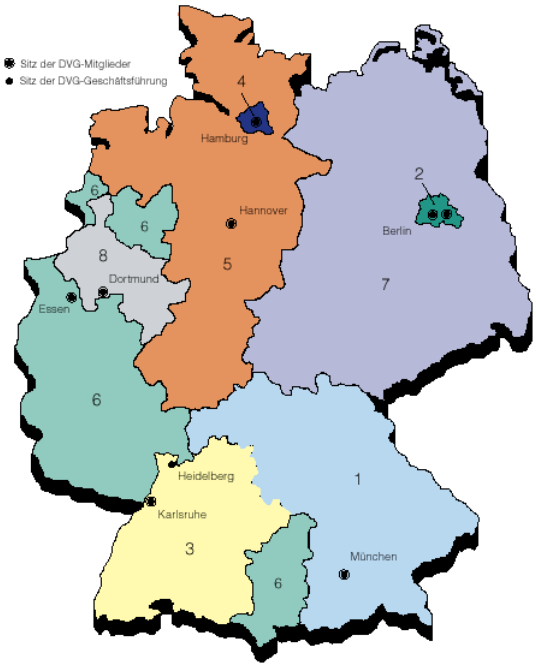
See the map of Germany and the 220 kV (green) and 380 kV (red) lines.

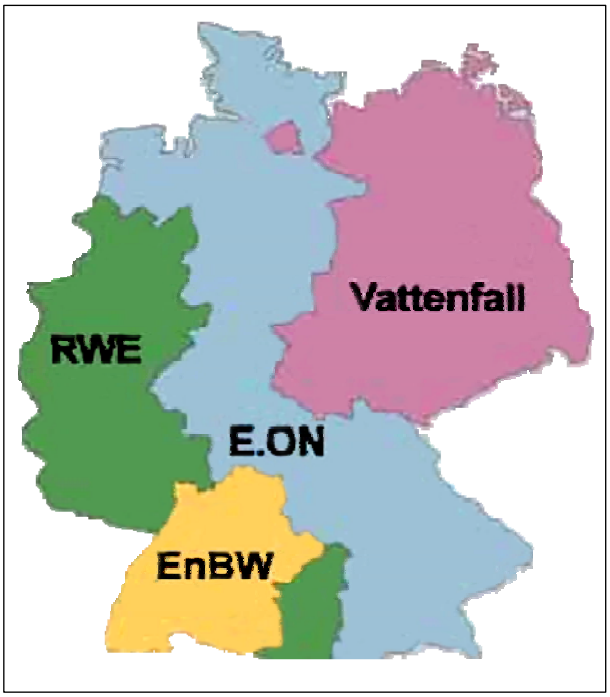


The power Grid 400-kV (red) and 220-kV (green) in Germany
(Source: Data and Facts - Stromnetze in Deutschland VDN 2002)
www.vdn.de

Restructuring of the Energy market in Germany

Due to the energy market liberalization process forced by the European Legislation a concentration process took place in Germany. Within around 5 years the dispatcher centres in Germany were reduced from 9 to 4 and the city works utilities have reduced the number of technicians and engineers by around 40% in the years 1996 to 2002.





State 1999		State since 2002
1	Bayernwerk AG	E.ON Netz GmbH(1+5)
2	Bewag AG	Vattenfall Europe AG (2+4+7)
3	EnBW Energie Baden-Württenberg AG	EnBW Transportnetze AG
4	Hamburger Electricitäts-Werke AG	RWE Net AG (6+8)
5	PreussenElektra AG	
6	RWE Energie AG	
7	VEAG Vereinigte Energiewerke AG	
8	VEW Energiewerke AG	

Dynamic change in German dispatcher utility industry
(Source: DVG Jahresbericht 1999 und Daten und Fakten - Stromnetze in Deutschland VDN 2002)

Power systems need production capacity reserves to stand outages and spontaneous customer needs. Therefore the plant capacity is higher compared to the real needed power. See the following figures of year 1999 and 2003.

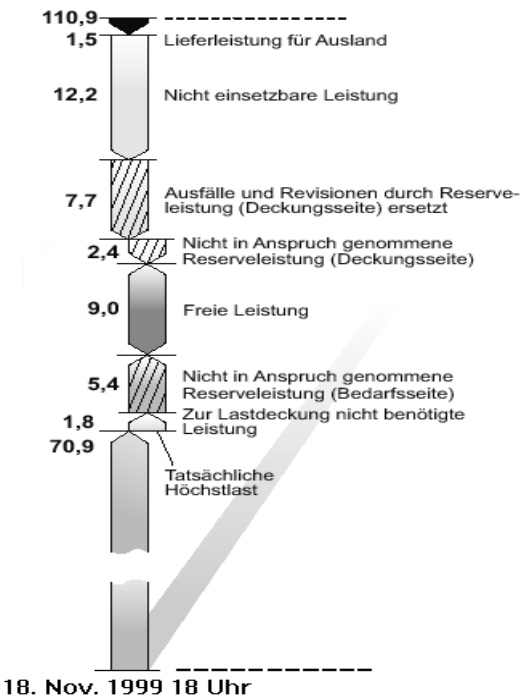
Customers' peak load was 70.9 GW.
Spinning reserve was 1.8 GW.
Demand side reserve was 5.4 GW.
Cold reserve 9.0 GW
Production side reserve 2.4 GW
Reserve not available 12.2 GW

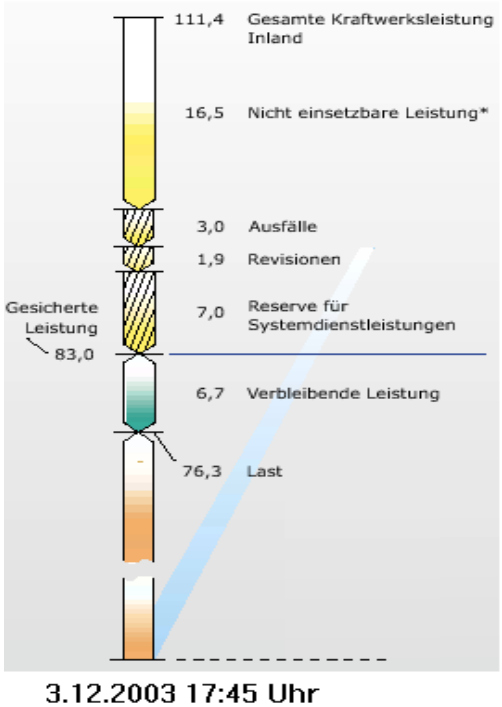
The production capacity overshoot is a result from the past when the increase of customer's consumption was around 3% every year. The capacity overshoot is estimated to be reduced during the next years. The planning of plants used to be done in Germany during monopoly market in terms of 10 years or even longer. This seriously has changed due to the liberalized market. See the peak loads and total power capacity of Germany in GW (1999 and 2003). Peak in year 2011 was about 82 GW.

Additional: Cut off of most of the Nuclear Power Plants by substitution with PV-farms and windfarms, on- and offshore. Figures in 2012 are around 27 GW of wind installation and 20 GW of PV peak. Electricity production is 20% by renewables. This is estimated to grow up to 30%-35% in year 2020.

1999: 70,9 / 110,9 = 63,9%

2002: 76,3 / 111,4 = 68,5 %

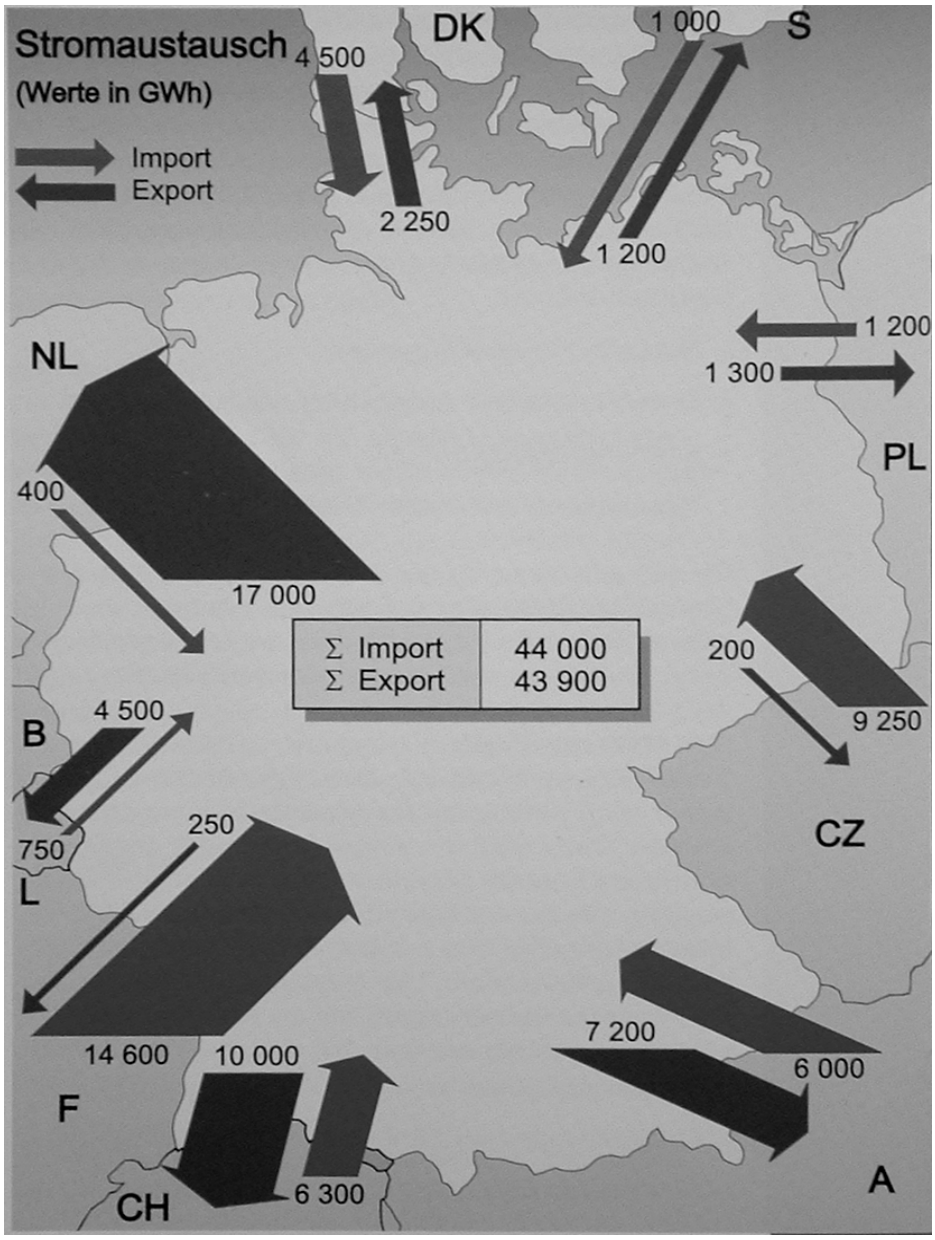




Power balance during peak load 1999 and 2003 Germany
(Source: DVG Jahresberichte 1999 and 2003)

1.2 Germany‘s electricity exchange with the surrounding countries

Due to the centralized position of Germany in Europe there is an intensive exchange of electrical energy through Germany. The main buyers of electricity are The Netherlands and Italy (passing through Switzerland). See the figures of year 2001. After cut off of Nuclear Plants the import has increased during times of low wind production. Cost effect: During peak wind production times a lot of coal plants are shut down. Market price drops!



Exchange of electrical energy of Germany in 2001 as example
(Source: Daten und Fakten - Stromnetze in Deutschland VDN 2002)

Exercise:

Calculate the costs/income of electrical energy exchange of Germany to France (Frankreich), The Netherlands (Holland) and to Switzerland (Schweiz) using a simple price modelling of 60 €/MWh.

1.3. European grids and electricity exchange

Electricity is being exchanged intensively in Europe. The bigger utilities are all European players. Biggest export countries are France selling about 70.000 GWh (EdF) und Tschech Republic selling around 10.000 GWh. Biggest import countries are Italy buying around 40.000 GWh and The Netherlands buying around 20.000 GWh.

The European power grids are interconnected to provide an international infrastructure for energy exchange. Goals are to be prepared if big plants have to be switched off and to use cheep energy production. Currently there are some 5 European power grid associations providing the technical and commercial infrastructure to exchange energy:

- **UCTE** **(**Union pour la **C**oordination du **T**ransport de l'**É**ecticité),
- **NORDEL** **(Nordic El**ectricity System)**,**
- **VES** **(Vereinigte E**nergie **S**ysteme),
- **UKTSOA** (**U**nited **K**ingdom **T**ransmisssion **S**ystem **O**perators' **A**ssociation)
- **TSO** **(**Association of **T**ransmission **S**ystem **O**perators in **I**reland)

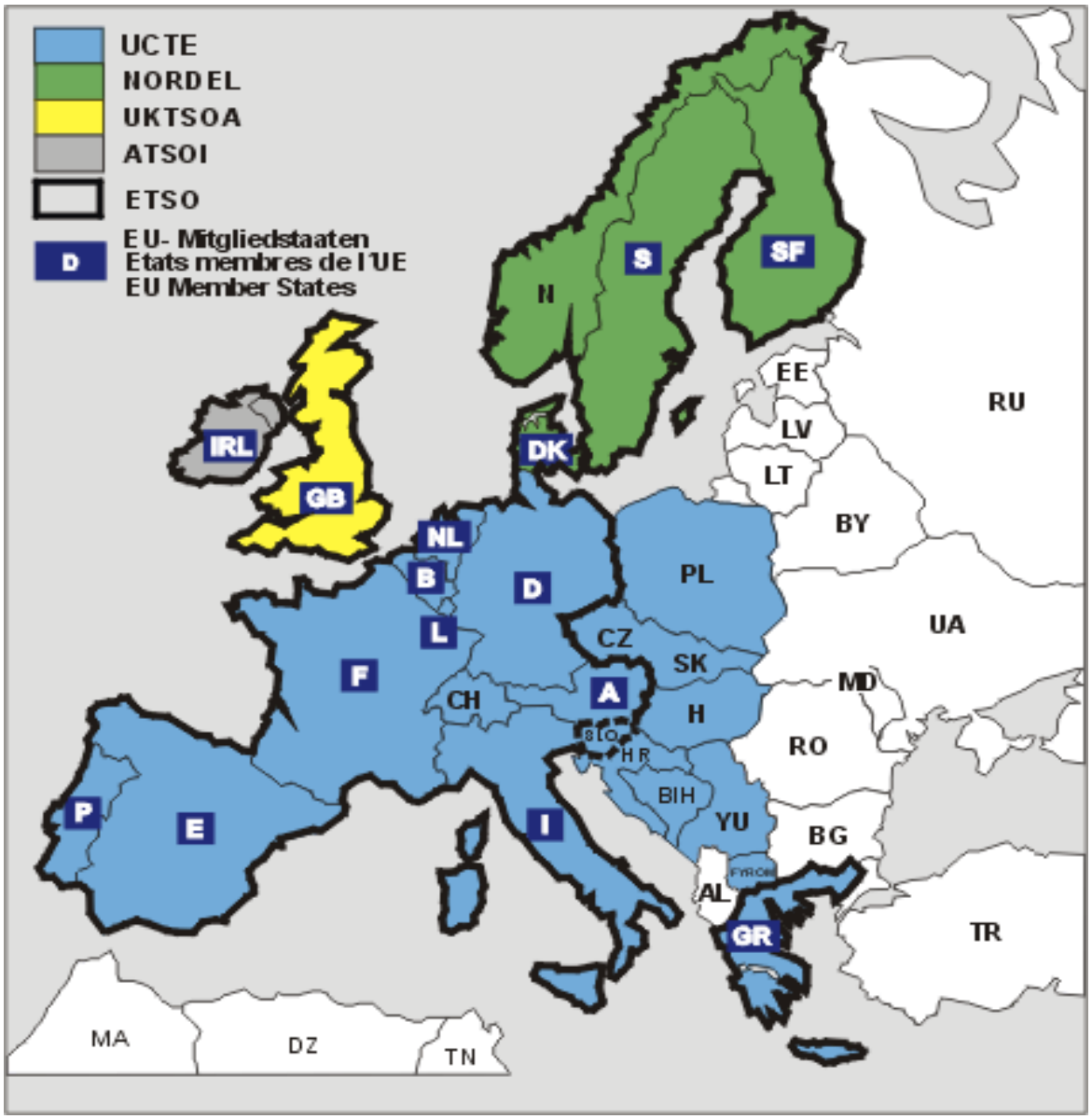
The overall regulation authority ETS (Association of **E**uropean **T**ransmission **S**ystem **O**perators) was founded in July 1999 in Frankfurt/Main to provide a platform for negotiations of the partners, currently UCTE, NORDEL, UKTSOA and ATSOI are members. The German representative is VDN (Verband der Netzbetreiber e.V.) im VDEW. See information and web site: www.vdn-berlin.de

UCTE

Year	***Peak Load***
1995	***258,6 GW***
1996	***264,1 GW***
1997	***270,1 GW***
1998	***276,2 GW***

Frequency drops

Grid	***Time***	***max Δf***
UCPT	***100%***	***< 50 mHz***
VES/EES	***40%***	***> 50 mHz***
	20%	***>200 mHz***
	5%	***>500 mHz***



European grids

The large west- and middle-European **UCTE-grid** was founded 1951 to obtain:

- high efficiency using production and transport capacity
- support for international electricity exchange
- improve reliability and quality of power supply

Partners are (2001):
Belgium, Germany, France, Greece, Italy, Yugoslavia, Luxemburg, The Netherlands, Austria, Poland, Portugal, Schweiz, Slovakia, Spain, Tschech Republic und Hungary.

In 2005 Bulgaria and Romania joined the UCTE partnership.

The **NORDEL (Scandinavian utilities)**: Denmark, Finland, Island, Norway und Sweden.

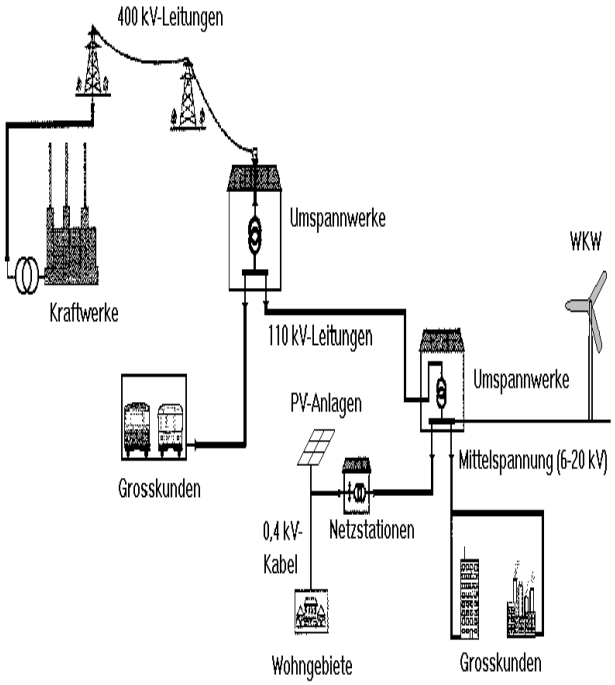
The **VES (Vereinigte Energiesysteme)**: Bulgaria, Rumania, Eastland, Let land, Lithuania, Russia, White Russia, Moldavia und Ukraine.

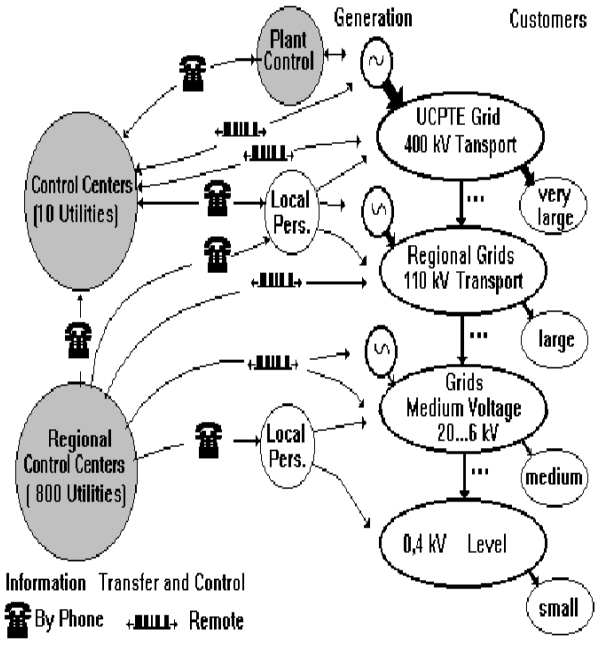
Future plans will link further countries of East-Europe to the UCTE-grid. Through the Spanish grid will be connected Algeria, Morocco and Tunisia. Future plans are to create a ring around the Mediterranean See to link renewable energies produced in the Sahara dessert and to transport electricity to central Europe.

Since March 2002 a European wide cash system is installed to charge for transported energy if border crossings happen. 1 € for each MWh has to be paid independently from the number of border crossings.

1.4 Power system structure and supervision

See the power system structure and supervision in next two figures that will be discussed in the meeting hours





1.5 Supervisory Control and Data Acquisition (SCADA)

SCADA

As the name indicates, a SCADA system is a computer based system to supervise and control a process, e.g. the power grid. It is a system that focuses on the supervisory level. The process can partly be supervised by remote control (RTUs) and with communication with personnel com links. SCADA as such it is a purely software package which provides the basic functions to supervise the grid e.g. power flow and voltage measurements, protection relays messages and the supply management. Typical software functions would be e.g. the updating of the process data base, event log file protocol and the presentation of displays to implement orders to the process. This software package is positioned on top of the hardware-software system of the computers to which it is interfaced. In general this would be via PLC's or other such commercial hardware modules.

The supervising operator would be responsible for the grid operations needed for maintenance, grid flow control and in the case of a fault the restoration of the network to normal operation.

Data Acquisition

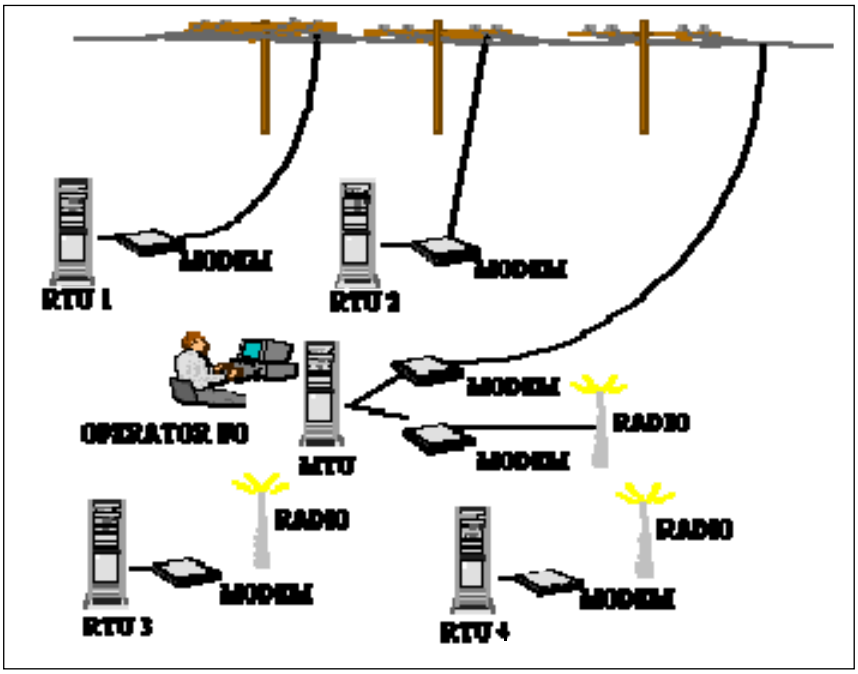
SCADA must be able to understand data from PLC's and other hardware distributed in the power grid, then to analyse and graphically represent this data to the user. SCADA systems must be able to read and write multiple sources of data using multiple industrial protocols. In the case of electrical grid network analysis protection relays events and measurements need to be collected such as voltage, current, frequency, active & reactive power and digital values of switching state,

SCADA systems include hardware and software components. The hardware gathers and feeds data into a computer that has SCADA software installed, usually a Standard PC. The computer then processes this data and presents it in a timely manner. SCADA also records and logs all events into a file stored on a hard disk and/or sends them to a printer. SCADA warns when conditions become hazardous by sounding alarms.

SCADA is a system that allows an operator to monitor and control processes that are distributed among various remote sites.

There are many processes that use SCADA systems including power, hydroelectric, water distribution and treatment utilities, natural gas etc. These systems allow remote sites to communicate with a control facility and provide the necessary data to control processes. For many of its uses, SCADA provides an economic advantage. As distance to remote sites and difficulty to access these increase, SCADA becomes a better alternative to an operator or technician visiting the site for adjustments and inspections. Distance and remoteness are two major factors for implementing SCADA systems. Nevertheless local inspections and maintenance are necessary from time to time.

See the structure of a SCADA system in next figure.



Generic SCADA System

There are four major elements to a SCADA system:

- The operator
- The master terminal unit (MTU)
- Communications
- Remote terminal unit (RTU).

The Operator

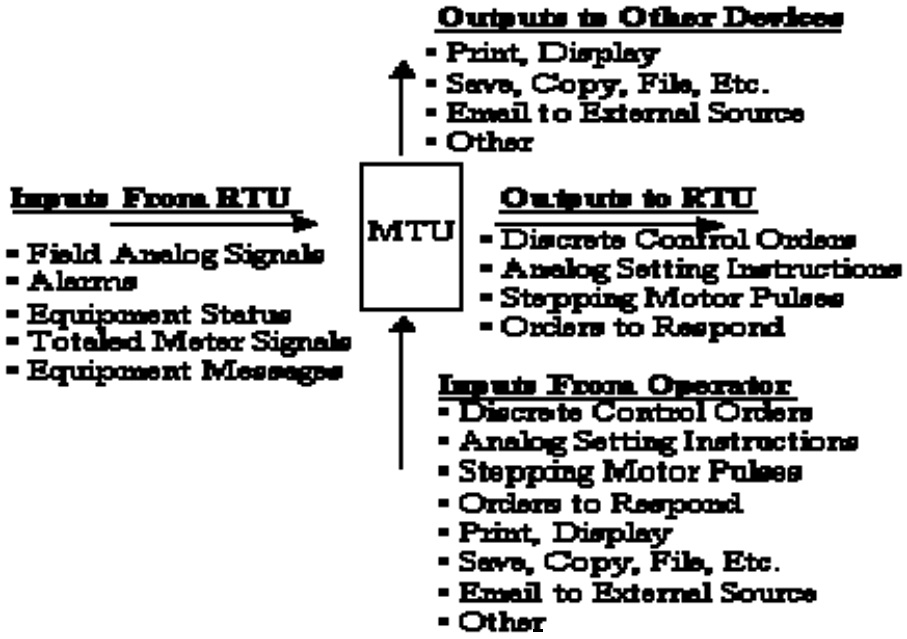
The operator exercises control through information that is depicted on a video display unit (VDU). Input to the system normally initiates from the operator via the master terminal unit's mouse and keyboard.

Master Terminal Unit

At the heart of the system is the master terminal unit (MTU). The master terminal unit initiates all communication, gathers data, stores information, sends information to other systems, and interfaces with operators. The relationship between MTU and RTU is analogous to master and slave. A cyclic polling mode is a common interaction between the MTU and the

RTUs. Other type of information exchange may be used such as spontaneous mode. Depending on the complexity or sophistication the MTU may employ heuristics embedded into its programming that allow it to make modifications to the system to maintain optimality. In the same fashion, the sophistication in the RTU may allow local optimization of functions.

The major difference between the MTU and RTU is that the MTU initiates virtually all communications by its programming. Concerning the traffic between the MTU and the RTU there are different modes. Communication maybe initiated by the MTU in a polling mode or by the RTU in a spontaneous interrupt mode. The MTU also communicates with other peripheral devices in the facility like monitors, printers or other information systems. The primary interface to the operator is the monitor that portrays a representation of valves, pumps, switchgear, breakers, etc. As incoming data changes, the screen is updated. Next Figure shows examples of inputs from the MTU and field devices.



Inputs & Outputs for MTU

Communications

SCADA systems are capable of communicating using a wide variety of media such as fibre optics, dial-up, or dedicated voice grade telephone lines, or radio. Recently, some utilities have employed Integrated Services Digital Network (ISDN) and DSL. Since the amount of information transmitted is still relatively small (less than 50K for common IEC protocols), voice grade phone lines, and radio work well. If other communication standards as IEC 61850 will be used, the transmission speed has to be improved seriously.

Remote Terminal Unit (RTU)

Remote terminal units gather information from their remote site from various input devices, like valves, pumps, alarms, meters, etc. Essentially, data is either analogue (real numbers) usually digital (on/off), or pulse data (e.g., counting revolutions of meters). Many remote terminal units hold the information gathered in their memory and wait for a request from the MTU to transmit the data. Other more sophisticated remote terminal units have microcomputers and programmed language controllers (PLC) that can perform direct control over a remote site without the direction of the MTU. One application for this would be the interlocking checks for switch ordering. Next figure shows an example of outputs of the RTU to the MTU and field devices.

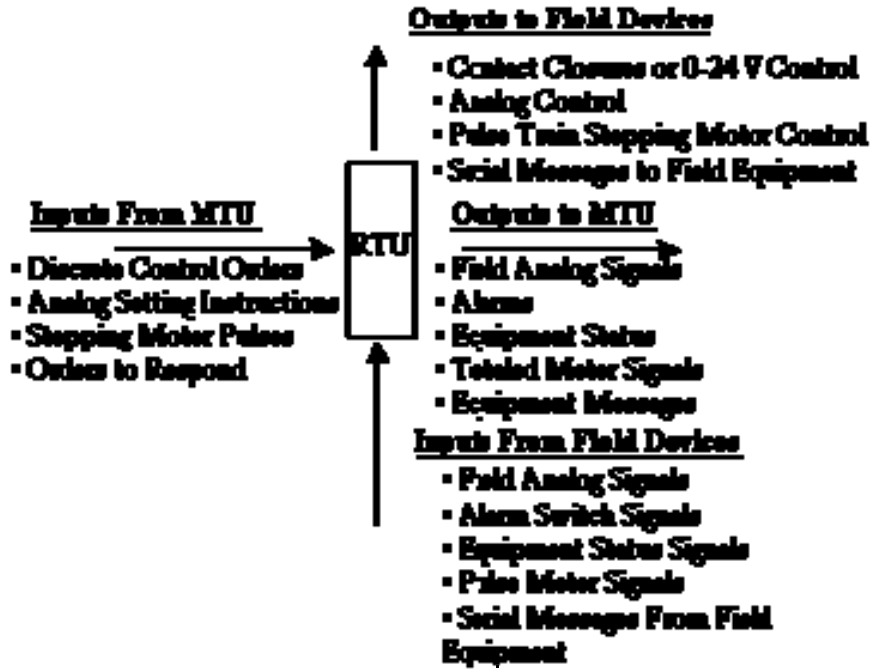


Figure: Inputs & Outputs for RTU

The RTU central processing unit (CPU) receives a binary data stream in accordance with the communication protocol. Protocols can be open, like Transmission Control Protocol and Internet Protocol (TCP/IP) or proprietary. There are relevant IEC standards. Data streams generally contain the information that is organized according to the seven layers Open Systems Interconnection Model (OSI Model). The OSI Model is used to set standards in the way information is exchanged with respect to protocols, communication, and data. The RTU receives its information because it sees its identification embedded in the protocol. The data is then interpreted, and the SCADA software directs the appropriate action at the site.

Requirements of a SCADA System

A system-wide supervisory control solution allows to enable better monitoring and control, faster reporting and analysis, easier direct and remote access, and smoother integration with business systems – all with improved efficiency and lower cost. Basically whether the needs are modest for control of a small grid or nation-wide control of an electrical power network, a central supervisory solution can make the job easier, faster and more efficient.

System-wide supervisory systems allows the control of overall mission including critical facility management as production monitoring, central dispatch, power management, quality control, environmental control, communications/networking, HVAC, security, and more. Typically, each of these dissimilar systems contains many individual functions and software subsystems.

By linking the equipment into a system-wide supervisory system and communications framework, you can usually:

- Reduce downtime and increase customer satisfaction
- Boosting equipment performance and better control of maintenance costs
- Drive real-time data collection and reporting for better decision making
- Have an automatic continuous monitoring, tracing and response
- Improve safety and security
- Reduce manual processes
- Facilitate on-line training to develop staff

The installation of SCADA has subsequently been seen as a means to satisfy a variety of increasing technical and commercial pressures such as consumer demands, regulatory requirements and to also satisfy the need to reduce operational costs.

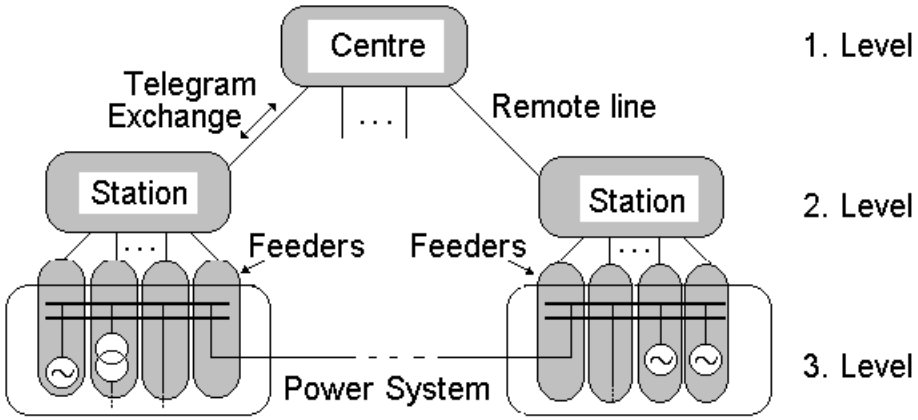
1.6 Remote line communication system

A real time supervision of the power system in a control centre bases on a remote line communication system. This is a very important part to fulfil the idea of remote data collection and process control. The remote line communication system connects the distributed substations to the central control system. The data is transmitted in a digital and serial form.

The communication between the subsystems in principle can be described in a seven layer model matching the IEC standards of ISO-OSI interconnections. But this is mainly a theoretical description of data exchange layers which will not be discussed in detail here because most of the practically used remote communication systems do not match this theoretical approach correctly because of its history.

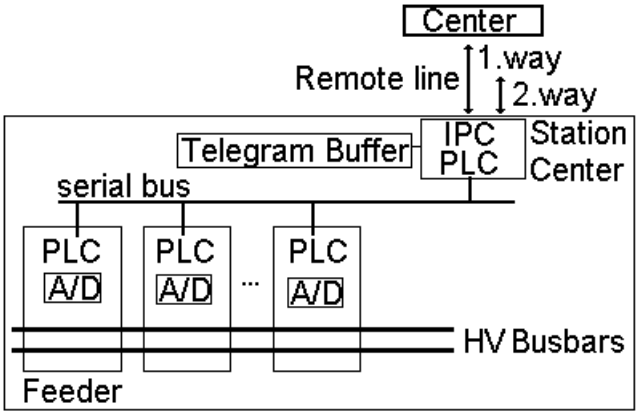
The physical communications media are wires, cables, broadcasting and satellite microwave links. More and more digital transfer on the base of fibreglass inside of the high voltage lines or digital microwave links each with special characteristics is being used.

The transmitted data as measurements of currents, voltages and loads are called PV which means process-variables. In principle three levels of data processing can be described, the (feeder-) device level, the station level and the centre level. The levels are shown in the next figure.



Structure of PV data processing: feeder-, station- and centre-level

In the feeder level the sensors for data collection are implemented usually by using PLC standard devices and interfaces using IEC standards. All analogue data are converted into digital form. In the station level the data are collected and gathered in data groups called telegrams. The telegrams are stored on the (sub-) station level control device (PLC or IPC) that also manages the data transfer to the control centre. The principle of data processing inside the station is described in next figure.



HV: High voltage PLC: Prog. Language Control-System IPC: Industry-PC
A/D: Converters from analogue to digital

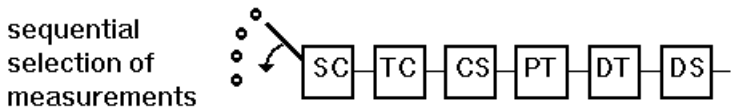
Data processing in a substation

It appears that parallel to the power network there is an information transfer network which is constructed in a radial type exchanging data between control centre and substations. A data exchange between the substations is not implemented.

Steps of processing

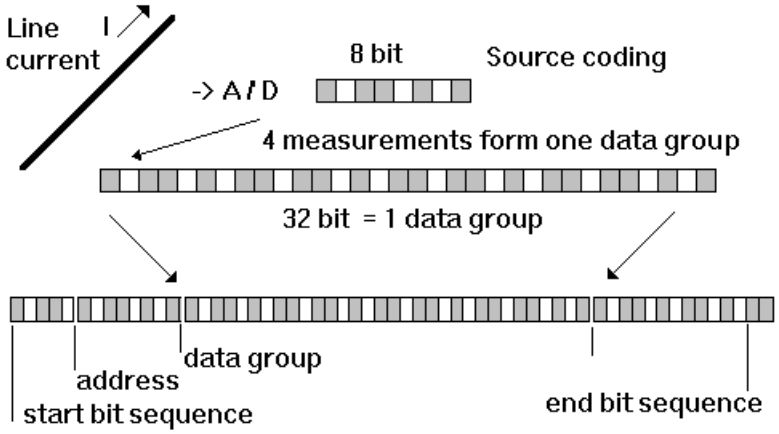
If we follow the steps of processing e.g. for a measurement of a current transformer then we can distinguish (next figure) processing steps as follows:

- **Source coding (SC): analogue-digital processing (e.g. 8 bit)**
- **Telegram coding (TC): PV data group creation and telegram construction**
- **Channel selection (CS): Serial sequences of a physical value send and receive**
- **Physical transport (PT): Physical medium transport**
- **Decoding (DT): Decoding telegram and PV separation of the data group**
- **Source decoding (DS): Conversion e.g. to an analogue data or to a float value**



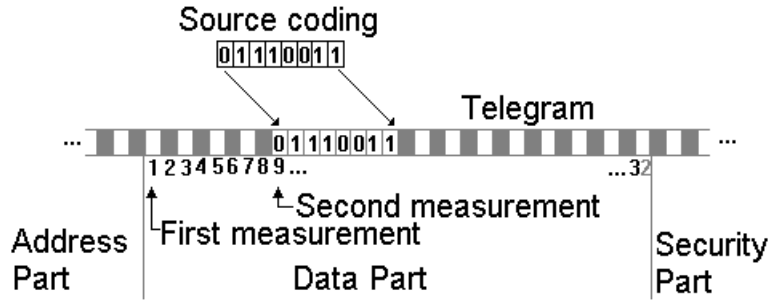
Steps of processing

Let us follow the steps of processing: One measurement passes the analogue-digital processing to form an 8-bit value that is stored in a data group of 32-bit. This is called source coding. A certain measurement has always a fixed position in fixed telegram (which has always the same address). Four measurements form a telegram data group. In addition to the data group information the address of the data group has to be added to identify the data group in the control centre. For a save transfer the telegram additionally needs security bits like parity-bits to identify faults that may occur during transfer. All steps to form a telegram are called telegram coding.



Telegram structure

Channel selection usually is done by the station IPC which selects first or second way of data transfer e.g. digital fibreglass channel or satellite link. After transfer the telegram arrives at the control centre. In the control centre it is very important to check if the transfer was done correctly without a fault. This is executed by again creating the security part of the telegram. The just created and the transferred security parts are compared. If no difference occurs the transfer is accepted as correct.
After telegram decoding the PV are separated and stored usually in the control centre data base for further data processing.

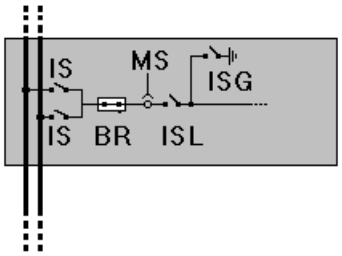


PV data

The process variables (PV) are from the physical point of view e.g. contacts of switchgear,

positions of devices, alarm contacts, regulation position (on/off), measurements like currents, voltages, temperatures and much more.

All PV are sampled by a cycle. Binary values like ON or OFF and analogue-digital converted values are put in the data group. Depending from the type of the PV we have to differ between 1-bit information (like contacts etc), 2-bit information (like switchgear, etc) and multiple-bit information (like measurements, etc). Please have a look at a substation feeder and its PV-information quantity in next figure:



PV of a Line Feeder

PV of a line feeder

			Information
IS	Isolator	to busbar 1	2 bit
		to busbar 2	2 bit
		to line	2 bit
ISG	Isolator to ground		2 bit
BR	Breaker		2 bit
MS	Measurements		
	Current transformer voltage		8 bit
	Current transformer current		8 bit
	Measurement active power		8 bit
	Measurement reactive power		8 bit
	Protection alarm state (each one bit)		1 bit
	(Differential, distance, current,		
	Low voltage,...) each one bit relay start		
	Each one bit relay execution		
	Real Time information		32 bit

Switchgear basically is coded by two bit, derived by the end positions. Therefore a "Working" or "Operating" switchgear can be identified as well as wrong information.

ON-Bit	**OFF-Bit**	**State**
0	1	OFF
1	0	ON
0	0	working
1	1	wrong

Coding of measurements is done in 8 (or 12) bit. Due to certain components some other bit ranges are used. Transformer tap positions (+13 ...0...-13) are transferred usually by 8 bit. Due to certain components like transformers some additional PV - information has to be transferred. Digital protection relays are able to process the impedance (to failure) and a real

time information additionally.

One data group contains 32 bit usually separated by 4 measurements or 16 switchgear positions or 32 single bit information or ... See the next table for information of the PLC standard device for substations:

Supply	**24 V DC, Buffer battery**			
Digital	**Groups of 8, 16, 32 Bit - 24 V DC**			
In-/Output	**24 V DC, 115/230 V AC potential free all IEC Standards**			
Real time	**1 ms resolution**			
Analogue	**Groups of 2, 4 independent channels**			
Input and	**4...20 mA, +/- 20 mA, +/- 10V, +/- 0.5 V**			
Output	**potential free, IEC**			
Cycle	**internal less than 1 ms for 1000 program steps**			
Information	**4096 Telegram ring buffer**			
Size	**Alarm direction**	**256**	**Messages**	
	(Substat. --> Centre)	**64**	**Counter**	**16Bit**
		64	**Measurement**	**12Bit**
		128	**Measurement**	**8Bit**
	Order direction	**256**	**Orders**	
	(Substat. <-- Centre)	**32**	**Values**	**12Bit**
Cycle range	**every 10 ms ... 32000s**			
Internal Counter	**every 10 ms, 100 ms, 1s**			
Communication-	**RS 232, WT, ISDN-network, IEC 245,**			**DCF-Time-**
Interfaces				**broadcasting**
Programming	**AWL, KOP, FUP, Step 5, etc.**			
Environment	**DIN/IEC 40040 (0...60°, Humidity class F ...)**			

Table: Technical data of a PLC feeder control device

2. Dynamic operational training

2.1 Why training for power system control?

The quality of the electrical power supply has reached a rather high level. The increased usage of loading limits of power system equipment and the cost optimisation result in the need for training for the control engineers. How can the efficiency of power systems and of power system operations be increased? How to optimize the power grid maintenance strategies? Answers to these questions lead to the idea of a tool for a risk free training to exercise and to develop skills. Training means a risk free exercise of all power system operations under normal and emergency conditions. The focus of the training goals may be very different depending on grid structure, voltage level and staff situation.
The training in a national dispatcher control centre is focused with respect to technical tasks like energy production, dispatching, load flow and voltage level optimization. Optimal power flow, minimizing losses under constrains of short circuit currents need special tools to solve. Experiences show that power system behaviour is influenced by both, deterministic and stochastic elements. If emergencies occur, the interpretation of the network alarm status, the event classification and the execution of proper control orders need a lot of operators' experiences. Missing experiences may result in operator' errors and may cause high cost. A frequent training will result in professional acting.

Medium voltage power systems are characterized by a need of reliability and security while the usage of equipment up to the limits is continuously increasing. Network operation should be supported by interlocking automatics and powerful software functions like colour topology indication and tracing. Due to the lack of remote lines there are only a small number of online measurements and the large amount of switchgear is controlled locally. As most of the medium voltage level problems result from unsymmetrical faults and the dynamic behaviour of loads and regulations a good coordination between control centre and grid staff operation is very important to avoid misunderstandings and accidents. A lot of training exercises appear also here to increase the professionalism.

How to increase the operators' skills? Operators are used to discuss problems theoretically by study cases. Consequently following the new technical possibilities the new dynamic training is a practical solution of the same basic idea. But there is a lot of progress because all discussions and exercises are executed on the operator's console and are dynamically simulated by software to perform a real-time power system presentation. Using the online training features integrated in the control centre the operators` skills can be trained very well.

Young engineers need to know the structure and the behaviour of the power grids. During the training they more and more become familiar with the network structure the network behaviour and the operation strategies. First doing normal and then complex and critical operations the operators' knowledge of power systems increase. As well does the knowledge about varied control system tools. Experienced engineers use the training as a tool to practice all network operations under normal and abnormal conditions. All emergencies can be trained. Even variations of preventive, corrective and restorative actions after serious changes in the network can be trained risk free without affecting the customers. Doing this the engineers can develop new strategies and increase their professionalism. Experienced operators from time to time need to train emergencies and to update their knowledge of how to handle the situation best.

2.2 General system design for online training in utilities.

The training is done on an original operator's workstation which is switched to the training mode. Therefore all SCADA MMI and control functions are available in the same way as during real grid operations. In the training mode all control orders are linked to a separate computer running the power system simulation software which calculates and presents the network reactions.

The basic concept is shown in fig. 2-1. See two parts of the control system. The upper part serves the online grid and operation. The remote lines communication is linked by the P-LAN and one dual main computer (MC1) with the SCADA- and MMI functions. After data processing the grid information address two operator consoles (OC1 and OC2) through the MMI-LAN to present the power system state. Online update and operation has to be executed permanently using OC1 and/or OC2.

The lower part in fig. 2-1 allows the online training during if training mode is switched on. The second dual main computer (MC2) also contains all SCADA and MMI functions. Instead to the remote lines it is linked to the network simulator (NS). NS is a software system working like an "artificial grid" producing an information flow similar to the real grid through the remote lines. Main control computer MC2 processes the artificial grid data while MC1 processes the real grid data. During a training session the control system works simultaneously with two processes, the real and the artificial grid. Real network operations are executed on operator consoles (OC1 and OC2) by MC1. Training network operations are executed by MC2 and OC3. The terminal down right (fig. 2-1) is used for as instructor console to supervise and influence the training state.

If training is not executed, MC2 is used for parallel grid data processing. OC3 can be used for additional parallel real grid operation providing a very high system redundancy.

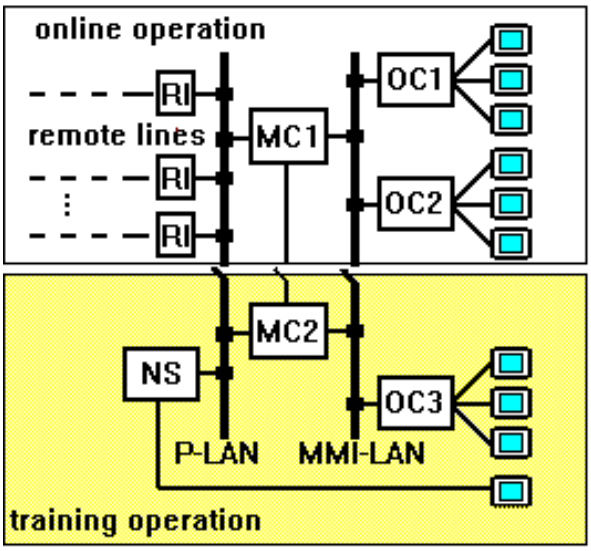


Figure 2-1: Control system with simultaneous online and training operation

2.3 Power system training in universities

The students' training concerning power system behaviour is used to be done by theoretical lectures and paper based practice. Power systems and its behaviour could hardly be rendered with mathematics in a descriptive way. Even for small networks in static and error free operations rather huge equations occur. The mathematical treatments become even more complex in case of faults.

The mathematical background of these calculations is rather complex and therefore most of the studies deal with simple network configurations. In practical power grid operations there is a strong demand to take the reciprocal effects of the network components into account. Main point is thinking about systems rather than on components. The effects can not be demonstrated in the real grids because of the high risks to customers and network components. The approach of simulation and training facilities very well support student's education if the simulation is authentic and done in real time. Two ways appeared to be helpful.

One is by presenting the network behaviour by a projector presentation. For this a PC-based standard control system like RESY-CIM and the network simulation is implemented on a portable PC (notebook). The PC is linked to a video projector by a standard graphic interface, e.g. VGA-out to DV-in. This configuration is shown in fig. 2-2. It is mainly used for power system network behaviour presentation and basic operations. Examples: how to change a busbar, how to influence voltages, minimize looses and optimize load flow, reaction of protection system in case of overload, reaction of a gas turbine in an island coming to overload and more.

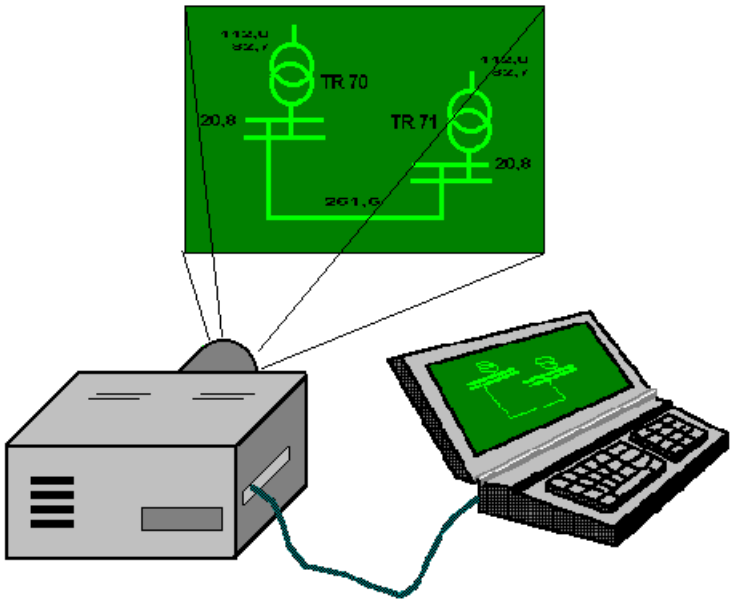
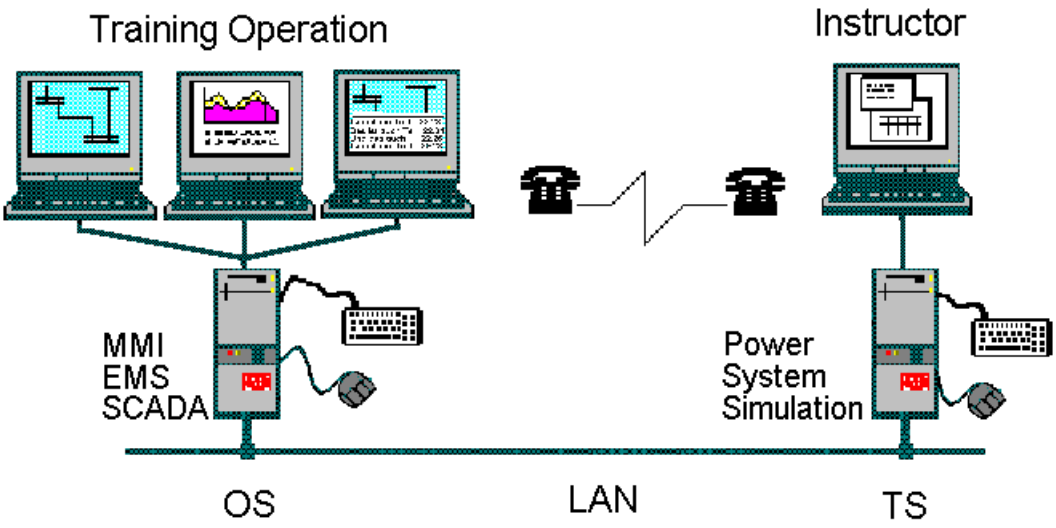


Figure 2-2: Offline configuration for network presentation

A second way to support students' understanding is a guided offline training, which needs more hardware. The hardware configuration is shown in fig. 2-3. This configuration is often used by universities´ and labs' application. Offline training means a stand alone system with no connection to any real power system. SCADA, EMS and MMI software is implemented in one PC, operation station (OS). New PC hardware construction allows running more monitors with the same graphic card presenting the power system status. Usually three monitors are used for displays such as station pictures, overview pictures and protocol lists on parallel. The power system simulation software is installed in the Instructor PC, also called Trainer Station (TS). Information exchange is done by a standard LAN link.

Figure 2-3: Offline configuration



PC Hardware:	PC – INTEL/AMD Processor
Operating system:	LINUX for TS and Windows for OS
Software system:	RESY-PMC (control system, MMI, SCADA and EMS) RESY-NES (power system simulation)

Students operate on console OS. All control orders given through the SCADA system pass to the network simulation software in TS. Network reactions, measurements and messages are calculated in TS station and then passed through the LAN to OS station and finally presented by the standard MMI functions. The instructor may optionally influence and supervise the training session.

If training is done in medium voltage level grids often the control orders can not be executed by remote lines. Control orders are executed by technical personnel working in the grid and contacted by a phone link. This can be simulated by a phone link installed close to OS and TS consoles.

2.4 Simulation software and general system design

Here is some basic information about the power system simulation software for the training. The network simulation software includes several detailed models for topology, generation, time-, voltage- and frequency-dependent loads, active and reactive power, network state calculation (power balance and frequency, load flow, short circuit, positive, negative and zero system), protection relay outputs, regulations and contact messages. Additionally a training session can be provided, controlled, recorded and replayed by the instructor. The instructor also can produce dynamic network events or sequences of events any time by a certain interface and menu. Fig. 2-4 gives an overview to the logical information processing and to the software tasks.

All operation commands (e.g. to breakers, other switchgear, transformers, generators etc.) executed and checked by the MMI pass through the main processing computer (OS) and is sent through the LAN to TS station. A task emulating the remote lines (RLE) receives the command. A receiver task (RCV) analyses the type of the command and passes it either to topology task (TOP) if switchgear is affected or to the load and generation task (LGT) if a generator control is ordered or to transformer tap task (TRF) if a tap shall be changed. The according task recalculates the state of the models. In any case the dynamic loads were always updated due to its time-load charts. If no command is given through the SCADA system, a timer with a cycle of one second starts the LGT task to update the loads. LGT task then starts the network calculations (NSC). NSC task calculates the actual network status, load flow and dynamics, taking existing short circuits, single earth faults or other events into account.

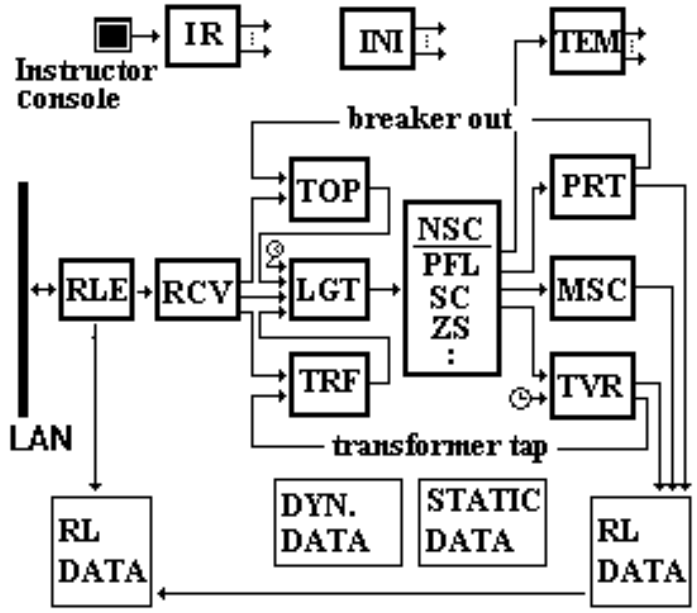


Figure 2-4: Overview to Power System Simulation Software inside TS

After NSC calculations of the complete network state is updated. Measurements, states and dynamics are calculated and stored in the data base. Then several tasks use that data to manage their duties. One task (TVR) analyses the busbar voltages and decides about changing the transformer tap positions. If so, a message is queued to TRF task to execute that in the next calculation cycle. Another task (PRT) checks the current flow with respect to overload criteria and network protection relays. If a breaker has to be opened, a message is queued to topology task to execute.

The according alarms are created and put into a telemetry buffer (RL DATA). This buffer is sent through the LAN to the SCADA system taking the remote line information flow parameters into account. Similar the measurements as bus bar voltages, active or reactive flow are put into the telemetry buffer by task MSC. Task TVR decides about the reaction of automatics, e.g. regulations, which are decentralized, controlling in the network frequency, voltage and load flow.

Another task (TEM) analyses if a trigger event has occurred. In that case, a prepared event file is started automatically. For example if a load of more than 90% appears for more than 10 minutes through a transformer, the "transformer oil high temperature" sequence is started followed after some minutes by the "transformer Buchholz protection" outage sequence. There are other instructor facilities for selecting, preparing, managing, recording and replaying the training session that will be not discussed here.

The whole software system is similar to a network of linked objects (tasks) which sends each other parameters and messages to recalculate the data and to decide about further reactions. The system is a mirror of the reality of interacting components and models in a way of an ever living system. If no control order appears, a one second cycle starts the whole calculation sequence by start of LGT task taking new values for the loads into account.

The control system software, EMS and SCADA parts, is constructed in modules in C programming language on LINUX platform. The MMI part is done in C on Windows platform. The power system simulation software is programmed in C on LINUX, also available on LINUX.

Some words to the dynamics in the simulation software. A simulation should be rather authentic and realistic. A poor simulation leads to problems. On the other side, any model is somehow limited and models need parameters to be declared. With respect to the inherent limitations:

- The simulation has to be authentic.
- Reactions of the power systems have to be calculated in real time.
- The reactions have to be processed in the control centre software like real data.
- All dynamic data that can be traced in the control room have to be calculated.

Although the progress in hardware is rather significant, a complicated and high detailed modelling needs CPU-time to be computed. Talking about real-time systems and dynamics there are several aspects to be taken into account:

- Remote lines and telemetry systems usually have cycles of about two to ten seconds.
- Real-time aspects require the calculation and presentation of net work information

during a cycle time. For this case, a simulator calculation cycle of one to two second seems to be sufficient.
- Network protection systems react in a shorter time e.g. in 100 ms or even less.
- The protection and indication dynamics were not presented in the control rooms, just the messages of relay start and relays execution including real time. Fast dynamics can not be seen in the control centre.
- Regulations of power and frequency have time constants of around one or some few seconds
- The CPU-time for all calculations depends from the workstation power and from the network size (number of components).

For load flow and regulation problems a cycle of around one second seems to be tolerable. In case of short circuit and protective relay actions a faster cycle is needed. Steps of 50 or 100 ms seem to be tolerable. If the overall calculation cycle can be done in one second, we have real-time simulation for load and regulation problems and a 1:10 extension for faster problems. Fortunately most of the network failures with one or two breaker falls only take a few hundred ms of real-time. If the trainer includes a short cut, the whole calculation of the events is done in background. After finishing calculations the event list is included in real time. The trainee does not see the background calculations and in most cases the trainer does not feel a significant time gap between mouse click to activate short cut and the messages.

2.5 Experiences in utilities and universities

The development of the training system started around 1988. At that time PCs were available on INTEL 286 base providing a frequency of 10 MHz. Today's (2007) state of the art PC of 3 to 5 GHz and memories of 1 to 2Gbytes one can say: Low cost industry PC and workstations can provide a real time feeling for grids up to a 1000 nodes.

The network simulation system has been developed since 1988 by the German former software company, repas AEG Automation, now Hereschwerke, and the University of Applied Sciences FH-Darmstadt, now Hochschule Darmstadt. Since 1989, prototypes have been used to support the students` theoretical lessons. In 1990, a lab "Power System Control & Operation Laboratory" was founded. Now (2007) there are a lot of installations in universities and utilities (around 30) using this training simulation for their applications. Students' education, continuing education and online training are the main applications. The system is also used for the factory acceptance tests before installing a new control system to a utility.

The following network exercises give an overview of tasks that are often used for beginners and for experienced operators.

- Exploring the control software features
- Testing the control functions and new data sets
- Remote line transmission and telemetry effects
- Standard control orders
- Parallel line effects, losses and voltage level
- Parallel transformer taps, circle currents and protection
- Investigation taps position automatics working in parallel

- Regulation of voltage with reactive power
- Voltage dependence of load
- How to influence the network losses and contingency status
- Communication to teams working somewhere in the network
- Check of relay parameters
- Strategies of failure location
- Strategies of network restoration after black-outs
- Economic aspects of power system operation
- Optimizing economic indicators

Experiences show that for beginners (undergraduate qualification) first should exercise easy examples and cases. How to do basic power system operations, how to influence the voltage? How to influence the flow? What are the effects of parallel network components? The reciprocal effects on power system components might be the second step.

Experienced power system operators sometimes have the problems of wrong or missing relays information from the network which makes it very difficult to interpret the network state and to locate failures. These exercises and generally the location of failures or power system operation in case of emergencies are the main focus by doing training in continuing education. Due to the history of emergencies and failures special training subjects can be treated during the seminars.

An additional application for training system appears during the change of the control system to the next generation. While the existing control system is continuously in use, the new one is build up first with just one new operation station connected to the network simulator. The simulator represents the original network model and produces a realistic environment. The new display layouts can be created and modified carefully including static and dynamic events. The alarm monitoring could be done very carefully. A new control centre generation often causes a change in handling the tools for operation and maintenance. These fields can also be trained well and risk free. The last step of system change is to switch over from the network simulator to the real remote lines when all operators feel experienced.

The coordination between on-line real grid operation and training is as this: In principle there is no difference between operating the real grid and operating the simulated grid: The same displays, the same functions, the same handling. To see the difference, usually the training simulation mode is indicated by a coloured bar in the displays. Of course, real power control has always the priority because the real grid control that has to be done continuously. If the training mode is needed, it runs in parallel, which could be done on a separate workstation or on one screen of the workstation. In this case, the operator can switch between two modes on the same workstation controlling either the real network or the simulated network. If there is more than one workstation, the second workstation with the same SCADA and MMI operates the training grid.

Finally a set of instructor functions have to be mentioned. It allows influencing the state of the components, recording of the information sequence during the training and a replay presentation. Using snapshots and replay function the whole sequence of events can be replayed and discussed again and again.

3. How to use the dynamic training simulator

3.1 First steps

When the system is delivered, it can be used immediately. Please connect the computers by LAN, plug in and switch on the power supply. The system has included a complete data set for an electrical power system to be controlled by the RESY PMC SCADA system. The network is described separately. All pictures and all process information are installed so that the system can be used directly. Of course the system is sophisticated and a brief introduction information has to be checked before starting to work. Please do it that way.

First plug in the components and switch on (230 V) . The actual starting procedure will be explained by a separate sheet.

3.2 Basic handling of the control system and some important functions

In the lab there are right now 5 identical training stations each with 3 monitors to work on plus one monitor to allow trainers' interactions, so one training station consists on 2 computers and four screens. A LAN connects the computers ready for remote control data communication. On all training stations the training can be executed independently because each simulator has declared its own grid.

The SCADA system presents all basic functions as display updating, log files, event lists, trend curves, operation etc. You will see an introduction given by the instructors.

System boot

Please get the current identifications and sequence to tart the system from your instructor.

After boot the system is ready for training when presenting the overview picture of the implemented power grid as given in figure 3.-1. The display shows the main stations, lines, transformers, loads etc.

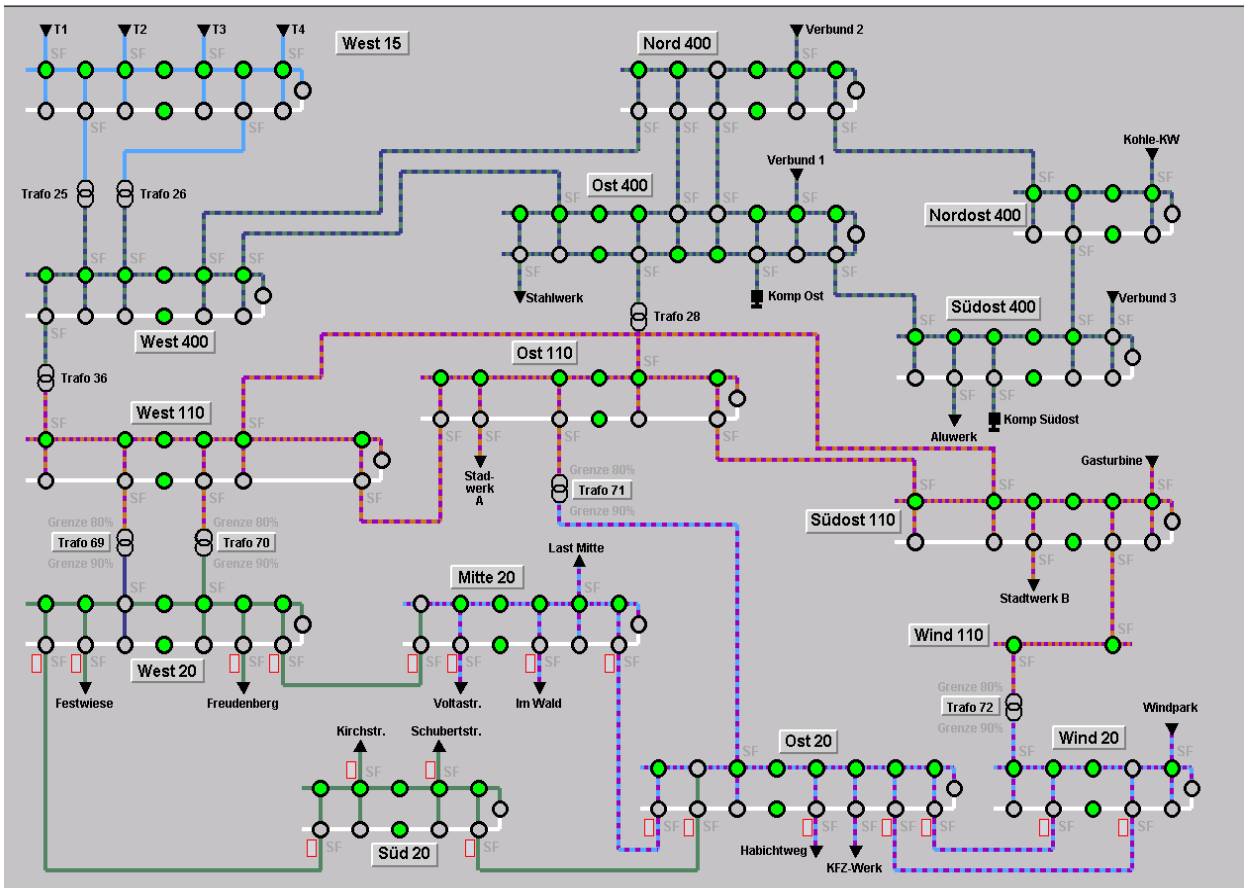


Figure 3.-1: Power system overview picture

End of training session

Please use the standard switch off sequence any time to stop the training session. Switch off the computers' power supply when the sequence has come to its normal end.

Menu

In the upper line a menu list to operate the control system is given on which the buttons can be selected by the curser and a mouse click, see figure 3.-2. Just do your experiences.

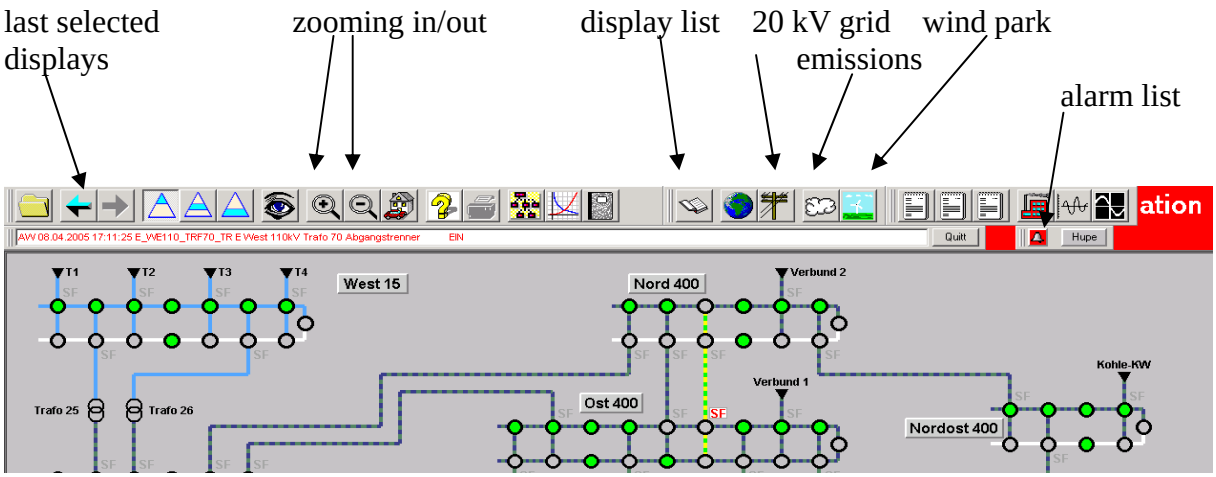


Figure 3.-2: Menu line

Basically all displays can be selected by the display list button or by picture-from-picture selection by a mouse click (left mouse button).

Select a picture by a single mouse click (left mouse button)
Select a component for operation by a double mouse click (left mouse button)

After selection of e.g. a breaker for operation a switchgear box appears. The selected device is presented by is PV-name, text and current state. Dangerous operations will be warned. See figure 3.-3.

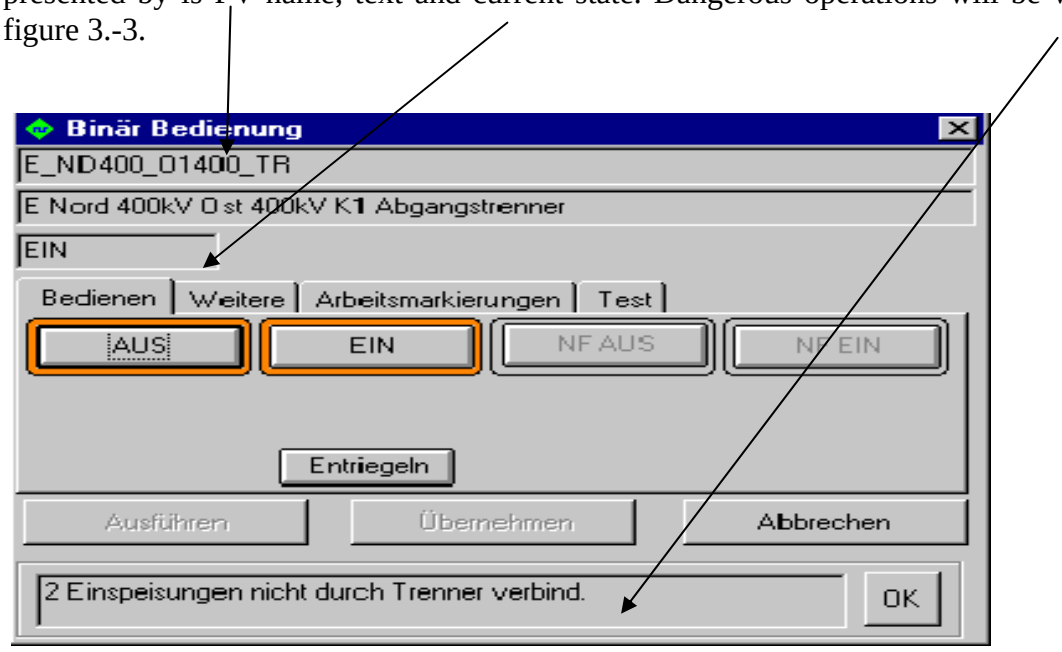


Figure 3.-3-: Switchgear box

The display list symbol (like an open book) gives the structure of the implemented displays to operate the power system, see figure 3.-4.

- Overview pictures (Netzbilder/Daten)
- Handling of neutral star points (MS- Sternpunkte)
- Regulations (Regelung)
- Emissions (Umweltdaten)
- Stations (Stationen 400kV, 110kV, 20/15kV)
- Wind park

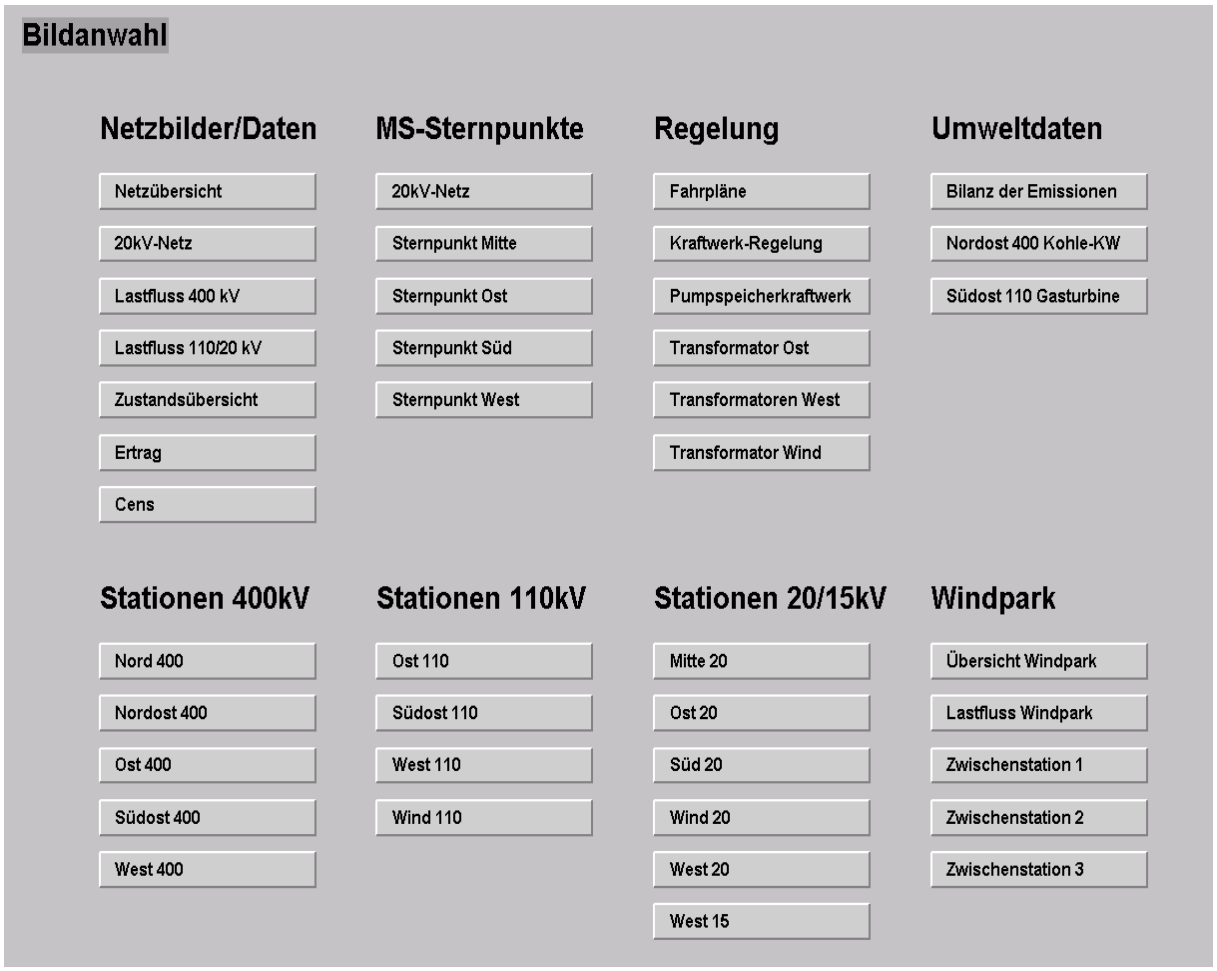


Fig. 3.-4.: Picture selection display

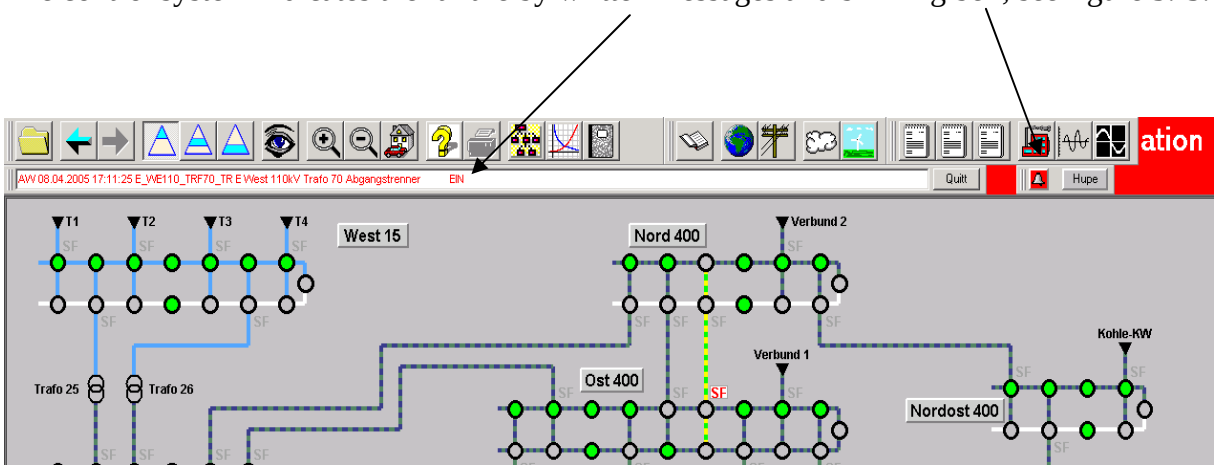
Just press the according button to enter the selected display.

Example: The button 

calls the power system overview display.

Alarms, alarm guidance and acknowledgement

Warnings and emergencies will address the bell and the alarm list. Example: Due to a failure operation the right hand side line from Nord 400 to Ost 400 which was connected in Ost 400 was grounded which causes a three phase short circuit.
The control system indicates the failure by written messages and blinking bell, see figure 3.-5.



In some protocols and event lists (log file) is additional information available.

08.04.2005 17:15:44	Sample	Sample	Allgemeine Mel	E Nord 400kV Ost 400kV K2 Erdungstrenner	E_ND400_O2400_ETR	EIN [-]	Administrator	sok
08.04.2005 17:15:44	Sample	Sample	Allgemeine Mel	E Nord 400kV Ost 400kV K2 Erdungstrenner	E_ND400_O2400_ETR	EIN [-]	Administrator	sbf
08.04.2005 17:15:44.657	Sample	Sample	Kommender Al	E Ost 400kV Nord 00kV K2 Dist. Generalanreg.	E_OS400_N2400_DI_GEN	KOMMT [-]		ala
08.04.2005 17:15:44.658	Sample	Sample	Kommender Al	E Nord 400kV Ost 400kV K1 UMZ Schutzanregung	E_ND400_O1400_UMZ_SA	KOMMT [-]		ala
08.04.2005 17:15:44.658	Sample	Sample	Kommender Al	E Nordost 400kV Kohle-KW UMZ Schutzanregung	E_NO400_KOHKW_UMZ_S	KOMMT [-]		ala
08.04.2005 17:15:44.662	Sample	Sample	Kommender Al	E Ost 400kV Nord 00kV K2 UMZ Schutzanregung	E_OS400_N2400_UMZ_SA	KOMMT [-]		ala
08.04.2005 17:15:44.662	Sample	Sample	Kommender Al	E Nordost 400kV Nord 400kV UMZ Schutzanregung	E_NO400_ND400_UMZ_SA	KOMMT [-]		ala
08.04.2005 17:15:44.957	Sample	Sample	Allgemeine Mel	E Ost 400kV Nord 00kV K2 Dist. Schutzauslösung	E_OS400_N2400_DI_AUS	KOMMT [-]		
08.04.2005 17:15:44.957	Sample	Sample	Kommender Al	E Ost 400kV Nord 00kV K2 Dist. Schutzauslösung	E_OS400_N2400_DI_AUS	KOMMT [-]		ala
08.04.2005 17:15:44.965	Sample	Sample	Allgemeiner Ala	E Ost 400kV Nord 00kV K2 Leistungsschalter	E_OS400_N2400_LS	AUS [-]		ala
08.04.2005 17:15:45	Sample	Sample	Kommende Me	E Ost 400kV Nord 00kV K2 Schalterfall	E_OS400_N2400_SF	KOMMT [-]		
08.04.2005 17:15:45	Sample	Sample	Kommender Al	E Ost 400kV Nord 00kV K2 Schalterfall	E_OS400_N2400_SF	KOMMT [-]		ala
08.04.2005 17:16:14	Sample	Sample	Gehender Alarr	E Nord 400kV Ost 400kV K1 UMZ Schutzanregung	E_ND400_O1400_UMZ_SA	GEHT [-]		alg
08.04.2005 17:16:14	Sample	Sample	Gehender Alarr	E Ost 400kV Nord 00kV K2 UMZ Schutzanregung	E_OS400_N2400_UMZ_SA	GEHT [-]		alg

After clicking the bell, automatically the event file is loaded. Clicking on an event in the list, automatically the station display will appear.

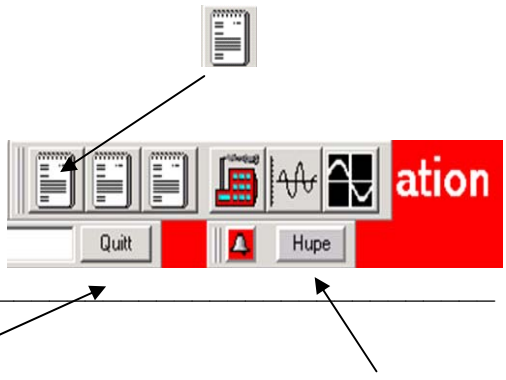


Figure 3.-5. Alarms and event lists

All alarms have to be acknowledged by “Quit” button. The acoustic bell can be stopped by the “Hupe” button.

If an alarm occurs, the SCADA system sends a message in the protocol and to the bell. First check the protocol and the overview picture. This will show information and /or indicators to understand the problems and to locate the station. What information has appeared? What is the new state? Is the grid state stable or is any failure development still going on?

Note: Don't quit the alarms until you haven't understood the problem. Select detailed information in the same or in other displays to fix the problem. Only if you are sure having understood the problem you may quit the alarm. You may quit the bell any time. Decide then about priorities to react.

3.3 Trainer interface

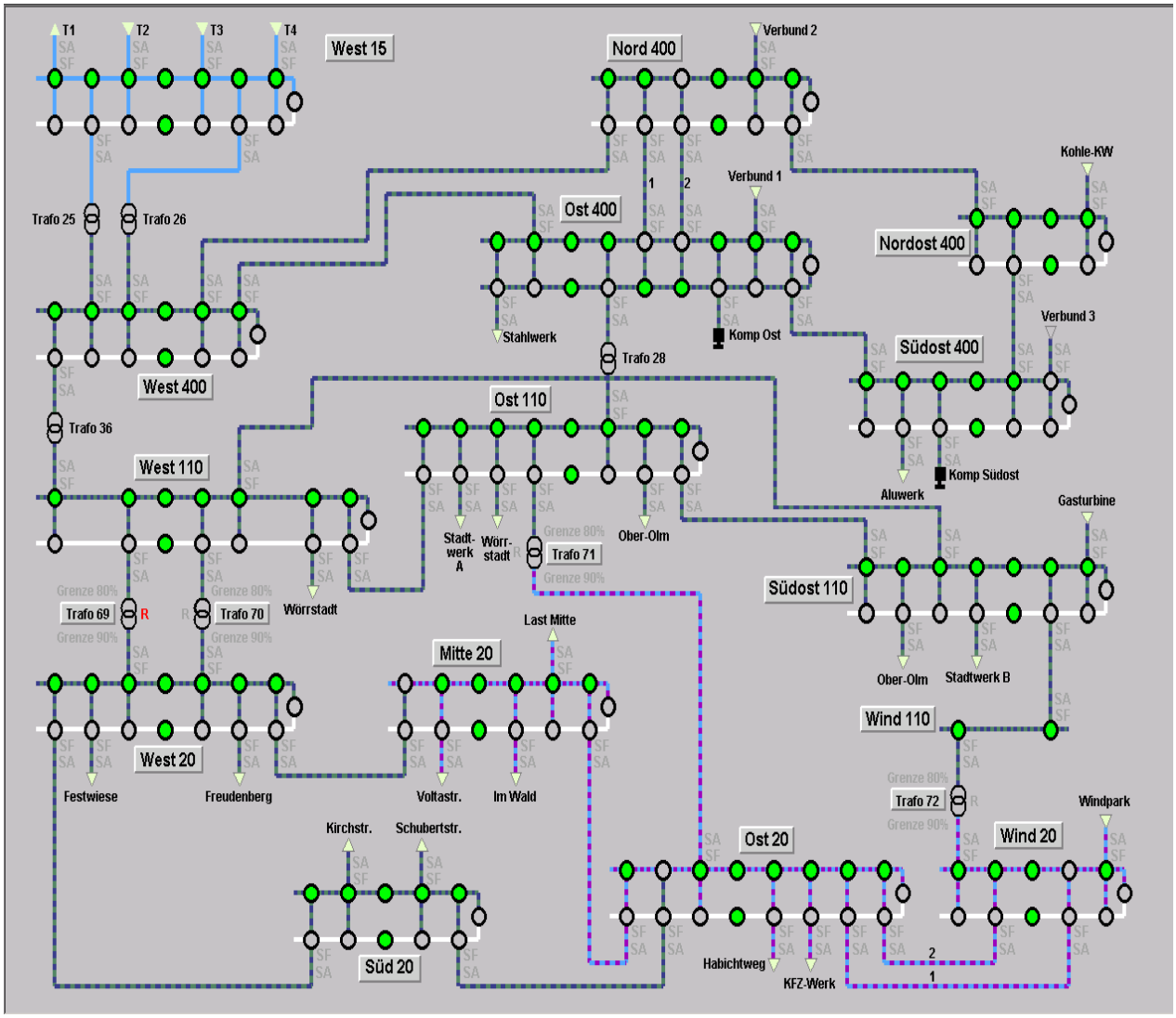
Is not described here.

4. Power system description

4.1. Power system network and components

Here is a brief summary and introduction of the network and the network components. For exercises' understanding it is strongly recommended to become familiar with the implemented power system and components. Additionally the first exercises will help to catch the main ideas.

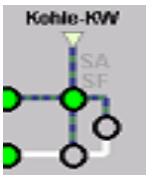
The power systems and all components belong to a grid company called "FHD-AG" which represents a private company. The electrical power system is constructed with the following voltage levels: 400 kV, 110 kV and 20 kV including distribution network plus a 15-kV station with four turbines for a hydro storage, see grid overview. Please note that the station names as "Nord 400" or "West 110" indicates the voltage level.



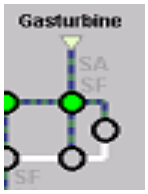
Power Generation and loads

The power system has three overhead lines as connections to border utilities to exchange active and reactive power, enough to feed all internal loads even if all internal plants are switched off. There are some internal plants to produce electrical energy:

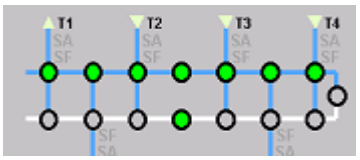
- A coal plant (Kohle KW) in station Nordost 400 is able to produce up to 500 MW feeding the 400 kV grids.



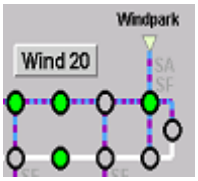
- A gas turbine (Gasturbine) in station Südost 110 is able to produce 100 MW peak. This turbine feeds into the 110 kV level.



- The hydro storage water plant (T1 ... T4) in station West 15 is able to store around 1500 MWh by a maximum power of +/- 500 MW.



- There is a wind park with some 20 windmills of different type. The peak injection is 22 MW.



The peak load of the whole network is about 1100 MW active power. During peak load the link to the border utility is important to support all loads as a normal topology.

Implemented RTUs

All 400-kV and 110-kV stations have a link to the control centre by RTUs and remote lines. Also the medium voltage 20-kV substations have RTUs and remote lines, the 20 kV distribution grid and its transformer stations to 0.4 kV don’t have remote connection.

All track line transformer stations and the local switch gear can only be operated by the network personnel that have to drive to the stations and to operate by control centre supervision. The grid operations are supervised by the control centre and a phone link to the grid staff. These communication phone calls and all switch operations done by the grid staff is simulated by the trainer.

4.2 Some important parameters of the electrical power network components

a) Lines and cables

400-kV overhead lines: 4x Al/St 240/40 Overhead line

Ohm Resistance	$R' = 32{,}0 \cdot 10^{-3}$ Ohm/km
Inductivity	$L' = 0{,}809\ 10^{-3}$ H/km
Capacity	$C_r' = 14{,}51 \cdot 10^{-9}$ F/km
Capacity to ground	$C' = 7{,}8 \cdot 10^{-9}$ F/km
Rated Current	$I_r = 2580$ A

110-kV overhead lines: 1x Al/St 380/50 Overhead line

Ohm Resistance	$R' = 75{,}7 \cdot 10^{-3}$ Ohm/km
Inductivity	$L' = 1{,}02\ 10^{-3}$ H/km
Capacity	$C_r' = 10{,}0 \cdot 10^{-9}$ F/km
Capacity to ground	$C' = 5{,}5 \cdot 10^{-9}$ F/km
Rated Current	$I_r = 840$ A

20-kV cable lines connecting substations: VPE Cable Cu 120 mm^2

Ohm Resistance	$R' = 149 \cdot 10^{-3}$ Ohm/km
Inductivity	$L' = 0{,}6 \cdot 10^{-3}$ H/km
Capacity	$C_r' = 230 \cdot 10^{-9}$ F/km
Capacity to ground	$C' = 230 \cdot 10^{-9}$ F/km
Rated Current	$I_r = 400$ A

20-kV cable lines distribution grid: VPE Cable Cu 95 mm^2

Ohm Resistance	$R' = 192 \cdot 10^{-3}$ Ohm/km
Inductivity	$L' = 0{,}63 \cdot 10^{-3}$ H/km
Capacity	$C_r' = 220 \cdot 10^{-9}$ F/km
Capacity to ground	$C' = 220 \cdot 10^{-9}$ F/km
Rated Current	$I_r = 360$ A

b) Transformation

400/110 kV TRAFO 28 and 36

Rated apparent power	$S_r =$	**150 MVA**
Rated current	$I_r =$	**230 A**
Inductive zero current	$I_0 =$	**0,3 A**
Copper losses	$P_{cu} =$	**700 kW**
Short cut voltage	$u_k =$	**18%**

110/20 kV TRAFO 69, 70 and 71

Rated apparent power	$S_r =$	**31,5 MVA**
Rated current	$I_r =$	**185 A**
Inductive zero current	$I_0 =$	**0,25 A**
Copper losses	$P_{cu} =$	**170 kW**
Short cut voltage	$u_k =$	**14,7%**

c) Regulations

110/20 kV voltages

TRAFO 69, 70 and 71 regulations
Voltage regulation to 20 kV bus bar within 27 taps (+13...0...-13)
Voltage steps around 1% each tap
Voltage regulation manual (OFF) or automatic (ON)
Target voltage (adjustable) 20.5 kV, Hysteresis 0,3 kV

400 kV voltages

KW2 voltage regulation
Voltage regulation to 400-kV bus bar
by reactive power injection (+500 MVAr ...0...-500 MVAr)
changing excitation current of generator
Regulation automatic ON or OFF

Frequency

Primary frequency regulation automatic (regulates Δf to zero)

Interlocked networks	**ON (only)**
KW2	**ON and OFF**
Hydro water plant	**ON and OFF**
Gas turbine	**ON and OFF**

Secondary frequency regulation automatic regulates
either Δf_{50} to zero or own power injection to target value

Interlocked networks	**ON (Δf_{50} only)**
KW2	**ON and OFF**
Hydro water plant	**ON and OFF**
Gas turbine	**ON and OFF**

d) Types of loads and loads

All loads are time-depended: active power P=f(t) and reactive power Q=f(t).

All loads are voltage dependent by a parameter (p, q) declaration:

Active load: $P(U) = P_r (U/U_r)^p$
Reactive load: $Q(U) = Q_r (U/U_r)^q$

U actual voltage
r rated (nominal)
U_n rated voltage
p, q Parameters describe type of voltage-load dependence (0.. 4)
p, q = 0 is a constant load
p, q = 1 is a constant current load
p, q = 2 is a constant impedance load

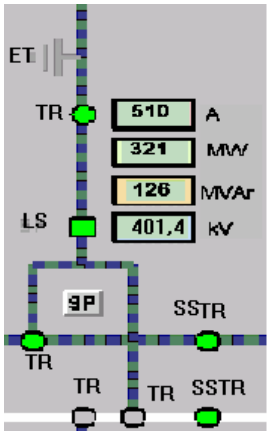
All loads are frequency dependent

Loads	loads	voltage [kV]	peak [MW]
	in Süd-West (Aluminum factory)	400	450
	in Süd-Ost (Iron factory)	400	350
	in Mitte 110 (Town)	110	125
	in Ost 20 (KFZ-Company)	20	10
	households	20/0.4	35

e) Reactive power compensators

For compensation of reactive power in two 400 kV substations are compensators accessable, each of rated ±100 MVAr. You can change values by means of the picture "Fahrpläne". Compensations are in station Süd-West - K1 and in Süd-Ost - K2

f) Naming conventions for switchgear



The switch gears have the following names:

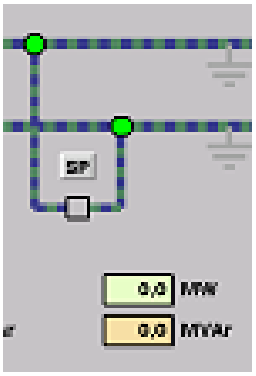
Breaker (Leistungsschalter, LS)

Bus-bar isolator (Sammelschinen-Trenner, TR)

Earth switchgear (Erdungsschalter, ET)

Line (feeder) isolator (Abgangstrenner, TR)

Busbar coupling switchgear (Längstrenner) (SSTR)



Bus-bar coupling (Sammelschienen-Querkupplung)

Measurement names

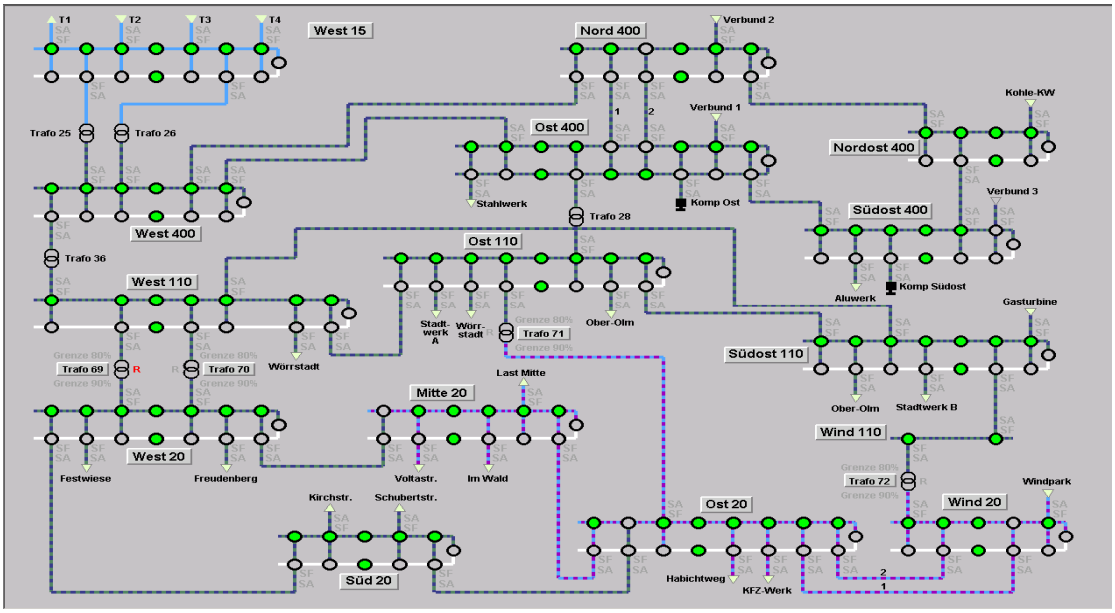
Each name starts with letter **T**. The second letter gives information about the kind of the measurement:

- **I** current measurement [A]
- **U** voltage measurement [kV]
- **P** active power measurement [MW]
- **Q** reactive power measurement [MVAr]
- **f** frequency measurement [Hz]

4.3 Explanations to the SCADA system displays

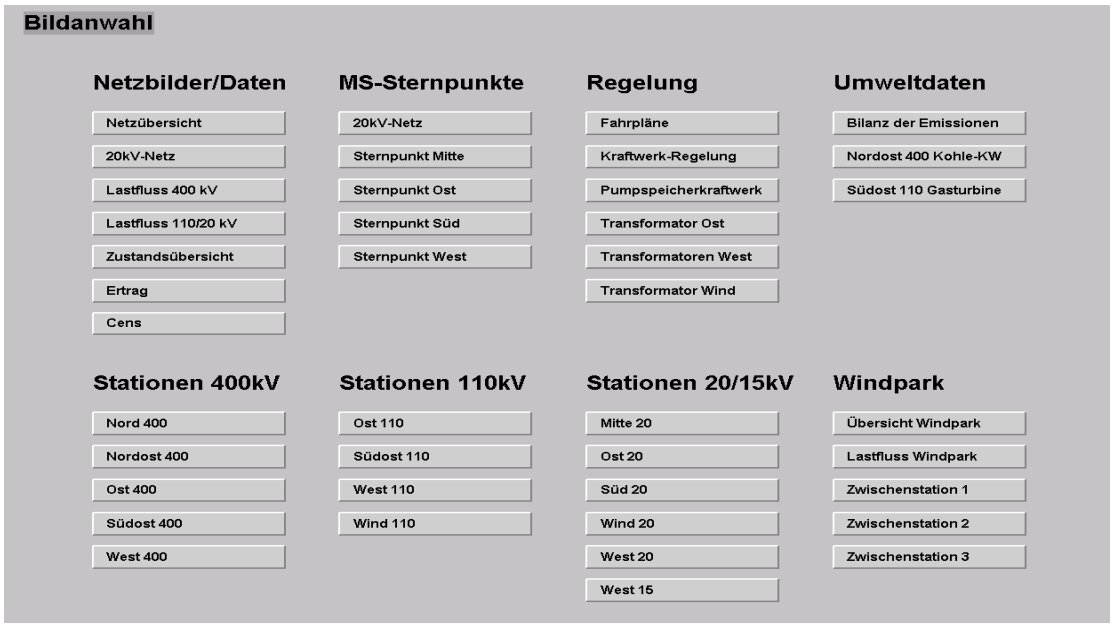
Network overview picture

The network overview picture (start picture) will be presented after start of the system automatically. It shows the topology and brief information of the topology and customer's energy supply. All stations are presented and the topology connections can be seen. A green mark indicates a connection. From this picture no process order can be emitted. By positioning the cursors to a picture variable (e.g. name button of a station) and after pressing left mouse key the other picture is selected.



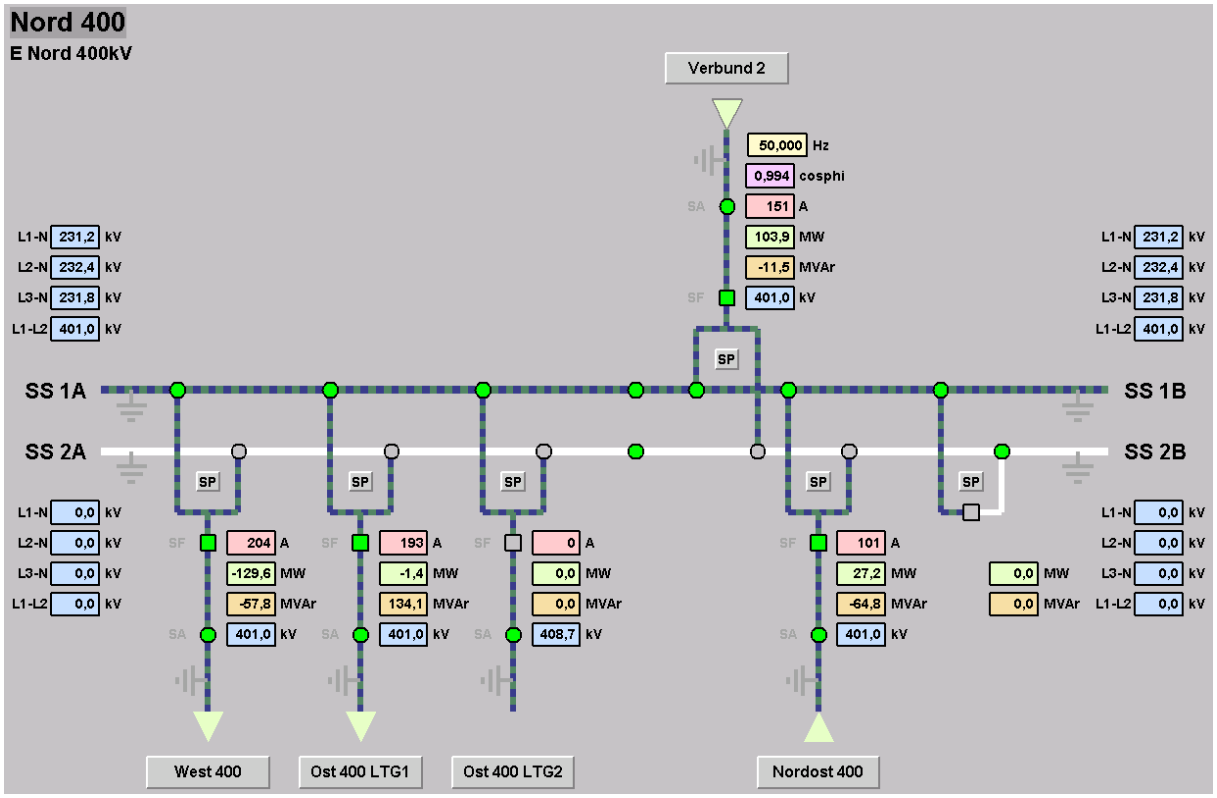
Display selection by menu symbol:





Station pictures

In this pictures al information of the actual station / substation are presented. The single feeders, breakers, switchgears are presented in more detail including their status and position. Generally a green symbol indicates ON position and a grey symbol indicates an OFF position. Lines, transformers and couplings are monitored by symbols and the actual voltage, current, active and reactive powers are presented by values. Most of the names lead to the next pictures. Breakers, isolators, state of automatics, target values for regulations, transformer taps and some other switchgear are declared to be controlled by picture operation.



Measurement signs

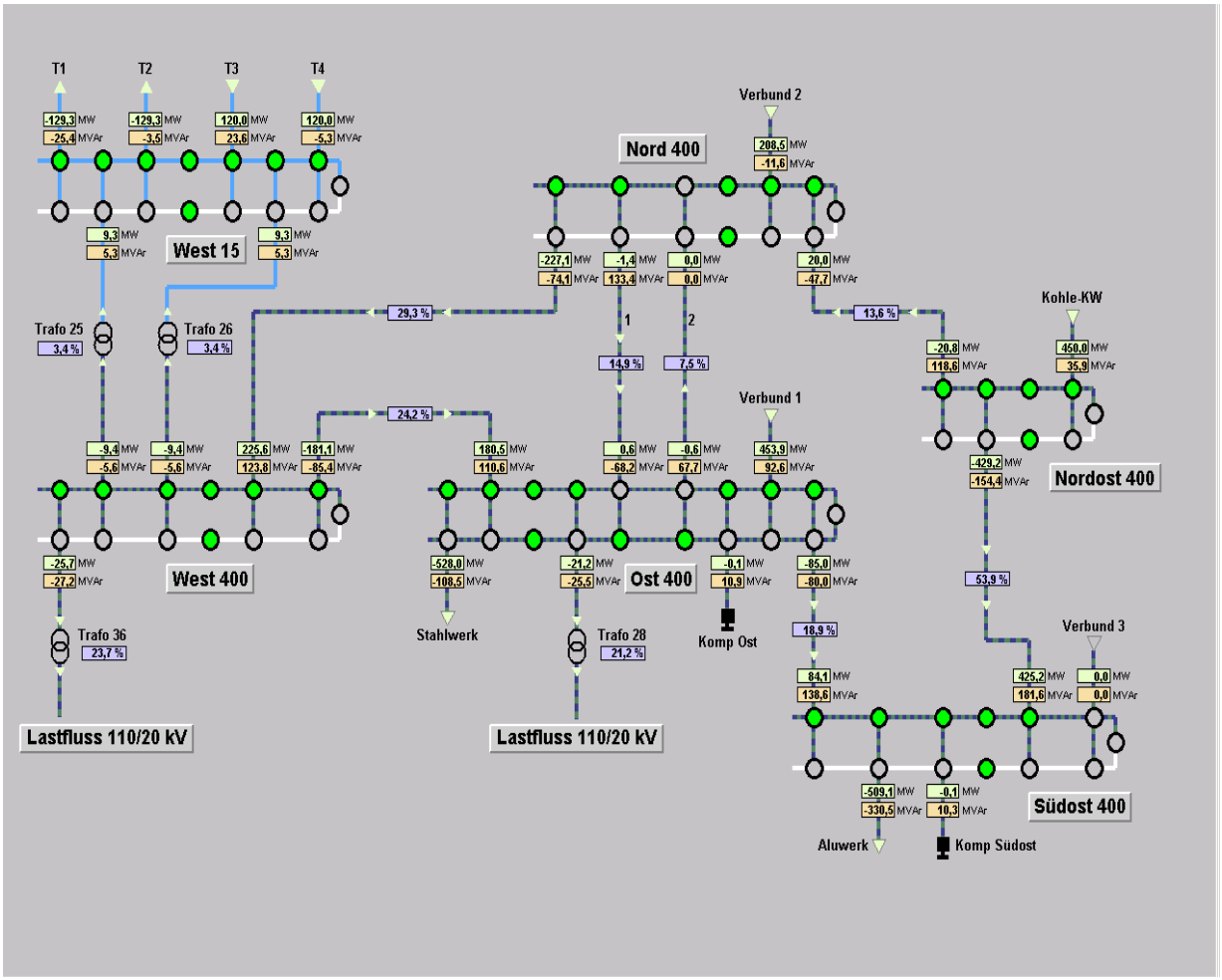
Power flow: (3 phase)	+ coming to busbar	active power flow or reactive inductive power flow
	- going from busbar	active Power flow or reactive inductive power flow

Voltage of feeders: Absolute value of phase to phase voltage
Voltage of busbars: Absolute value 3 phase to ground and 1 phase to phase

1 Line current: Absolute value of phase current

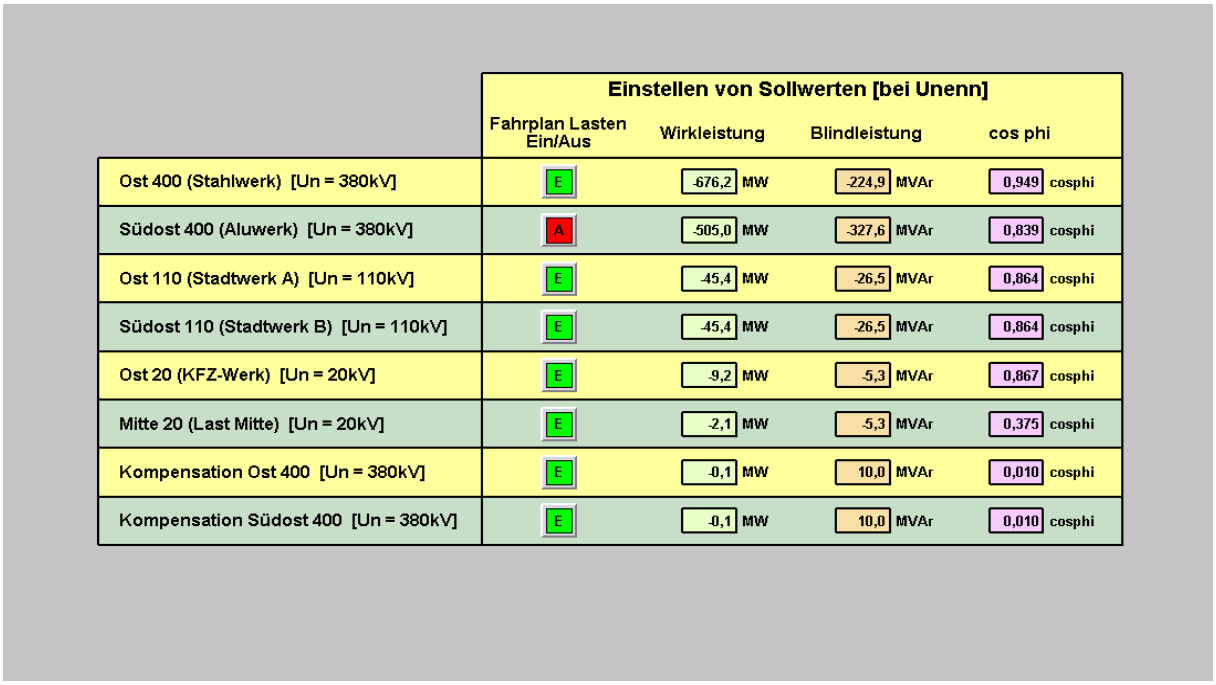
Power flow "Lastfluss 400 kV" and "Lastfluss 110/20 kV"

Two pictures present the power flow. One is for the 400 kV and the other for the 110/20 kV level. These two pictures are for information not for sending control orders. The values belong to that feeder that is close to the symbol. In addition the load percentages of the network components are monitored as well as the direction of active load by a triangle.



Load values "Fahrplanbild"

In the picture "Fahrpläne" all values for reactive and active power of the plants are presented and can be changed. Before changing a load the time schedule has to be switched off (Fahrplan aus). This is even possible for loads. It allows a study mode for future load and topology.



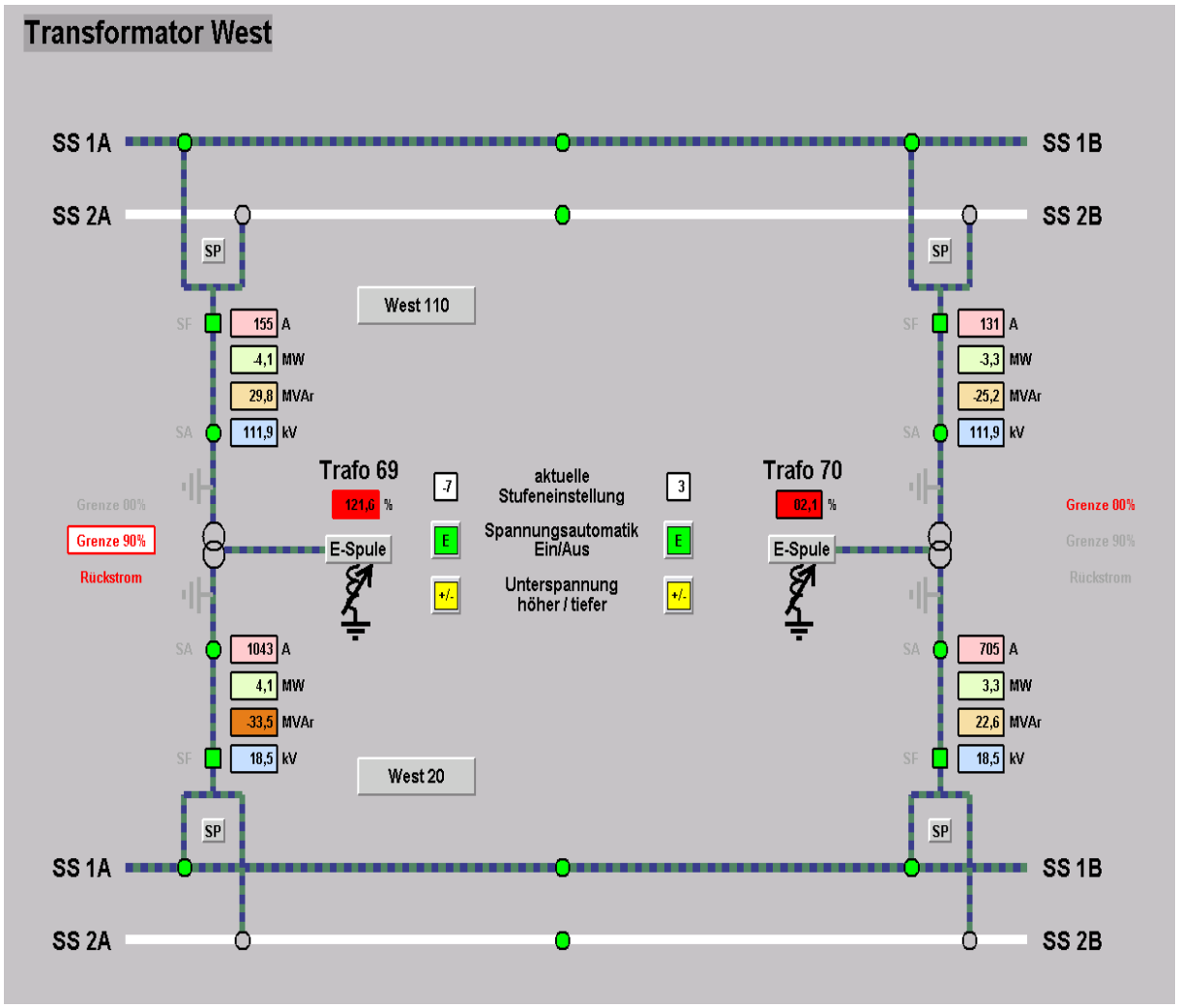
	Einstellen von Sollwerten [bei Unenn]			
	Fahrplan Lasten Ein/Aus	Wirkleistung	Blindleistung	cos phi
Ost 400 (Stahlwerk) [Un = 380kV]	E	-676,2 MW	-224,9 MVAr	0,949 cosphi
Südost 400 (Aluwerk) [Un = 380kV]	A	-505,0 MW	-327,6 MVAr	0,839 cosphi
Ost 110 (Stadtwerk A) [Un = 110kV]	E	-45,4 MW	-26,5 MVAr	0,864 cosphi
Südost 110 (Stadtwerk B) [Un = 110kV]	E	-45,4 MW	-26,5 MVAr	0,864 cosphi
Ost 20 (KFZ-Werk) [Un = 20kV]	E	-9,2 MW	-5,3 MVAr	0,867 cosphi
Mitte 20 (Last Mitte) [Un = 20kV]	E	-2,1 MW	-5,3 MVAr	0,375 cosphi
Kompensation Ost 400 [Un = 380kV]	E	-0,1 MW	10,0 MVAr	0,010 cosphi
Kompensation Südost 400 [Un = 380kV]	E	-0,1 MW	10,0 MVAr	0,010 cosphi

Costs "Kostenübersicht"

In this picture a commercial balance with respect to the cost situation is presented from the utitities' point of view. Outages and high losses can influence the balance extremely.

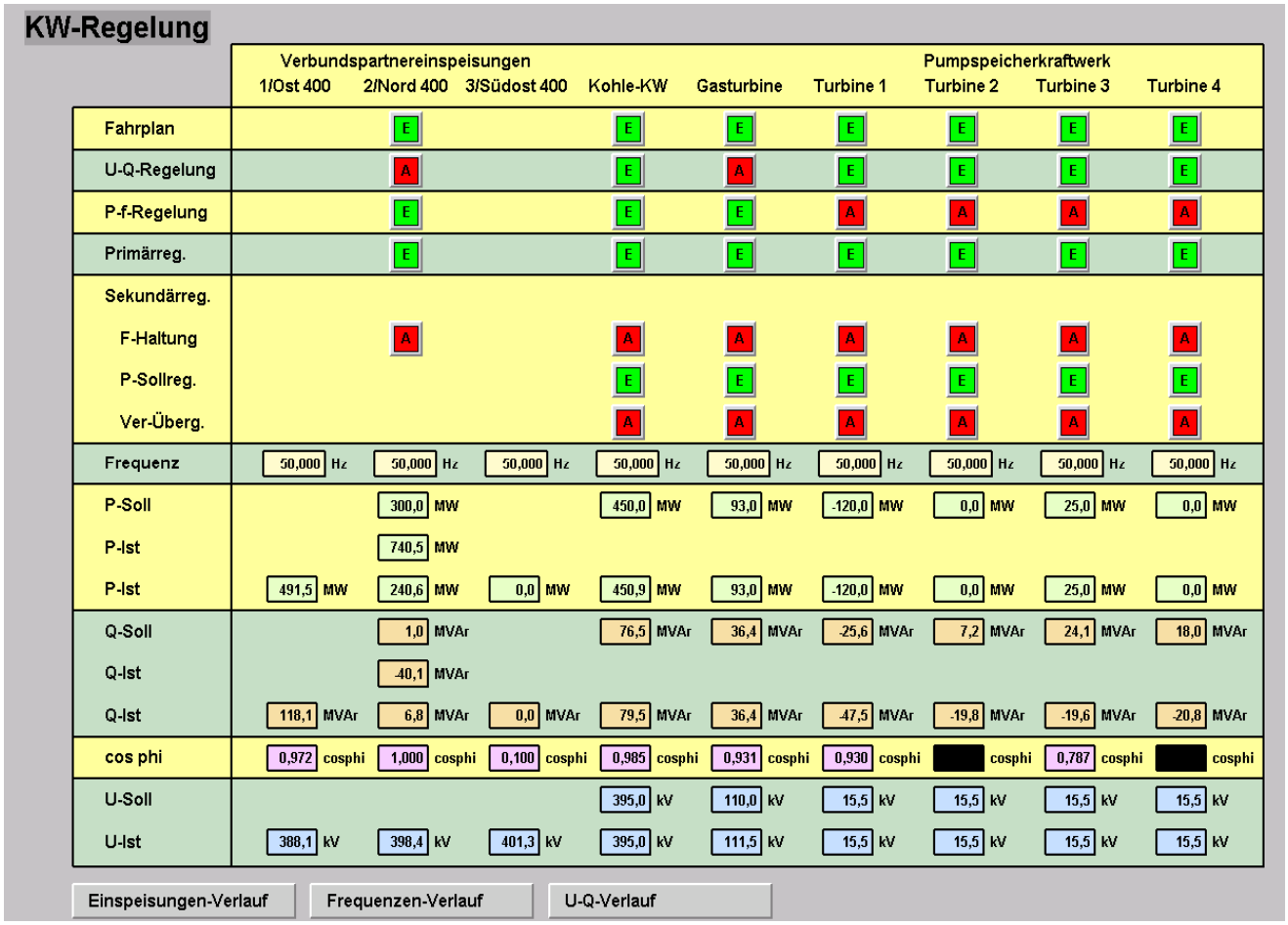
Transformer regulation "Trafo-Regler-Bild”

There are some pictures especially to control the transformer regulations from 110 to 20 kV, one for each 110/20-kV transformer station. E.g. one picture monitors the transformers Trafo_69 and Trafo_70 and the other display shows the Trafo_71. The voltage regulation automatic can be switched from ON to OFF (EIN/AUS) and transformer taps can be ordered + or – changing the windings and by this the voltage transformation. The regulation works with respect to the 20 kV - level and regulates e.g. for 20.3-kV ± 0.2-kV (hysteresis).



Power plant regulation control " KW – Regelung ”

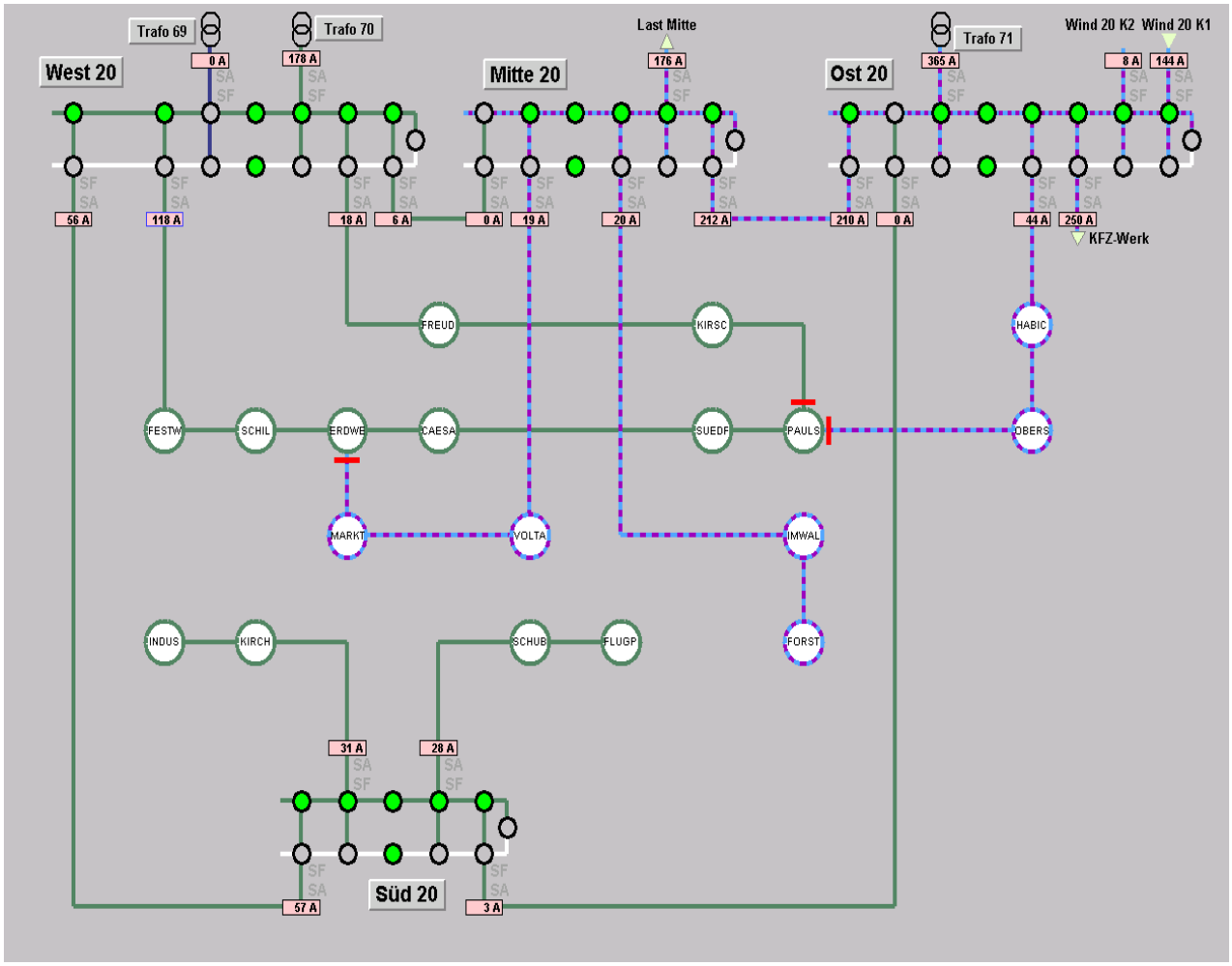
The picture allows controlling the frequency and voltage regulation of all Generators like Kohle-KW. The voltage is controlled by reactive power injection, the frequency by primary and secondary active generation regulation. Green E means ON (Ein) and red A means OFF (Aus). Regulations will be described during the meeting hours.



20 kV distribution network "20-kV-Streckennetz"

In this picture an overview of the 20-kV-distribution network is monitored.

Please note: In MV grids only in the 20 kV main stations the switchgear can be controlled by SCADA and RTUs. In the track lines the switchgear of the transformer stations (20kV/0.4kV) are operated locally and manually by the grid team. The operation orders are given by the control centre staff and means of phone calls. After changing the switchgear locally no automatic update is done because of missing RTUs. So it has to be updated in the SCADA data base. This is done manually in the control centre by using the function "Nachführung - Handeingabe".



5. Exercises

All network exercises are described in detail here starting with this brief summery. The training sessions can be prepared from the instructor console. After starting the system the static as well as the dynamic data are loaded. The instructor can select and load other networks status files concerning topology, load and other items that are stored before in the trainer's function "snapshot". These dynamic network status files can be stored any time the simulator runs. The network status files include network topology, status for voltage regulation, schedule for the plants and loads and all dynamic items to fix a complete network status. By loading a new network status file, some alarms may occur due to the change from one status to another. The alarms can be quitted by the control system functions.

From the operator's console the network now is ready to be controlled and supervised. The instructor can influence the network status any time. By this he can influence also the alarm status of the network. Alarms, breaker falls, disconnecting loads and plants, malfunction of regulations, locking of remote lines, single earth connections or short circuit can be influenced... Even lists of events can be activated. These functions are described in the trainers´ interface in more detail.

The students' network tasks aim to increasing the understanding of electrical power networks as well as to educating with respect to control system functions in addition to theoretical lessons. For some practical reasons a phone link is recommended to simulate the communication coordination to grid personnel. By this the operator training is very close to the real network duties.

The network allows operating a three voltage level power system including dispatcher tasks as well as tasks to manage distribution level problems. This network is artificially designed with lines, transformers generators that are installed in real networks. Due to the time to handle this network training system, the network should be complicated enough to allow practical tasks on one side. And on the other side it should be simple enough to understand the interactions of the components. Anyway the network can be changed for special tasks to be simulated.

The students were guided by several tasks and exercises through the course supported by the supervisors. It is strongly recommended to follow the guideline of tasks and exercises. Some of the exercises need knowledge of the technical background. Therefore the main background is addressed before in the "Theoretical Sessions". If additional technical background is needed please feel free to ask your questions or get the references to books or papers from the supervisors.

Please note: During the meeting hours and exercises your group shall open a doc-file and shall make hardcopies (screenshots) from important pictures, starting situations, trends, results etc. to prepare a lab-report (one report per group) containing the exercises and results. It is strongly recommended to write the report soon after the exercise hours and to send it via mail to the supervisor.

Please use the name structure ”No-report-name1-name2-name-3” where No is the current number of the group’s report.

The reports will be reviewed and sent back with comments.

1. Page:

Lab-report No.

Exercise No. xx to No. yy

Date

Group:	Name 1	Matr.no.
	Name 2	Matr.no.
	...	

1. Getting familiar with SCADA control system

Goal: Getting the general idea of the SCADA- and Training system

Discuss the following questions before start of the computers.

- What is a SCADA system and what are the basic functions?
- What are the advantages of training?
- Please feel free to ask your personal questions!

For your written report:

- Sketch out the hardware of the training station briefly and describe the software packages.
- Answer the questions above.

2. Basic operations of the SCADA system

Goal: Getting knowledge of the SCADA's basic features and of handling

2.1. Start the computer(s) of your training station.

2.2 Ask for a short introduction how to operate the control system, to select displays, to open a doc.-file and to do screenshots. After that you should be able to select all displays by mouse click, to handle the main operational sequences. And you should know how to start and stop the system.

2.3

2.4 Open a personal directory like "SS2010-miller-…" and a doc-file for every session like "1-session"

For your written report:

Describe the sequences how to operate the system.

3. Displays presenting the power system state

Goal: Getting knowledge of the display information presenting the power system state

Get familiar with the overview-, station-, line- and device-displays. Make some screenshots.

Select the displays by using the picture display (access by menu-button “open book”) or by “picture from picture” selection and mouse clicks (single click left mouse button)”.

For your written report:

Describe

- How can you select the pictures?
- What different types of picture (-groups) exist in the control system and what are they needed for? E.g. station displays for switching operations, load flow display for dispatcher operations, …
- Give a hierarchical tree structure of the displays in accordance with increasing details.
  E.g. general system overview without topology, load flow display with condensed topology and measurements, station displays with detailed topology and all measurements
- What grid control operations occur usually (daily)?
- What grid control operations occur during emergencies?
- Which display/picture gives information about total system losses and the according costs of losses?
- Which displays present the active and reactive power flows best?
- Which group of displays presents the line currents?

4. Getting more familiar with the power system

Goal: Getting more familiar with the construction of the power system, e.g. voltage levels, transformers, stations and components...

- Which different voltage levels exist in the implemented power system?
- Determine the 400kV-line with the highest relative load (%). How big is the line load (%)? How big are the line losses (MW)?
- Which transformers are constructed between the voltage levels?
- What are the main power plants, which primaray resource do they use and where are they located? State the current electrical output of each (MW).
- What are the main loads and where are they located? State the current electrical load of each (MW).
- What types of reactive power compensators exist and where are they located?

Make some hardcopies presenting the stations' and grid structures.

For your written report: Give brief information about the network of the utility company (electrical and commercial aspects, customers,...) by answering the questions above.

5. Getting familiar with the topology and measurements

Goal: Getting familiar with the displayed topological information, topology symbols, colour indication and measurements.

Switchgear
The states of open or closed switchgear create the colour of the lines and busbars and indicate the topology of the power system. The colour shows the connected network components. Colour is given by the one or two mayor generators.

Topology
Please see: How is the topology information presented in the displays? Describe:

- The topology information in overview pictures.
- The topology information in load flow pictures.
- The topology information in station pictures.
- What does a single or a double colouring mean?

Measurements

- What kinds of measurements are displayed in the pictures?
- What is the regular update cycle for the measurements?
- What is the meaning of the algebraic sign (+/-) of the displayed measurements (MW/MVAr)?
- How can operators get a more precise resolution (more digits) of a measurement?

For your written report: Answer the questions above.

6. First power system operation.

Goal: Do your first online switching operation.

Select a feeder with an open end line in the 400-kV network e.g. in station “Nord 400”. Select the feeder of the feeding end of the line and put the curser to the breaker. It usually can be switched off without any risk. Explain your choice and ask your instructor before doing the operation. What effects of the operation do you estimate? Think on change of active and reactive power flow.

Execute a double click on the breaker and see the switching box. Make a hardcopy.
How do you identify the selected switchgear? How do you identify the current state?
Select the new state and do the operation by pressing the button “ausführen” and see the results.

See the button “SP” which means switching program for the feeder. It allows selecting a feeder for a complete bus bar operation. Learn to use it by doing and getting experience.

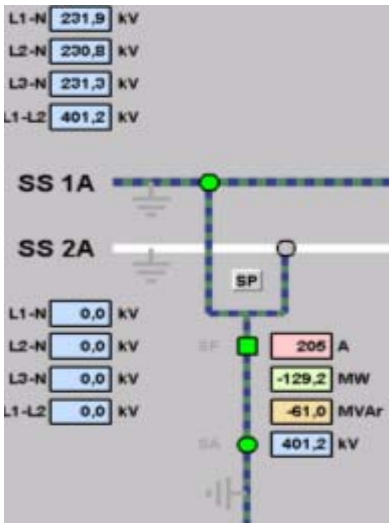
Important: Learn how to lock a switch gear / a feeder? How can you do that? It is important for staff safety if staff has to do local maintenance.

For your written report:
Explain the difference of a beaker and an isolator. What to operate first
a) for closing a feeder and b) for disconnecting a line?

Answer the questions above. Describe your experiences using hardcopies of station pictures and of the protocol.

7. Interpretation of measurements

Select a 400-kV station, e.g. station Ost 400. First check the displayed measurements of a feeder, of a busbar coupling and of the busbars. Note what the arithmetic sign of the displayed measurement (+/-) indicate:



Sign	Units	Physical meaning
(+)	MW	active flow going to bus bar
(-)	MW	active flow going from bus bar to line
(+)	MVAr	reactive inductive flow going to bus bar
(-)	MVAr	reactive inductive flow going from bus
(+)	kV	apparent voltage always positive value Busbars: 3 Line - ground, 1 Line – line Feeders: Line - line
(+)	A	apparent line current always as a positive value

For your report:
Give some examples (snapshots) and discuss it.
How can you get a precise resolution of the measurement?

8. Balances

8.1 Balances for feeders

The apparent power S of a feeder (electrical flow) can be calculated by the equation

$$S = \underline{U}_{ph1}\,\underline{I}_{L1}^{*} + \underline{U}_{ph2}\,\underline{I}_{L2}^{*} + \underline{U}_{ph3}\,\underline{I}_{L3}^{*}$$

S	Complex 3-phase power (apparent power)
$\underline{U}_{phi}$	Complex phase-i to ground voltages (i = 1, 2, 3)
$\underline{I}_{Li}^{*}$	Conjugated complex phase-i current (i = 1, 2, 3)

To solve the equation, all three phase to ground voltages, line currents and angles have to be measured. Usually these measurements are not displayed in the control room. As the electrical system is assumed to be symmetric, the values of just one line current and one phase to phase voltage is transmitted. Additionally the active and reactive flow values are displayed. These measurements have to fit to each other. Check that by calculating the mismatch MM :

$$S_{UI} = \sqrt{3} \cdot U_v \cdot I_L \qquad S_{PQ} = \sqrt{(P^2 + Q^2)}$$

$$MM = \mathrm{Abs}\,(S_{UI} - S_{PQ})$$

U_v	Phase to phase voltage measurement
I_L	line current measurement
P	Active power measurement
Q	Reactive power measurement

For your report: Do snapshots and select one feeder of a 400-kV line, one of a 110-kV line and one of a 20-kV line and use the displayed measurements to calculate the MM of each of the selected feeder. Discuss and qualify the results.

8.2 Balances for bus bars

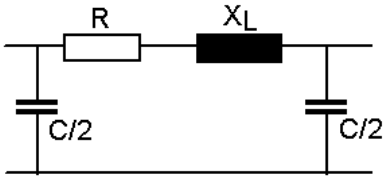
The rule of Kirchhoff demands a balance of current at an electrical node. Because current is displayed without sign and angle, the check is done by calculating the mismatch of the active and reactive power for MM_P and MM_Q :

$$\mathbf{MM_P = \Sigma P_i \quad (i= 1,2,... \text{ all feeders})}$$
$$\mathbf{MM_Q = \Sigma Q_i \quad (i= 1,2,... \text{ all feeders})}$$

MM_P	Mismatch of active power
MM_Q	Mismatch of reactive power
P_i	Active power measurement of feeder i
Q_i	Reactive power measurement of feeder i
i	Feeder No.

For your report: Select one bus bar of a 400-kV station, one of a 110-kV station and one of a 20-kV station and the displayed measurements to calculate the mismatch. Make hardcopies, calculate and qualify the results.

8.3 Balances for lines



One equivalent network for a line is usually described like this:

R is the line resistance, X_L the line inductivity and C the line Capacity. In this exercise we will see the effects of these line elements.

Switch on the load flow display "Lastfluss 400" and see the active and reactive power of a loaded line and compare the measurements. The line consumption of active power (losses P_v) depends mainly on element R. Give the equation.

The line consumption of reactive power (inductive consumption Q_L) depends on element X_L. Give the equation. Likewise, the line production of reactive power (capacitive consumption is equal to inductive generation) depends on X_C. Give the equation.

Imagine a line with balanced reactive power: The values of reactive power on both ends are the same. This line load of balance of reactive power is called "natural load".

If the production of inductive power (in the line capacity) is higher than the consumption (in the line inductivity) the sum of the reactive power measurements is positive. Line has lower load than natural load. If the consumption of inductive power (in the line inductivity) is higher than the production (in the line capacity) the sum of the reactive power measurements is negative. Line load is higher than natural load.

Your tasks and for the report: Switch on the load flow displays. Select one 400-kV line, one 110-kV line and one 20-kV line and compare the displayed line measurements. For each of this lines calculate the balance of the active power P and reactive power Q. Compare and explain the results:

$dP = \Sigma P_i$ (both ends) gives the active losses
$dQ = \Sigma Q_i$ (both ends) gives the production/consumption of reactive power

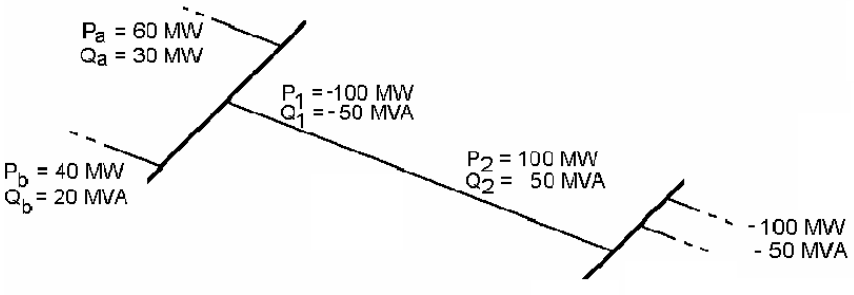
dP	Difference of active power
dQ	Difference of reactive power
P_i	Active power measurement of feeders
Q_i	Reactive power measurement of feeders

Remember: X_L is ωL and X_C is $1/\omega C$
The complex impedance of an inductivity is $\underline{Z}_L = j\omega L$
The complex impedance of a capacity is $\underline{Z}_C = -j/\omega C$
Remember the effect of compensation

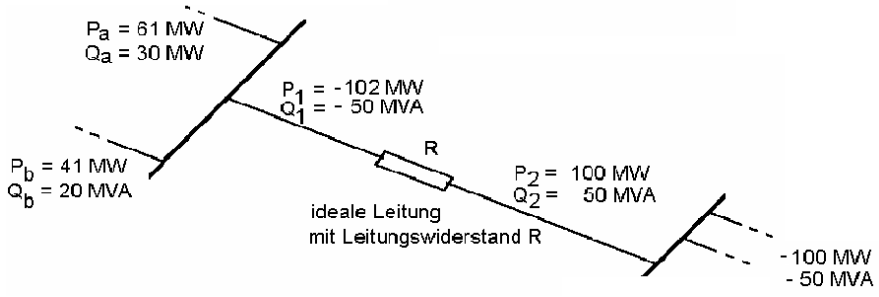
Make hardcopies and calculate and explain the results. Check more lines in all voltage levels and identify the line state. Can you identify any line close to natural load?

Follow the effects of the network equivalent elements of a line

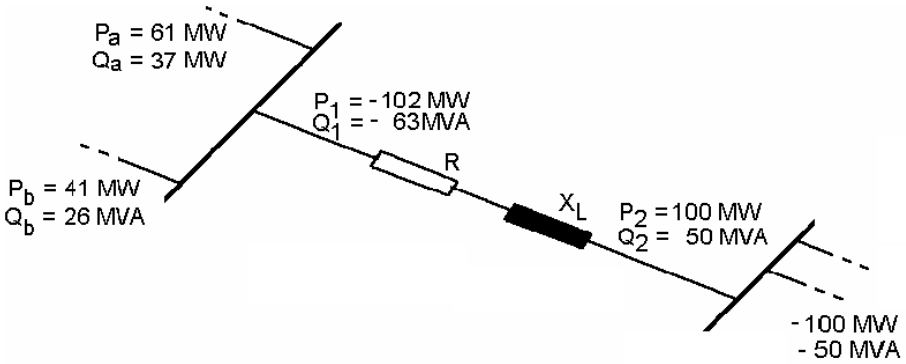
a) Ideal line – no elements



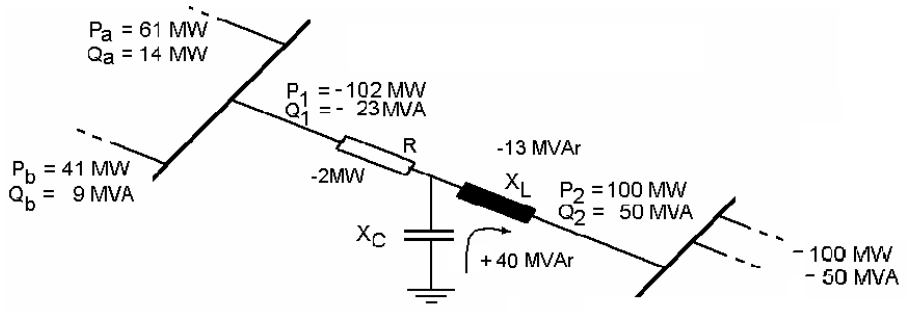
b) Effect of resistance R



c) Effect of inductivity X_L

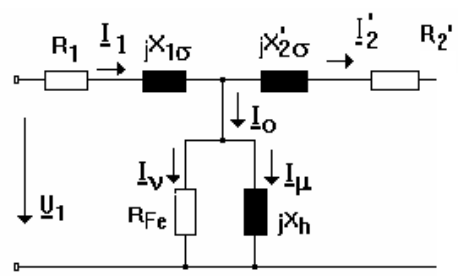


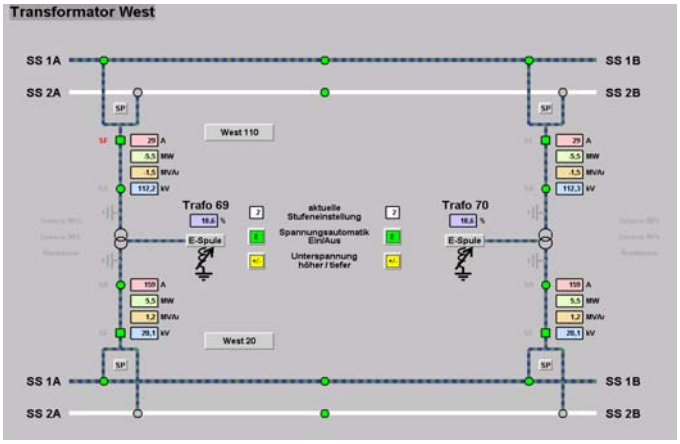
c) Effect of capacity X_C



8.4 Balances for transformers

The equivalent network of a transformer is usually described like this (left picture):





This exercise is about to observe general effects, not to calculate values here. That will be done in a later exercise.
R_1 and R_2' are the winding resistances, X_h the main coupling inductivity and R_{Fe} the representation of the iron losses caused by iron circle currents and $X_{1\sigma}$ and $X_{2\sigma}$' the representation of the inductive flux losses. In this exercise we will see the effects of these line elements.

Switch the station display West 20 and see the active and reactive power of both - a loaded transformer and a not loaded transformer. By measurements of a loaded transformer we can calculate the effects of

ΔP is effected by $R_k = R_1 + R_2'$
ΔQ is effected by $X_k = X_{1\sigma} + X_{2\sigma}'$

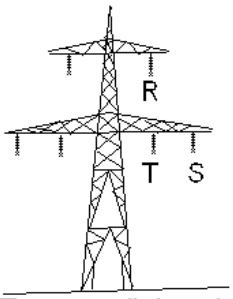
By measurements of an unloaded transformer we can calculate the effects of R_{Fe} and X_h. See the unloaded transformer measurements, e.g. in West 110. Important: **A double-click** shows a higher resolution of the measurements which is necessary to see the difference dP and dQ.

Your tasks and for the report: See the measurements, make hardcopies and explain the measurements.

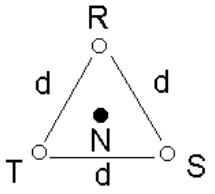
9. Overhead lines and effects

- Getting familiar with the representation of lines/cables and network equivalents
- Long line effects and compensation

9.1 Representation of lines and parameters



Two parallel real 3-phase System



Overhead line with equilateral spacing

Power pylon and overhead lines - R, S, T is a 3-phase System

Line parameters and π-model

The line parameters for circuit analysis are inductance, capacitance, and resistance and if taken into account also the equivalent leakage resistance of corona sparkling in a π-model. The derivation of formulae for the calculation of the quantities is not given here. It is intended here to quote and discuss their applications. The line-neutral inductance L for equilateral spacing is

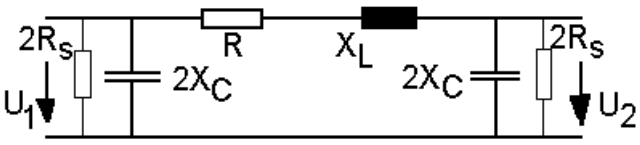
$$L = \frac{\mu_o}{8\pi}\left[1+4\cdot\ln\left(d/r-1\right)\right] \text{ H/m}$$

r **radius of a conductor**
d **equilateral spacing**

The line-neutral capacitance for equilateral spacing is

$$C = \frac{2\pi\varepsilon_o}{\ln\left[(d/r-1)\right]} \text{ F/m}$$

r **radius of a conductor**
d **equilateral spacing calculated by** $d = \sqrt[3]{d_{12}\, d_{23}\, d_{31}}$



The parameters proportionally depend on the line length. The equivalent circuit of a line is shown in next figure. It is sufficient for modelling a line up to about 250 km line length. The resistor representing the sparkling corona effect R_s is often neglected. Because it looks like the Greek character π it is called the π-form of the line equivalent.

π-parameter representation of a line

Note: Using $\omega = 2\pi f$ and f for frequency

$X_L = \omega L$ and $X_C = 1 / (\omega C)$

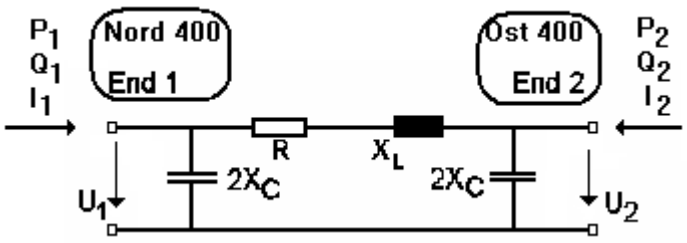
Some typical 400-kV overhead line constants at 50 Hz (per phase) for a conductor of 4x258 mm^2 (British)

Resistance	R	(Ohms/km)	0.017
Reactance	X_L	(Ohms/km)	0.27
Capacitance of an Overhead Line		(nF/km)	15
=> Susceptance	$1/X_C$	($Ohms^{-1}$ x10^{-6})	10.58…5.0
=> Charging Current	Ic	(Amps)	0.945
=> Natural load	P_{Nat}	(MW)	620
Thermal rating		(MVA)	2200 ... 1580 (5°C ... 18°C)

A hundred km overhead line usually has approximately
R = 2 Ohm, X_L = 20 Ohm, X_C = 4500 Ohm

Exercise: Determination of parameters using display data

The π-parameters of a line can be determined approximately using the displayed measurements for both line ends. Therefore two snapshots of the same line has to be done, one for a no load situation and a second one for a (nearly) rated load situation.



Network equivalent of a line

Consider the line Nord 400 – Ost 400

a) Provide a no load situation on end 2 (e.g. in Nord 400 open) by switching off the line end in Nord 400 and see the measurements of reactive power Q_C on feeding side, this allows to approximately calculate X_C:

$Q_C = 3 \cdot U_{Ph}^2 / X_C$

$X_C = 3 \cdot U_{Ph}^2 / Q_C$

U_{Ph} phase to ground voltage in Ost 400
Q_C reactive power in Ost 400

Using $X_C = 1 / \omega C$ calculate the line capacity C in [mF].

Using $C' = 14.5$ nF/km as the line capacity per km and

$C = C' \cdot L_L$

Calculate the line length L_L.

b) Provide a load situation of the line, this allows to approximately calculate R and X_L. See the measurements

Difference of active losses: $- P_1 - P_2 = dP = 3 \cdot I^2 \cdot R$

Difference of reactive losses: $-Q_1 - Q_2 + Q_C = dQ = 3 \cdot I^2 \cdot X_L$

By using mean value of line current calculate R and X_L.

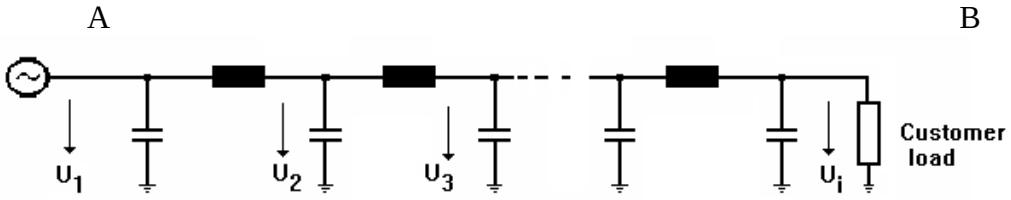
$S^2 = P^2 + Q^2$ and $I = S / (\sqrt{3} \cdot U_V)$ at both line ends and
$I_m = (I_1 + I_2) / 2$
$R = P_V / (3 \cdot I_m^2)$

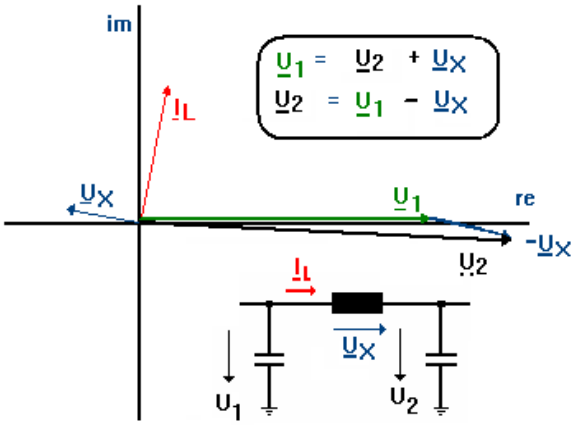
For your report:

Do the hardcopies, calculate the Parameters R, X_L and X_C, answer the questions and discuss the results.

9.2 Exercise: Long distance overhead lines and effects

Often the distances between the best location for generating plants and the customers are rather long, hundreds of kilometres or more. UHV-overhead lines are the cheapest solution to connect them. Depending on the load of the line electrical effects appear, e.g. a low loaded line shows a voltage increase at the receiving end B.





This effect appears due to a capacitive line current passing through the line inductivities.

Feeding light loads the capacitive consumption of a line exceeds the inductive VAr consumed by the inductance. There is a physical effect which makes the voltage rise that is called Ferranti-effect. This effect causes problems to generation (to produce the needed reactive power) and to the insulation (to stand the rise of voltage).

High loaded lines with inductive consumption show the opposite effect, voltage drops at the receiving end B.

At very long lines the voltage rise/drop can be massive. A length of 1500 km at 50 Hz corresponds to a quarter wavelength line of extreme voltage increase.

Series capacitors SC, see next figure, would normally be installed to reduce voltage rise and to effectively shorten the line length electrically. In addition (inductive) shunt reactors Sh, also see next figure, are switched on at times of light load to absorb the capacitive load. For long lines (400 km and more) in general it is common to divide the system into sections with compensation at the ends of each section.

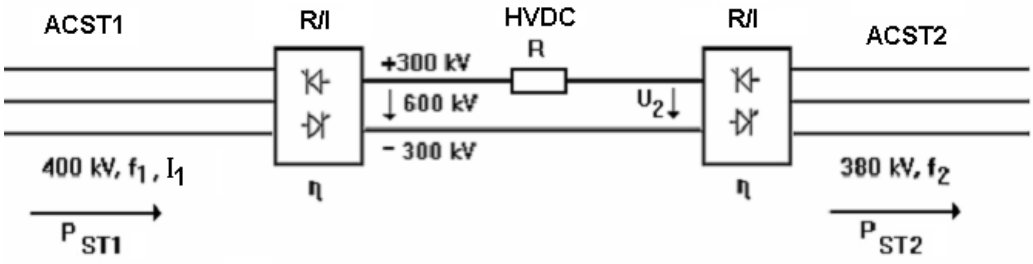
FACTS (flexible AC Transmission Systems) work on the base of power electronics to adjust the reactive power to the actual needs depending on the line load. Voltage and flow can be adjusted to the needs. In case of very long distances (400 km or more) a cheaper solution is HVDC. HV-DC-lines are used to cross very long distances providing two advantages:

1) Decoupling the 3-phase systems in terms of reactive power and short cut power.
2) Additionally: DC-lines do not need any compensation.

Cost expensive elements are the rectifiers and inverters.

Example

Two 3-phase AC systems ACST1 and ACST2 having different voltages U_1 and U_2 and different frequency f_1 , f_2 are connected by an HVDC line System. See the date in the figure.



In one situation ACST1 send an active power P_{ST1} = 433 MW to the rectifier/Inverter R/I from left hand side. The efficiency of the R/Is are η = 0,96. The resistance of the HVDC line is R = 6 Ω. Calculate the electrical system:

1. What line current I_1 flows in ACST1?
2. What power P_{HVDC1} is feeded into the HVDC line?
3. What line current I_{HVDC} flows into the DC line at DC voltage- 600 kV?
4. What active losses ΔP_{HVDC} appear in the DC line?
5. What voltage U_2 appears?
6. What power P_{HVDC2} is feeded into the R/I right hand side?
7. What power P_{ST2} is feeded into the ACST2 system?
8. What is the total efficiency of the transmission?

Solution

1. $I = P / (\sqrt{3} \cdot U_V) \rightarrow I_1 = 433\ MW/(\sqrt{3} \cdot 400kV) = 625\ A$

2. $P_{HVDC1} = P_{ST1} * \eta = 433\ MW * 0{,}96 = 415{,}68\ MW$

3. $I_{HVDC} = P_{HVDC} / U = 415{,}68\ MW/600\ kV = 692{,}8\ A$

4. $\Delta P_{HVDC} = I^2 R = 2{,}88\ MW$

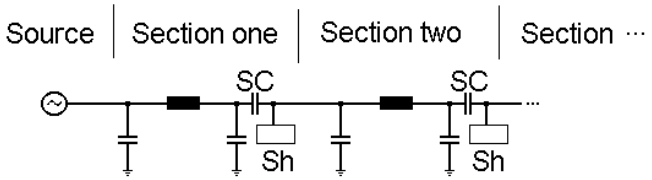
5. $U_2 = 600\ kV - I*R = 600\ kV - 4{,}157\ kV = 595{,}84\ kV$

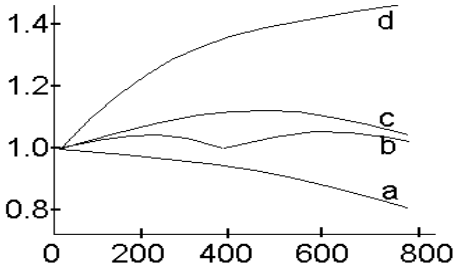
6. $P_{HVDC2} = P_{HVDC1} - I^2 R = 415{,}68\ MW - 2{,}88\ MW = 412{,}8\ MW$

7. $P_{ST2} = P_{HVDC2} * \eta = 412{,}8\ MW * 0{,}96 = 396{,}3\ MW$

8. $\eta_{Tot} = P_{in}/P_{out} = 396{,}3\ MW / 433\ MW = 91{,}5\ \%$

Sectioning of long lines and compensators



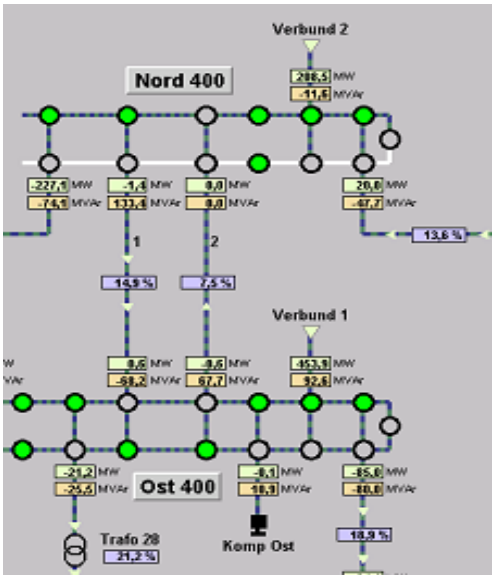


The voltage variation along a long line (of approximately 750 km) is given in next figure.

Curve d: No load and no compensation.
Curve c: No load and a single compensation at the end.
Curve b: No load and compensations at the end and at centre (middle).
Curve a: Natural load and compensations at the end and at centre (middle).

Exercise: voltage rise on a long overhead line

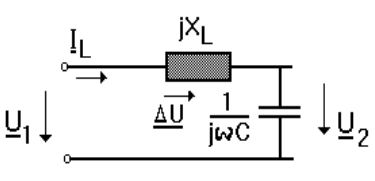
Open picture “Lastfluss 400kV”. Provide a no load situation of the lines as shown in next figure by switching on the line 1 in Nord 400 to busbar 1 (use SP button, “einschalten auf SS1”), in Ost 400 to busbar 2. Switch on the line 2 from Ost 400 to Nord 400 in Ost 400 also on busbar 2. Make sure that there is no other line or load on busbar 2 in Ost 400 besids the two lines. Line 2 is open in Nord 400.

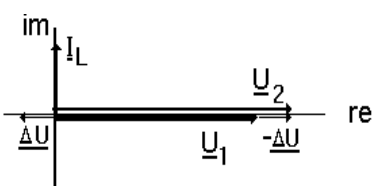


By this topology you have created a long line which consists of line 1 and line 2 in series connection. Now follow the voltage measurements starting from Nord 400 U_1, Ost 400 U_2 and U_3 and again Nord 400 U_4 in the station displays. Calculate and explain the increase of voltage along the line (Ferranti effect).–

Now use the compensator “Komp Ost” in Ost 400 to decrease the voltage. You have to switch Komp Ost to busbar 2 and to set the compensation value manually in the display “Fahrpläne”. First: Switch off “Fahrplan Lasten” by pressing Button “E/A” and switch to “A”. Start with -10 MVAr and do steps by -20 MVar up to -100 MVAr. What compensation MVAr is needed to achieve U_1 voltage in Ost 400 also on busbar 2?

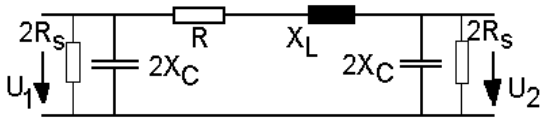
For your report: Calculate and explain the increase of voltage along the line. Give answers to the questions above. Explain the Ferranti-effect by the network equivalent in words.





9.3 Natural load exercises

Remember the parameter representation of a line (network equivalent).



The characteristic impedance Z_o is known as the surge impedance. When a line is terminated in its characteristic impedance the power delivered is known as the natural load. In other words: The capacitive consumption of a line depends from the voltage and the inductive consumption depends from the line current, in natural load condition, the capacitive load of the line is compensated by the inductive transfer load. For a single line:

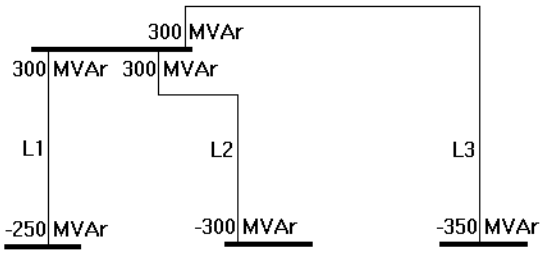
$$\mathbf{Z_o = \sqrt{(X_L \cdot X_C)} = \sqrt{(L/C)}}$$

Reactive consumption of capacitors: $Q_C = U^2 / X_C$
Reactive consumption of impedance: $Q_L = I^2 \cdot X_L$

U	phase to ground voltage
I	current of a line conductor
X_L, X_C	Line parameters

Note that a three phase line has three times the reactive load compared to the single line.

A line below natural load shows capacitive consumer behaviour (L1). A line above natural load shows a inductive consumer behaviour (L3). A line in natural load conditions (L2) shows neutral consumer behaviour: Then there is no change in reactive flow along the line.



Some examples: power grid lines of below / balanced / over natural load conditions

Q_1 Reactive power measurement in station 1
Q_2 Reactive power measurement in station 2

If $Q_1 + Q_2 > 0$: as line L1 - capacitive behaviour (below natural load)
If $Q_1 + Q_2 = 0$: as line L2 - neutral behaviour (natural load)
If $Q_1 + Q_2 < 0$: as line L3 - inductive behaviour (over natural load)

Exercise:
Follow the simulator's reactive power measurements on both end of some lines (e.g. 3 of each voltage level) and check the lines' state. Do some hardcopies of the according load flow displays. Make a list of the lines, calculate the reactive power balance and decide about the lines' states. Where is a higher chance to find lines in natural load condition concerning voltage levels and why? What can be done to put a line in or over natural load condition?

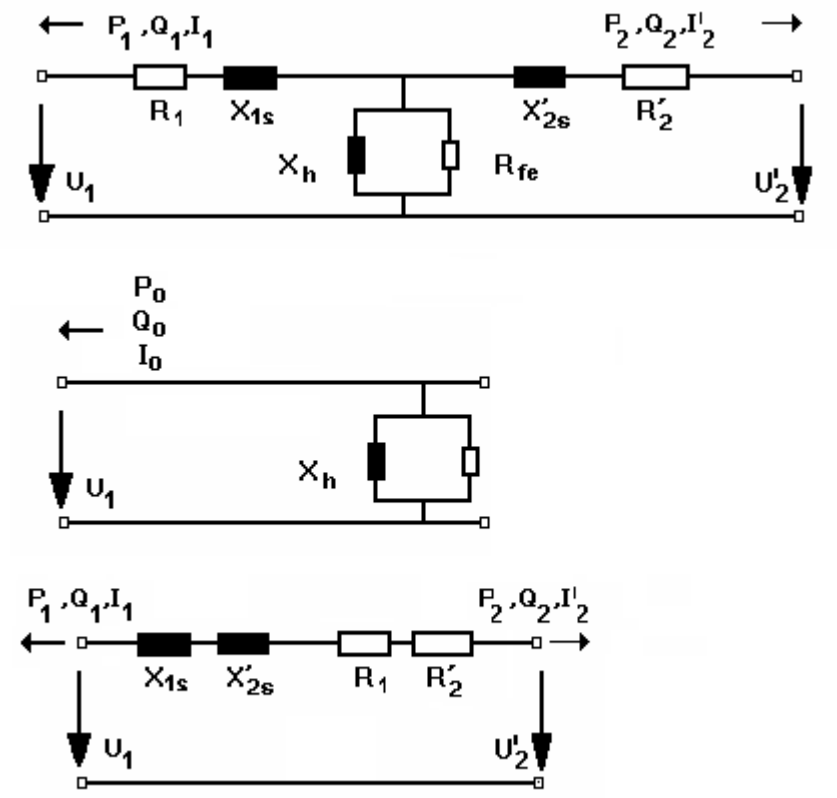
For your report: Hardcopies, answer of questions and list the results of the exercise above.

10. Transformers and parameters

As the losses of electricity transfer ($P_V = 3\ I^2\ R_L$) depend on the current (I) to the power of 2 and on the line Resistance (R_L) distance transfer is done using high voltages HV and UHV. Common voltage levels being used in Europe are UHV (400 kV), HV (110 kV) and MV (20 kV… 6 kV). Therefore transformers have an important role. High efficiency, low losses, voltage regulation, flexible windings using taps and a life cycles of 50 years or more are provided by modern transformer technology.

10.1 Representation of transformer

The representation of transformers in networks and for power flow calculation is a π or a T-network equivalent. See the T-equivalent in next figure. The upper network equivalent is the full parameter set. The middle one is a simplified network equivalent for no or low load situation and the lower one for high load transfer. Simplifying is allowed due to the values of the parameters: The horizontal elements (transfer) are just a few Ohms, the vertical elements are values of some thousands Ohms.



Transformer T-Network equivalent and simplifications

Meaning of the parameters and measurements:

R_{fe}	Resistor representing active losses of the iron core
X_h	Impedance representing main inductivity
R_k	(= R_1+R'_2) Resistor representing copper losses (windings)
X_k	(= $X_{1\sigma}$+$X'_{2\sigma}$) Scatter impedances
U_1	Primary line to ground voltage
P_o	No-load active losses
Q_o	No-load reactive Vars
I	Current measurement
dP	Active losses calculated by: abs(P_2+P_1)
dQ	Reactive (inductive) losses: abs(Q_2+Q_1)

Usually we assume: $R_1 = R'_2$ and $X_{1\sigma} = X'_{2\sigma}$. Note that the measurements are from the 3-phase system (current of one conductor, three phase power and phase to phase voltage). Because the horizontal parameters are low compared to the vertical parameters it is allowed to write

				Measurements from
$3 \cdot U_1^2 / R_{fe}$	=	P_o	(1)	no load condition
$3 \cdot U_1^2 / X_h$	=	Q_o	(2)	no load condition
$3 \cdot I^2 \cdot R_k$	=	dP	(3)	high load
$3 \cdot I^2 \cdot X_k$	=	dQ	(4)	high load

By the measurements elements can be calculated using the above four equations. Equation (1) and (2) uses measurements of a no load situation, equation (3) and (4) uses measurements of a high load situation.

Taking current as I_1 and phase to ground voltage U_1 from the primary side, the parameters are calculated on the primary side. Taking I_2 and U_2 from the secondary side, the parameters are calculated on the secondary side.

10.2 Exercise to calculate the transformer parameters

Consider the two identical parallel 110/20-kV transformers (TR 69, TR 70). Provide the following situation: One transformer is loaded, the other is connected to HV-voltage but transfers no power because of it's open end on the 20 kV side. Make a snapshot of the measurements. Using the measurements you can approximately calculate the parameters of the T-model. Please use the precise resolution of the measurements (by click on the measurement). If not the parameter results are inaccurate and useless.

For your report:

- Precise measurements and hardcopy and calculation of the parameters.
- Calculate the power factor of both feeders (HV and MV)

11. Isolating and grounding a 400-kV overhead line to prepare for maintenance



Usually the UHV and HV overhead lines are checked cicely by helicopter flights to identify any problem caused by birds, corrosion or damages of conductors or insulators etc.

For maintenance the line has to be switched off and grounded.

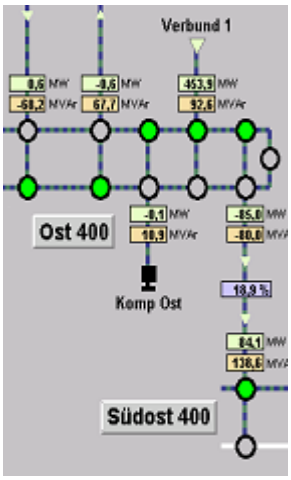




Most of the switching operations are planned for above mentioned maintenance reasons. For these tasks usually written switching orders are created and scheduled. When the task comes to action, the system operators (load dispatcher) have to decide:

- Is it feasible to switch off the line taking actual or future load flow into account?
- What will be the effects concerning load flow?
- Will the power system be in any danger if other outages may happen?

Exercise: Consider line Ost 400 - Süd-Ost 400 to be isolated and grounded for maintenance.



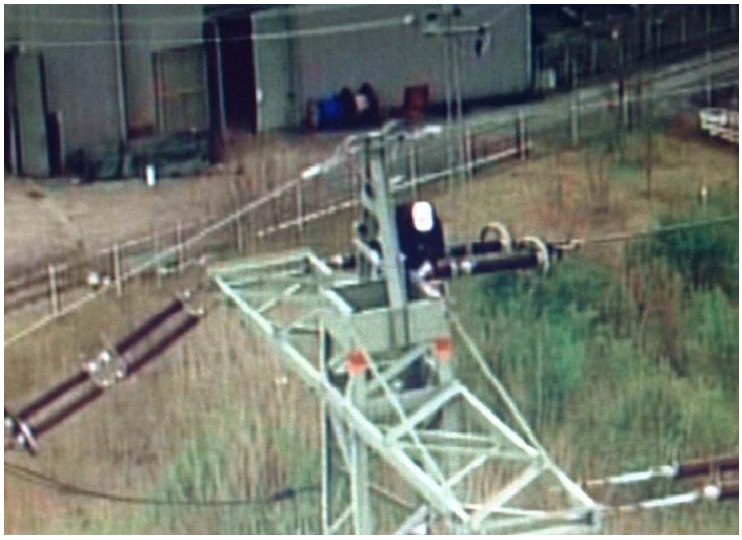
Check the load flow. Is the heavy load “Aluwerk 1 and 2” able to be feeded by line from Nordost 400? Think about security of power supply (n-1 security). What can be done additionally? Finally make your decision and discuss the decision with your instructor.

After that you may open the line. Verify the effects. After having switched off the line do the grounding and locking. The you have to pass the "disposability" to the grid-team by a phone call (simulated together with the instructor). Mark the line as "disposed" on both ends. How is that done?

For your report: Do a hardcopy. Write down the questions and the solutions. Give the answers.

12. Deadly accident

Read the note of a newspaper: ***"Alzenau-Albstadt. A young electrical worker was deadly injured during maintenance work. He appeared was stroke by sudden death after touching a 400-kV high voltage overhead line near Albstadt yesterday at 12.00 p.m. The man worked on the line and was belted. Colleagues and a called ambulance recovered the man. An immediate reanimation finally failed. As the police speaker informed, it is proofed that the line has been switched off and grounded by the control centre. Police and experts went to the line for reconstructing the accident. It has to be verified so the speaker that all security instructions have been executed."***





Think about the accident and make your opinion on the final causes. Think about the five security rules of electrical working:

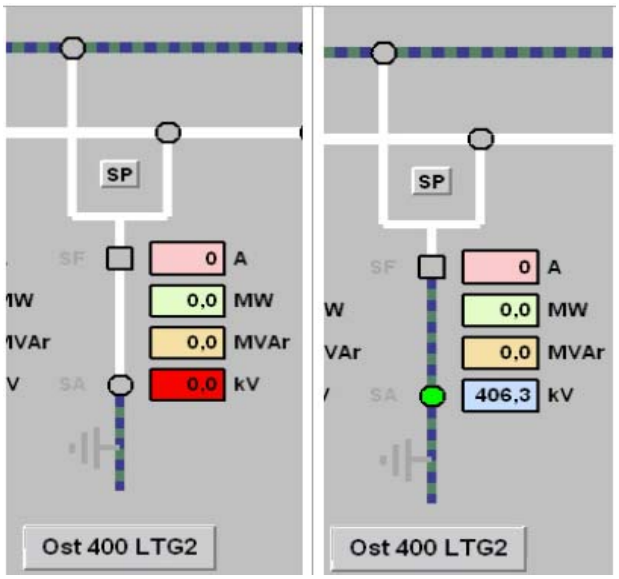
1. Switch off
2. Protect against reconnection
3. Check for zero voltage
4. Grounding
5. Covering of near devices under voltage

For your report: Discuss the accident. What has been neglected? What steps have to be done by the control centre team? What steps have to be executed by the grid team?
Write down a correct sequence of actions including the passing of disposability.

13. Location of the current & voltage transformers

This exercise is to identify the location of the current/voltage transformers as sensor elements to obtain the measurements.

Do as follows: Consider a feeder of an open line end, e.g. line Nord 400 - Ost 400 LTG2. First all switchgear of the feeder in Nord 400 are off. The line is set to voltage from the other end in Ost 400. You may locate the positions of the current and voltage transformers by a switching operation of the line isolator. In off position the voltage is zero. The appearing of the voltage 406,3 kV indicates the measurement transformer locations.



off position of isolator on position

Location of the current and voltage transformers

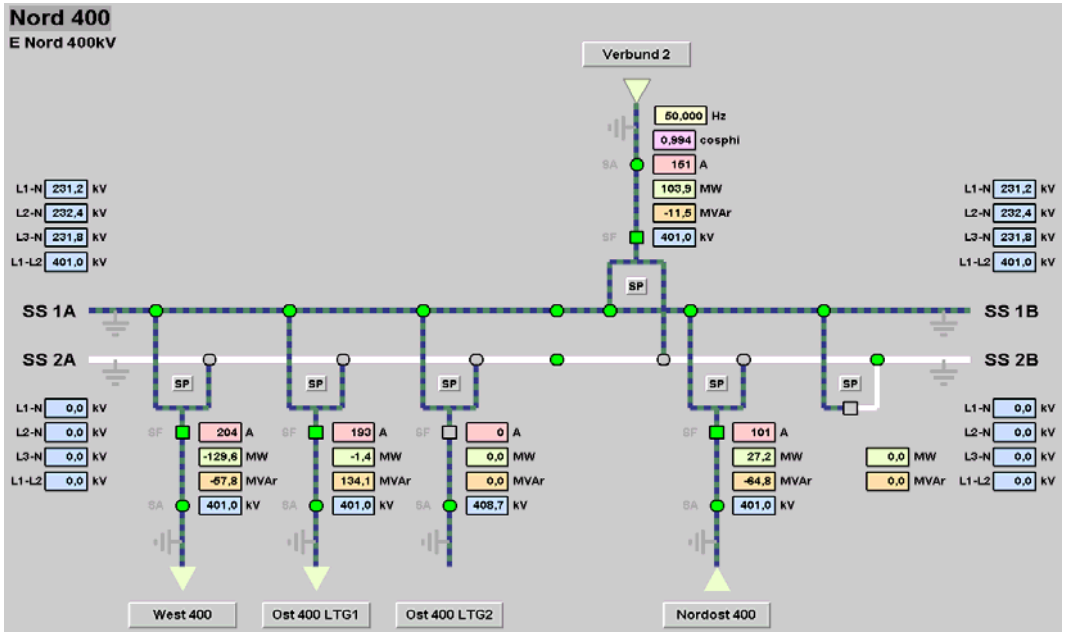
For your report:
- Make a snapshot.
- Mark the position of the transformers.
- What locations for a short circuit are critical?

14. Exploring the Interlocking System und changing a busbar

The Interlocking is done to protect the equipment against wrong operations. For security reasons there are usually three levels of interlocking levels: in the feeder, in the station and in the control centre. The basic rule is simple:

> **A breaker can operate anytime to switch the loads on or off. An isolator is allowed to operate only if the breaker is open because it cannot switch loads.**

Substation interlocking systems are more sophisticated. The solutions and the levels of protection depend on the data availability. Usually the feeder level interlocking system only takes its own (feeder's) data into account. Station level interlocking uses all feeders' data including the busbar coupling data and centre level takes all remote data into account. All levels are needed because switch operations can be done from feeder PLC motor drives, from a substation local operation console or from the control centre SCADA system.



Exercise 14.1: Select a feeder e.g. of a line and explore the interlocking system. Take also the switchgear to ground into account. Is a wrong operation always locked? What mal operation is not protected?

For your report: Give examples of locked operations and answer the questions. Describe the results.

Exercise 14.2: Execute a busbar change in station Nord 400 without switching off a line or load by using the busbar coupling.

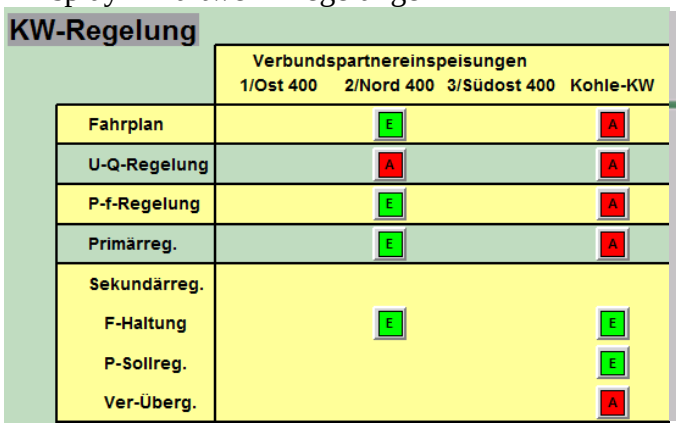
For your report: Give a complete list of the operations in sequence and do hardcopies of the start and the end topology.

1. Operation: Close breaker of busbar coupling
2. Operation: Close isolator feeder West 400 to 2nd busbar
3. Operation: Open isolator feeder West 400 to 1st busbar
4. …Close…Open…Close…Open…

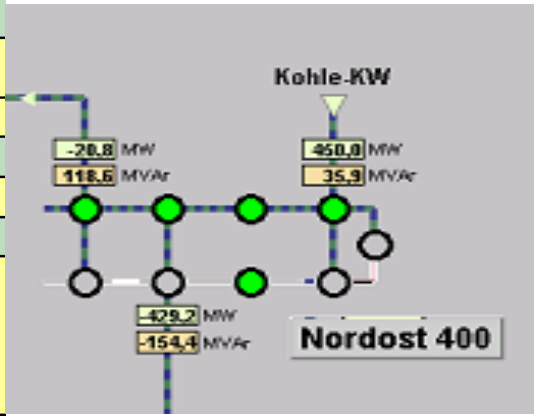
15. Impact of power injections on the voltage.

During this exercise the impact of busbars' active and reactive power injections on the voltage is explored. Power plant Kohle-KW in station Nordost 400 is used for this exercise. We investigate the busbar's voltage increase and drop during power plant generation changes.

Display "Kraftwerk-Regelungen"



Display "Lastfluss 400 kV"



To exercise, do settings for the regulations in the display "Kraftwerk-Regelungen", see figure:

1. Switch off the regulations (to red A) with the exception of "P-Sollreg." which is the secondary regulation of manually given power target values (to green E).
2. Open a trend curve display from top menu "Kurvenanzeige" and by using drag and drop technique implement the Kohle-KW's active and reactive power values (target and online measurements) and the busbar voltage in the curve display.
3. Then set the reactive power injection target Q "Q-Soll" to 0 and vary the active power injection target "P-Soll". Give the plant's regulations some time to change from one target value to next.

Variation of active power P injection when reactive power is constant 0 MVAr:
Note the 400-kV bus-bar voltage and complete the list.

P [MW] target values	400	200	0
U[kV]			

4. Now sequentially change reactive power injection Q while active power injection P is set to zero. Note the bus-bar voltage and complete the list. Give the plant's regulations some time to change from one target value to next.

Variation of reactive power injection Q when active power is constant 0 MW:

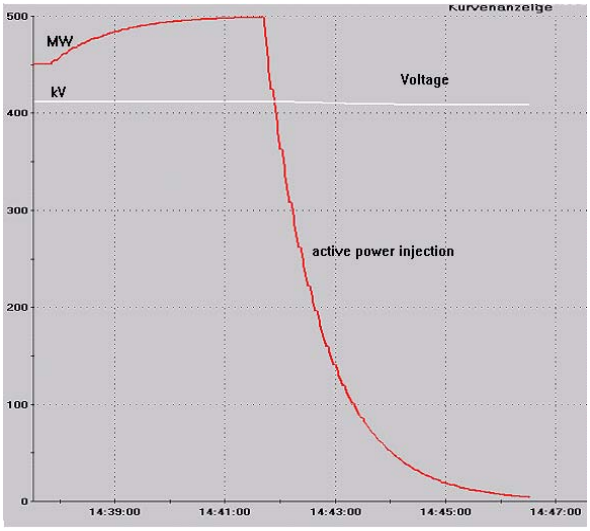
Q [MVAr] target values	0	200	400
U[kV]			

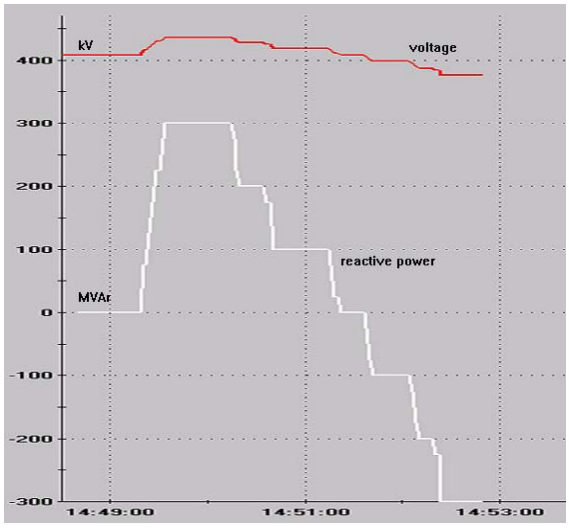
For your report:

- Do hardcopies of the measurements similar to the next figures fill in the tables your measurements and answer the questions:

Change of P at Q = 0 MVAr			
P [MW]	400	200	0
U [kV]			

Change of Q at P=0MW			
Q [MVAr]	400	200	0
U [kV]			



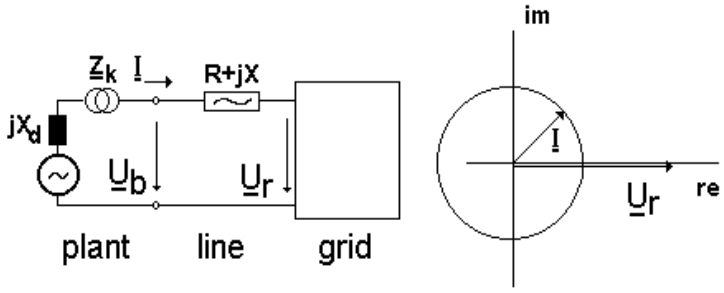


- Which type of injection is better to manipulate the voltage?

- How is it done in practice? What are the limits?

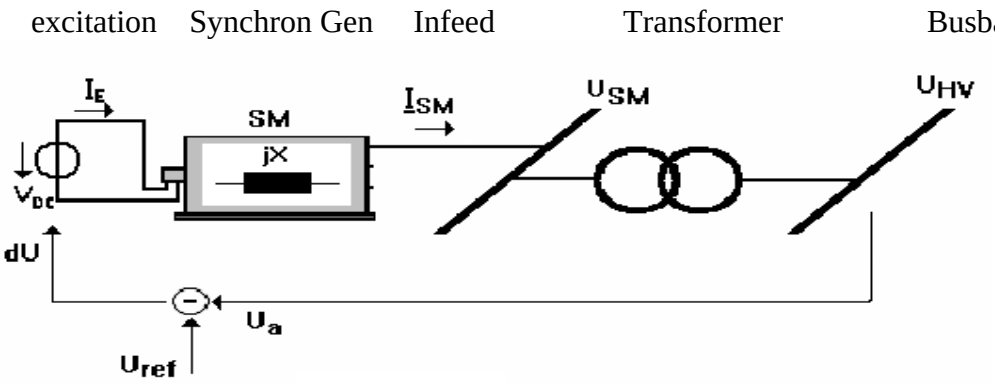
- Calculate the sensibilities: Vp = dU/dP ; Vq = dU/dQ

- Explain the results on the basis of a simple network equivalent and a complex diagram. $\underline{U}_b$ is busbar voltage, $\underline{U}_r$ is the rated grid voltage assumed to be fix. Take into account that jX is much greater than R in 400 kV.



16. AVC - Automatic Voltage Control

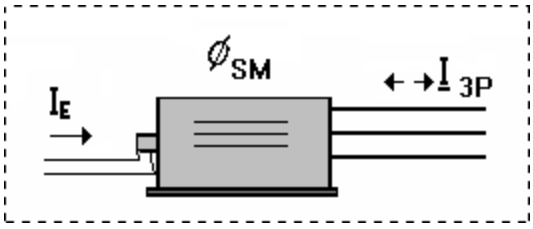
This exercise is to investigate the automatic voltage control system. The automatic system changes the reactive power injection ($\underline{I}_{SM}$) of a generator to achieve the target voltage on the busbar as we have done in exercise 15 manually. This AVC regulates the exciting current I_E (excitation DC) of the SM to control the magnetic flux inside the generator, so that the target voltage on the busbar will be reached. U_{HV} voltage is measured and compared to U_{ref} as reference target value. The difference dU is used to change I_E to control U_{HV}.



Note that a synchronous motor/generator (SM) needs a magnetic balance. In principle a SM is like a coil (inductivity L) which is connected to a voltage source taking an inductive current I_L to create the magnetic flux $\Phi_{(V)}$ which depends from the voltage U. For a single coil the following equations are used:

Inductivity	$L = N^2 \Lambda$
Inductance	$X_L = \omega L$
Inductive current	$\underline{I}_L = \underline{U}/jX_L$
Magnetic flow	$\Phi = L\ I_L / N$

N Windings
Λ Connectivity characteristic of the magnetic circle, [As/V]
U Voltage
ω $2\pi f$
f Frequency



The magnetic flux Φ_{SM} of a SM can be created either from the (fixed) three phase side by I_{3P} or from the rotating excitation DC side by I_E.

In a simplified model the magnetic flux balance Φ_{SM} of a SM can be calculated:

$$\Phi_{(IE)} + \Phi_{(3P)} = \Phi_{(SM)}$$

$\Phi_{(IE)}$	**Magnetic flux created from DC-side**
$\Phi_{(I3P)}$	**Magnetic flux created from 3-phase-side**
$\Phi_{(SM)}$	**Needed magnetic flux of the SM for a certain voltage**

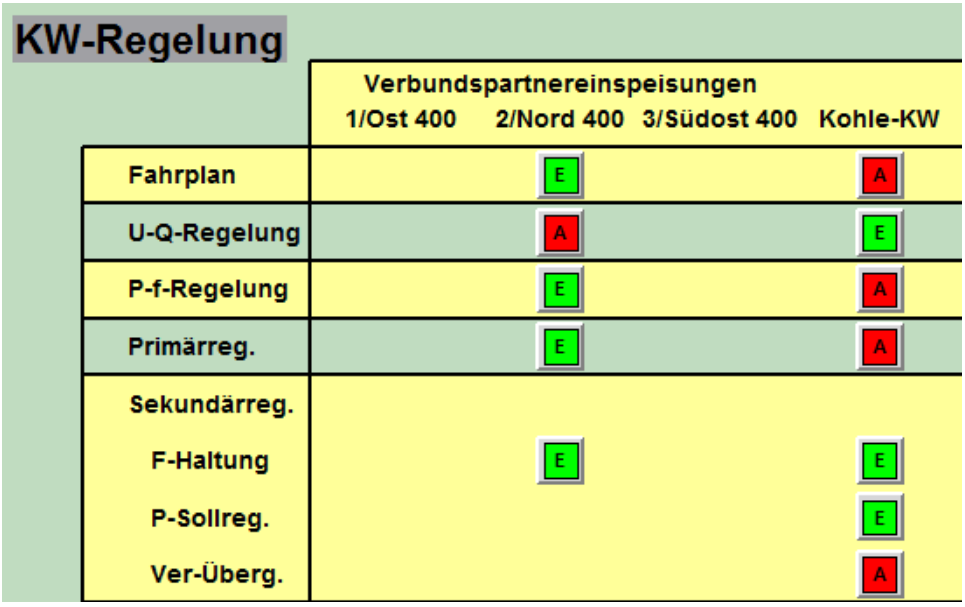
By **increasing** I_E accordingly $\Phi_{(IE)}$ increases. As the balance has to be fullfilled $\Phi_{(I3P)}$ **decreases**. That reduces the reactive current production/consumption from the 3 phase side which is equivalent to production/reduction of the SM reactive power.

Low I_E means low excitation. The SM behaves inductive (like a coil). It consumes inductive reactive power. Balanced I_E results that the SM has no reactive power exchange to the grid.

High I_E means high excitation. The SM is over exited; it sends inductive reactive power to the grid. It behaves capacitive (like a capacitor). This is equal to production of inductive power.

Limits of I_E are given by stability (low I_E) and temperature (high I_E).

Exercise:



In the display "W-Regelungen" switch on the "U-Q-Regelung" of Kohle-KW" which is the AVC controller. To what voltage does the automatic control? Now give following target voltage values as listed in the next table. Note the reactive power injection Q, which will be controlled by AVC

U_{target} [kV]	**380**	**390**	**400**	**410**	**415(?)**
Q [MVAr]					

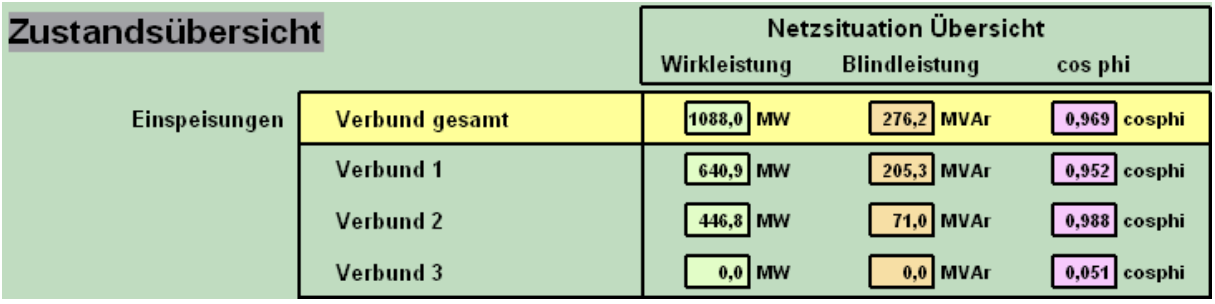
For your report:

- See the AVC behaviour, note voltages and the according reactive power injections and fill in the table.

17. Usage of compensators for reactive power balance

We already discussed the usage of compensators concerning long line effects and capacitive compensation. Here is another usage of compensators concerning reactive power and adjusting the power factor.

Consider the display “Zustandsübersicht” displaying the sum of the tree injections from the utility partner (three stations) and the total power factor cos phi in the line “Verbund gesamt”.



What electrical components do effect the utility's reactive power injection?

The power factor cos φ is defined as

$$\cos\varphi = P/S$$

P **Active power**
S **Apparent Power** $S = \sqrt{P^2+Q^2)}$
Q **Reactive Power**

Note the actual injections of active and reactive power and of the grid's losses and calculate cos φ. Adjust the power factor manually up to 0.999 by changing the injections of the compensations in the two 400 kV-stations "Ost 400" and "Südost 400" using the display "Fahrpläne". The exercise is successfully done if the reactive power is smaller than + or – 10 MVAr. Make a hardcopy of the results.

For your report:

- What technical components affect the grid's reactive power (consumption and production)?
- Note the measurements and the results.

18. Voltage dependence of loads

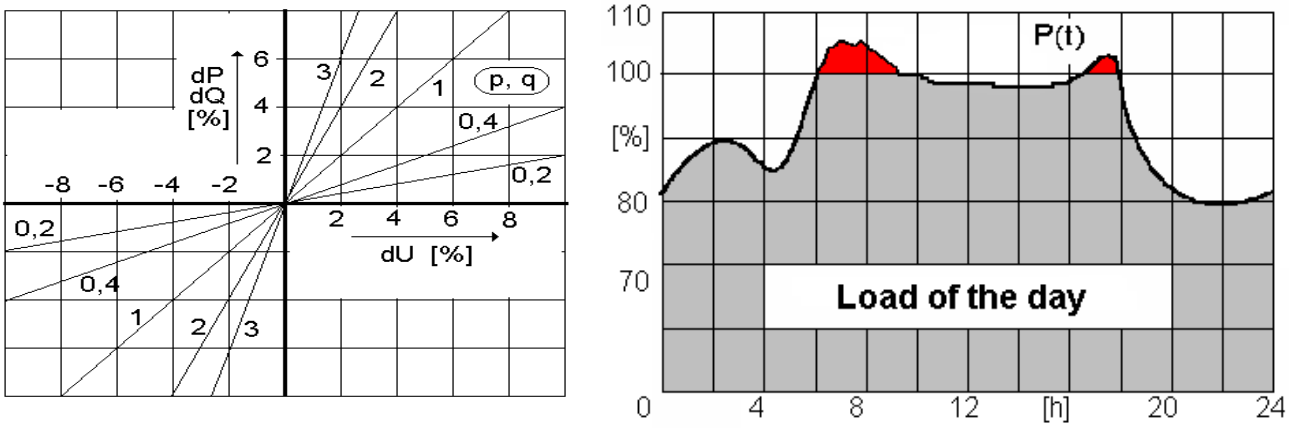
Demand side management is an important task in MV distribution grids mainly because of electrical and commercial optimization. Any peak loads appearing during the day need additional peak generation to cover. Additional generation is expensive the more if it is used just a short time of the day. Expensive investments for new generators can be avoided by reducing the peak load. Switch off is a common solution for heating or air condition loads which may be switch off for some minutes without causing troubles. Another solution is the drop of the voltage. As loads' consumption depends from the voltage a drop of the voltage (adjusted by the tap transformers) covers the lack of generation by reducing the loads consumption. See the behaviour of load depending from the type in next figure (left side). Load types can be classified by their reaction after a change of the voltage. Two examples:

- A resistor R consumes active power P_R depending from the voltage U which can be described by the equation $P_R = U^2 / R$
- A regulated load L takes constant power (if regulation works). The equation is $P_L = U^0 / R$

Any load can be described by the exponent of the voltage. Professor FUNK has described the spontaneous load reaction after a change of the voltage by the following equations:

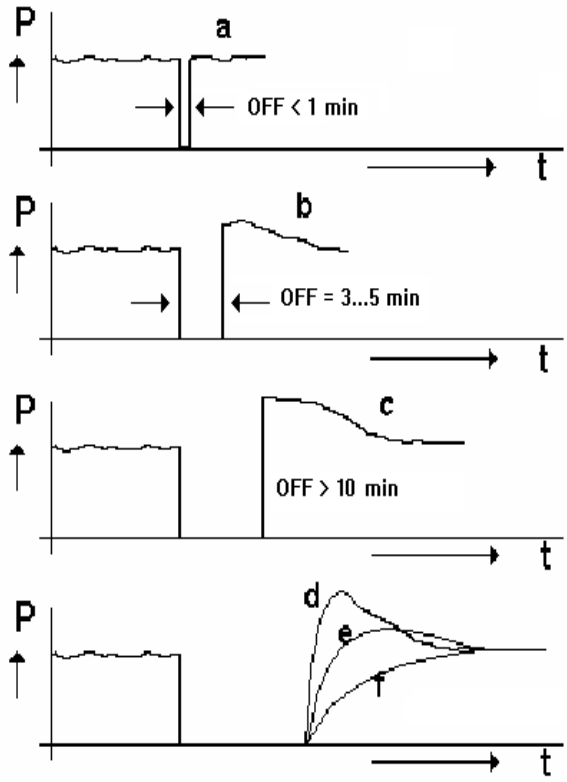
$$P_{act} / P_{rated} = (U_{act} / U_{rated})^p$$
$$Q_{act} / Q_{rated} = (U_{act} / U_{rated})^q$$

P_{act}	Actual active power
P_{rated}	Nominal rated active power (at rated voltage)
Q_{act}	Actual reactive power
Q_{rated}	Nominal rated reactive power (at rated voltage)
U_{act}	Actual voltage
U_{rated}	Rated voltage
p	Exponent for active power
q	Exponent for reactive power



The peak of the load (106%) can be covered by reducing the voltage by 3% - if load type exponent is 2.

We will see the voltage dependence of loads by some experiments. Real busbar loads show exponents that result from a mixture of the different load types connected to the busbar.



Measurements in real grids show different load behaviour.

Curve a is typical for households and a short switch off time. The reappearance of the load is suddenly nearly the same as is has been switched off.

Curve b and c are typical for households. The longer the switch off time is the higher is the reappearance load which exceeds suddenly up to 150% or more compared to the switched off load. This is caused by temperature controlled electrical devices (freezer, air condition etc.)

Curve d, e and f are typical for industry loads. Nearly independent from the off-time the reappearance of the load is gradual and smooth because of restoration of the production processes.

The exponents p and q can be calculated using the measurements of loads (P_1, Q_1 and P_2, Q_2) at two different voltages (U_1 and U_2) by the following equations:

$$\mathbf{p = \log(P_1/P_2) / \log(U_1/U_2)}$$
$$\mathbf{q = \log(Q_1/Q_2) / \log(U_1/U_2)}$$

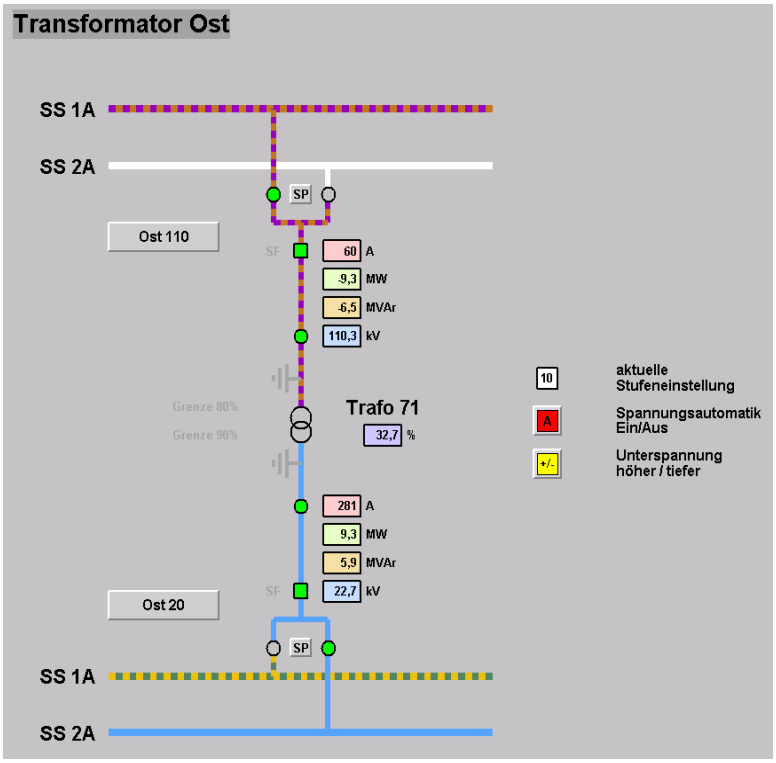
The voltages of the feeding busbar can be changed by e.g. topology, reactive power injections or by changing the tap of transformers.

Exercise: Investigate the types and calculate the parameters p and q of the loads
- 400-kV load „Stahlwerk“ in Station "Ost 400"
- 400-kV load “Aluwerk” in station „Südost 400“
- 110-kV load “Stadtwerk B” in Station "Südost 110"
- 20-kV total transformer load in Station "West 20"

For your report:
- Describe how you have changed the voltages.
- Note the load and voltage measurements and the results after the change.
- Answer the questions:
  - What type of loads present factors p, q = 0?
  - What type of loads present factors p, q = 1?
  - What type of loads present factors p, q = 2?

19. Voltage control on customers' side

Usually the voltage on the 0.4 kV customers' side is approximately 230 V. In Germany the range (VDE recommendation) for sockets' voltage allows +10% ... -10%. Where is the customers' voltage adjusted and controlled? This voltage is usually adjusted by the 110-kV/20-kV transformer by changing tap positions. There is another transformer 20-kV / 0.4-kV on the supply way down to the customer. This transformer has a fixed tap position which cannot be changed by the SCADA system operator. So it's the 110/20-kV substation transformer that adjusts and controls the busbar voltage and therefore also the sockets' voltages.



Usually a target voltage (e.g. 20.5 kV) on the medium voltage side is adjusted regulated by a controller changing the taps. Tap changing causes a change of windings and the transformation, usually by 1% per tap. Typical tap transformers have 27 taps (-13…0(neutral)…+13). Fluttering of tap changing is avoided by **a delay time** of 10 or 20 seconds between two tap changes. Usually the regulation can be switched in automatic AVC (Automatic Voltage Control) or in manual mode.

ON is indicated by a green E and OFF is indicated by a red A. Taps can manually be changed in OFF position by double-click of yellow +/- button.

Exercise: Investigation of the voltage regulation behaviour of a substation transformer.

First put the 20-kV grid into the following status: TR 69 feeds West 20, Mitte 20 and Süd 20. TR 70 and TR 71 are switched off. Automatic voltage regulation of TR 69 is off. Voltage of TR 69 is manually set to about 21.5 kV. Create a trend curve indicating the voltage and the tap No.

After this preparation the voltage controller is switched on. Measure the course of the voltage U depending from the time t and see the U = f(t) diagram. When the transformer doesn't change anymore, the target voltage is reached. Then adjust the tap to -11 and repeat the measurement again.

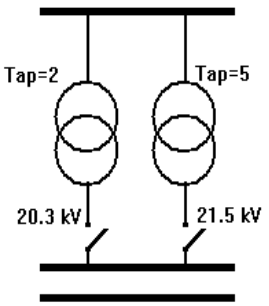
For your report:
- Discuss the diagrams U = f (t).
- Measure the control range and the delay time.
- See also the dependence of power

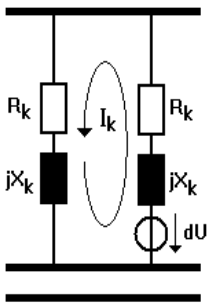
20. Parallel tap transformer

This exercise deals with HV/MV transformers, e.g. 110kV to 20-kV. Usually a transformer station has more than one transformer to supply the MV-busbars. On 110-kV side they are connected on the same busbar. From time to time the transformers have to be switched in parallel on the MV side also. Before parallel switch operation some criteria have to be fulfilled to get a balanced load and to avoid problems.

- **Same tap position if same type of transformer**
- **Same medium side voltage if different types of transformers**

Problems will occour, if the voltage transformations are not equal. In parallel operation mode a difference of transformation will result in a circle current, see figure.



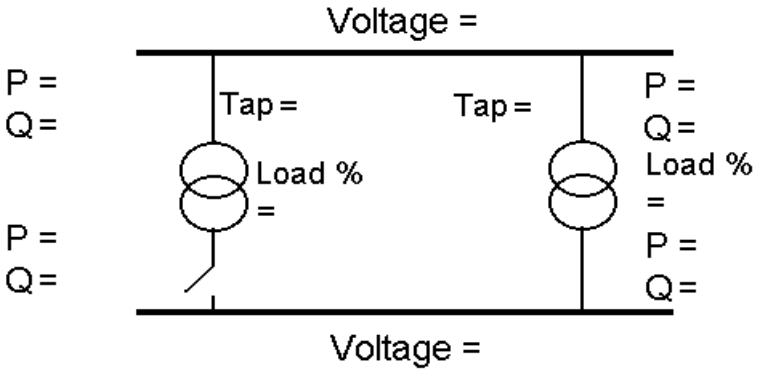


Different tap positions (2 and 5) result in different transformation (left figure). After parallel switching the voltage difference dU (in example equal to 1.2 kV) causes a circle current I_k.
The circle current may exceed some 100 A and causes losses and transformer overload because it has to be totalized with the load current.

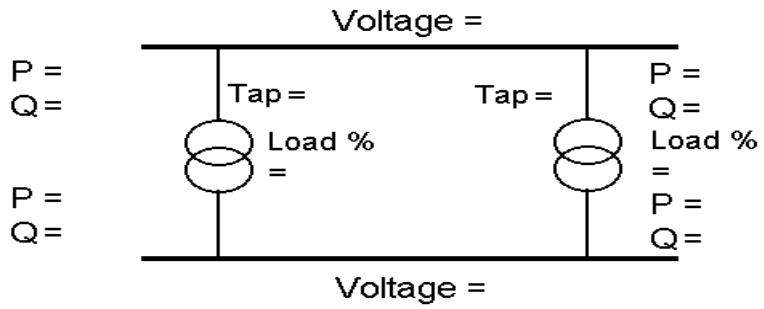
Exercise:

Use substation West 20 to switch two identical transformers TR 69 and TR 70 to parallel operation (a) and (b). First check the criteria to get a balanced load. Write down the measurements or do hardcopies:

(a) Before parallel

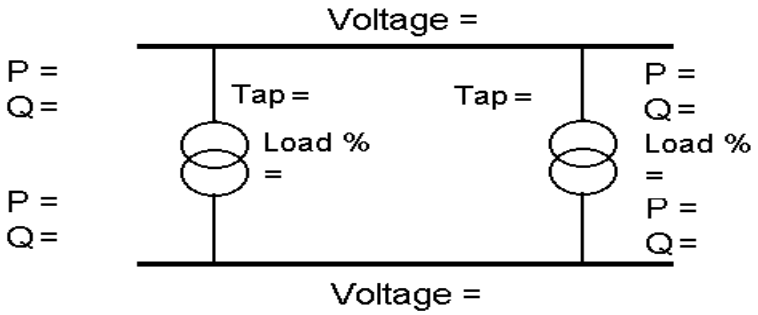


(b) After parallel switching



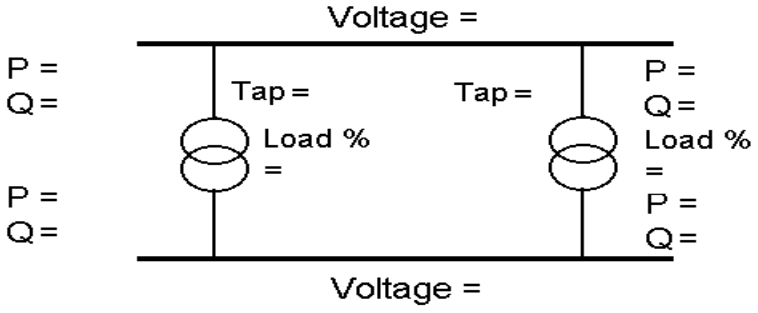
We now investigate the effect of a slight asymmetric transformation. So, in parallel operation change the tap positions to have a difference of 4 steps (c). Write down the measurements or do hardcopies:

(c) Parallel switching and tap difference of 4 (TR 69 +2 steps up / TR 70 -2 steps down).



We now investigate the effect of a heavy asymmetric transformation. So, in parallel operation again change the tap positions to have a difference of 10 steps (d). Write down the measurements or do hardcopies:

(d) Parallel switching and tap difference of 10 (TR 69 +3 more steps up / TR 70 -3 more steps down).



For your report:
- Write down the measurements
- Discuss the results
- Give an interpretation of the change of active and reactive flow

21. Parallel transformer & regulation fault sequence exercise

Usually, if transformers are in parallel operation, the voltage regulation of both is switched off. This is because of reciprocal influence. If voltage regulation is important, one of the regulations is turned into Master-mode, the other into Slave-mode.
If the coordination fails there is a danger of a circle current when major tap changes appear. We will exercise a failure like this. In substation West 20 two transformers are installed, they are switched in parallel. First we put the power system into the following status:

1. **TR 69 and TR 70 work parallel on the same tap.**
2. **Regulations of TRAFO 69 is ON and of 70 is ON.**

During parallel operation a regulation fault appears. Ask your trainer to execute this sequence**.**

For your report:

- Follow the sequence of events.
- Do some hardcopies.
- Describe and discuss the events and the reasons of the events.
- Give solutions to stabilize the situation, to stop the escalation and to solve the problem.

22. Optimal power flow - topology and losses

Optimal Power Flow (OPF) is a goal especially in dispatcher centres (e.g. national dispatcher centre) to achieve maximum transfer on low losses. There are a lot of parameters which influence the losses. Some examples: Differences of voltages in the same grid cause mainly additional reactive power flow which causes additional losses. Using parallel lines reduce the losses. Any transfer of power which can be substituted by local equipment will result in lower losses. Finally all components, injections, tap positions and topology as an interacting system of components have to be optimized. As this is a complex task, usually software called OPF is used in the control room to calculate a list of operations to optimize the power flow. The mathematical background is a non linear set of equations taking constrains into account. After solution is an iterative calculation process and the results are injections, tap positions usually for a fixed topology.

Exercise:

We will manually investigate the network losses depending from the topology and try to optimize the network state. Do a restart of the training system to have the start topology. After start of the system select the display "Zustandsübersicht" and note the total active load of the power system “Summe Last”, the active losses “Wirkleistungsverluste im Netz” and the costs of the losses “aktueller Verlust €/h” by doing hardcopies . Calculate the percentage of the total losses compared to the total load.

Now think about reduction of the losses.

- Parallel switching of network components (e.g. lines) to equalize the load flow.
- Avoid transfer of active power by using local equipment (if possible)
- Avoid transfer of reactive power by using local equipment (iIf possible)

Before switching discuss the expected results with your trainer.

After the parallel switching operation again note the total load of the power system and the losses. Recalculate the total percentage of the losses. Compare and discuss the results.

For your report:

- Hardcopy, notification and calculation of the above mentioned values.
- Discuss the disadvantage and limitations of parallel switching.

23. Protection system and three phase short circuit.

23.1 Introduction

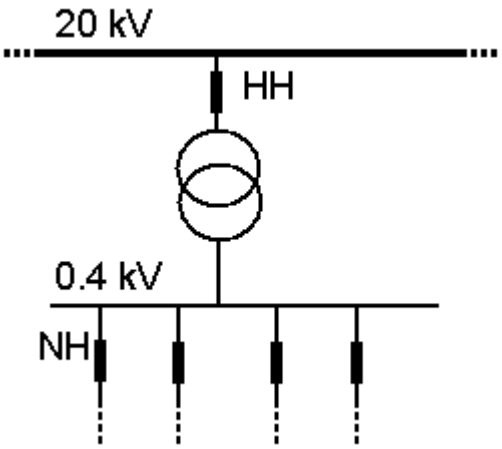
Due to thunderstorm effects on overhead lines, switching operation faults and other technical, human or environmental effects short circuits appear from time to time in the power systems. Protection components have to be installed. Protection components can not avoid the appearance of faults, they are installed to identify the fault and to selectively and fast switch off the affected components (e.g. line) to keep the remaining power system in operation. Selectivity is a strong demand. One protection components usually can not fulfil this demand. It is the system of the protection components and the coordination that fulfil the demand of selective reaction.

In general we have to distinguish between direct protection elements like fuses which are directly installed in the high voltage grids and indirect protection elements as relays which identify short circuit parameters (e.g. high current, low voltage) and react by tripping the breaker.

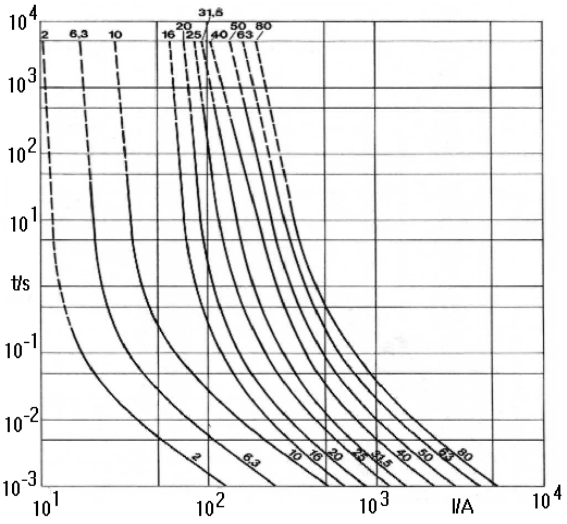
Fuses as direct working protection elements react by melting material due to high currents. Depending from the rated values and voltages the characteristic of the fuses can be seen in the next figure. Generally we distinguish

- Low voltage high energy fuses (NH fuses) and
- High voltage high energy fuses (HH fuses).

HH fuses take over the short circuit protection on the high voltage side of a local network transformer (20-/0.4-kV) and limit the current in case of a the short circuit. NH fuses are installed on the low voltage side feeders of a local network transformer (20-/0.4- kV) and limit the current in case of a the short circuit.



Transformer station and fuses



HH Standard fuses for 24-kV application

The current limiting effect of fuses is important: High short-circuit currents rise is limited by working fuses so the current does not reach the peak value. Currents are limited by the extremely short disconnecting time which is shorter than 10 ms.

According to IEC - VDE 0670, part 402, the time to current characteristics of HH fuses for transformer protection has to fulfil mainly two goals: Firstly a relatively high melting current in the 0.1 s area to stand the transformer's rush current by switching on. Secondly a relatively low melting current is also required, however, in the 10 s area to ensure switching off in case of transformer winding faults.

Protection relays

Circuit parameters supervised by the protection relays are:

- high currents
- low voltage
- low impedance
- low or high frequency
- differences of currents on both ends
- gas stream in oil
- …

Depending from the parameters to be supervised and the characteristic of the relay type we distinguish

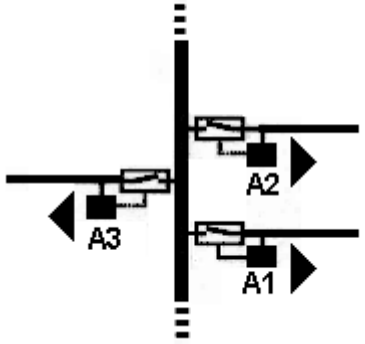
- Current depending protection relays (AMZ relays)
- Current independent protection relays (UMZ relays)
- Distance protection relays (Impedance relays)
- Frequency relays

Note the three states of a relay:

Normal state: If all supervised parameters are in normal conditions the relay is not excited.
Excited state: If one ore more of the criteria are fulfilled the relays react by an exciting state which can be described as ready to react. The final reaction of the relays can be delayed for a certain time that is dynamically calculated depending from failure parameters or set manually.
Executed state: The relay executes by tripping the breaker.

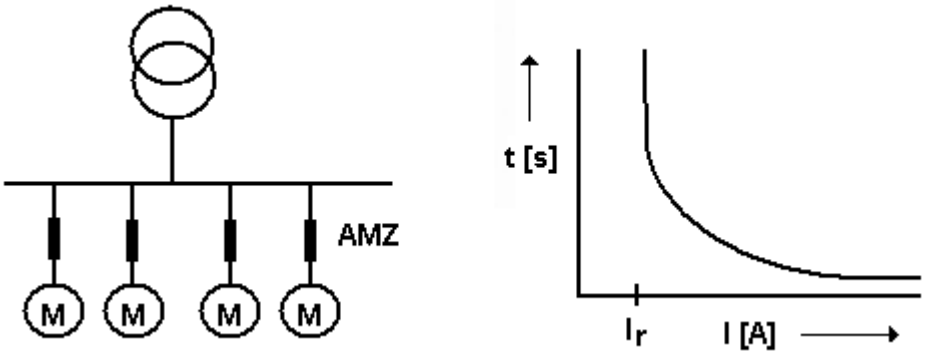
Direction identification sensor

Usually protection relays allow identifying the direction of the short circuit current. Note: It is the direction to the line, not to the busbar which excites the protection relay.



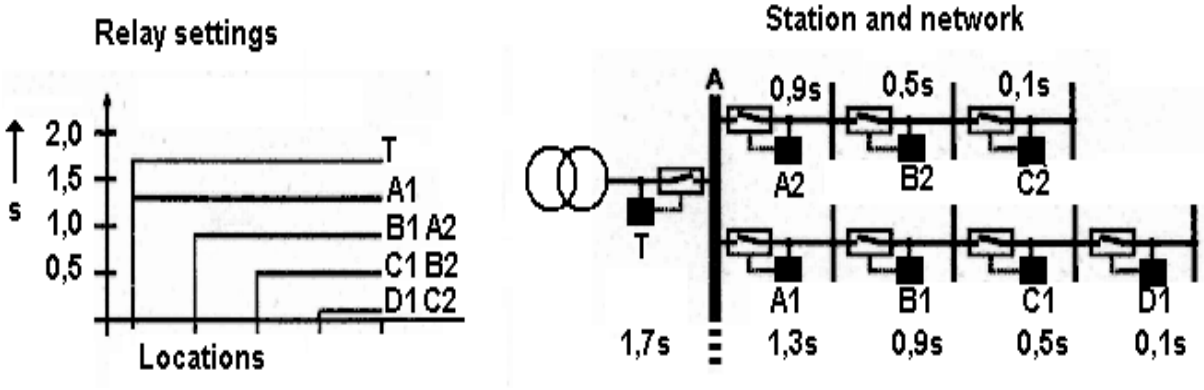
Current depending protection relays

The current depending relays (AMZ) can create a selective power cut off by measuring the short circuit current during appearance and reacting after a certain delay time. The delay time is an adjustable parameter (time) between the exciting and the reaction to trip the breaker which is calculated during short circuit appearance. Usually an inverse current-to-time characteristic is used. An application example is a set of drives in an industry production company.



Current independent protection relays

The current independent relays (UMZ) can create a selective power cut off by using different delay time reactions that can be set manually as parameter of the relay. The relay is excited if the current exceeds an adjustable limit independent from the value. The delay time is an adjustable parameter (time) between the exciting of the relay after occurrence of a short circuit current and the reaction to trip the breaker which is set manually. An application example is a set of drives in an industry production company.

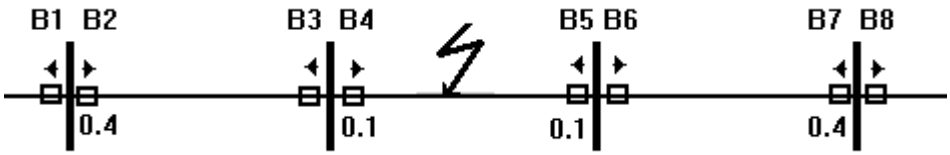


Distance protection relays

The current independent relays (UMZ) can create a selective power cut off by using different delay time reactions. But as it can be seen in the figure above, there are some disadvantages in operation.

- long track lines lead to long delay time settings
- in case of parallel track lines the feeding transformer will get the longest delay time plus one step more.
- The closer the short circuit appears to the feeder point the bigger the short cut current. This result in short delay time for low currents and long delay time for high currents which affects the equipment.

Selective and fast reaction is achieved by distance relays which dynamically measure the impedance. Low impedance result in fast reaction and high impedance in longer delay times. Additionally a relay usually supervises just the direction to the line, not to the busbar. By adjusting these parameters it is possible to achieve a selective protection reaction including a backup if the right one fails. The delay time is calculated online during failure detecting. The closer the failure (measured by impedance Z = U/I) is the faster the reaction will come. Usually coordinated steps of 300 ms are used starting from 0.1 s. see the figure:



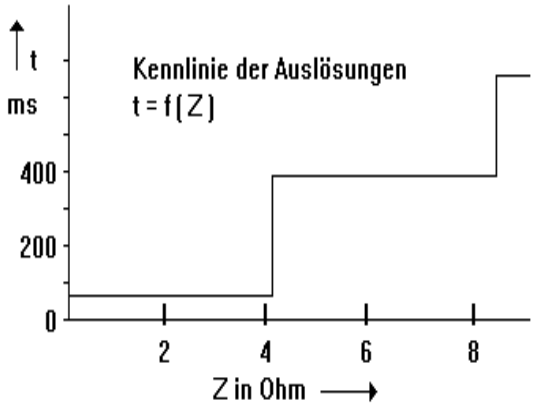
Which relays will be excited in case of a short circuit? B2, B4, B5 and B7.

Which relays will react? Answer: B4 and B5 react in 100 ms because of low impedance. B2 and B7 react in 400 ms because of higher impedance and only if the fast one fails.

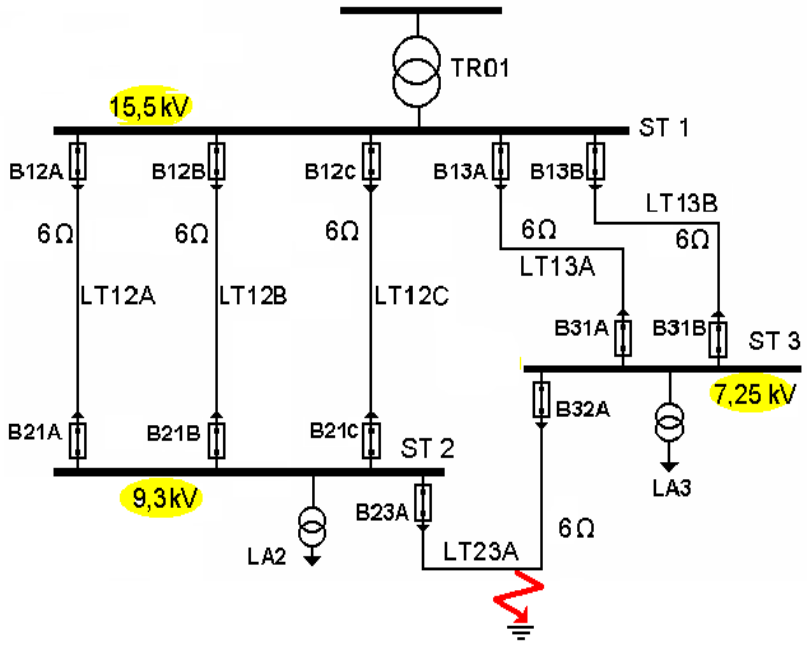
Another example:

In a net area a 3-pole short circuit appears for line LT23A centrically. The voltages of the bus bars are listed in the net picture. All lines have the impedance Z = 6 Ω and direction oriented distance protection relays at both line ends.

1. Calculate the short-circuit currents (in the simplified method discussed during the lecture) and the delay time of the distance protection relays.
2. Fill in the table, which relay trips when?



Parameter settings of the protection relays
(Same for all relays)

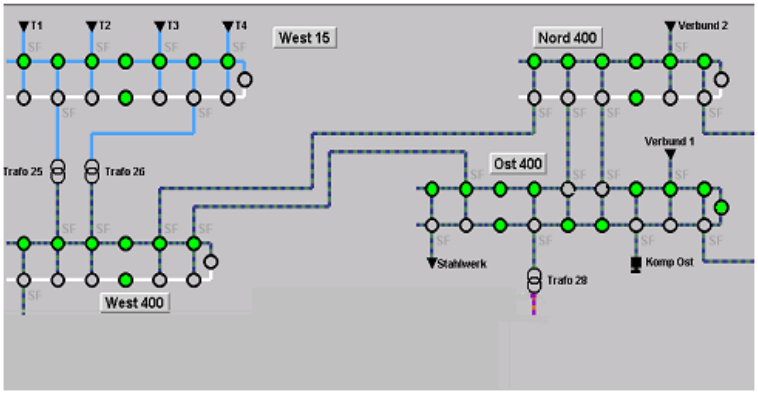


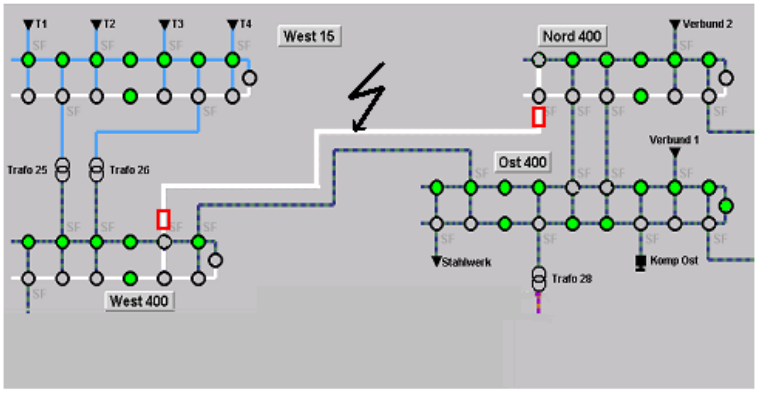
Bay/Relay	Voltage[kV]	Current [kA]	Impedance [Ω]	Delay time [ms]
B12A				
B12B				
B12C				
B13A				
B13B				
B21A				
B21B				
B21C				
B23A				
B32A				
B31A				
B31B				

23.2 Exercises

23.2.1 A 3-pase short circuit on a 400 kV line

See the figure: In line Nord 400 to West 400 a short circuit appears. Within 100 ms the breakers on both ends of the line were opened by the protection relays (distance - impedance relays). This is a correct selective reaction of the protection system.





Main requirements on protection systems are:

Selectivity: Clearly discriminate between normal and abnormal operations.

Reliability: Even after long interval between consecutive operations.

Reclosure: As most faults are temporary

Selective and fast protection reaction is needed because

- Arching faults can vaporize equipments and lead to fire and explosion.
- Overheating of components during abnormal current flow.
- Damage to utilization equipment due to unacceptable voltage levels.
- Loss of synchronism due to blockage of power.

23.2.2 A 3-pase short circuit on a 400 kV bus-bar.

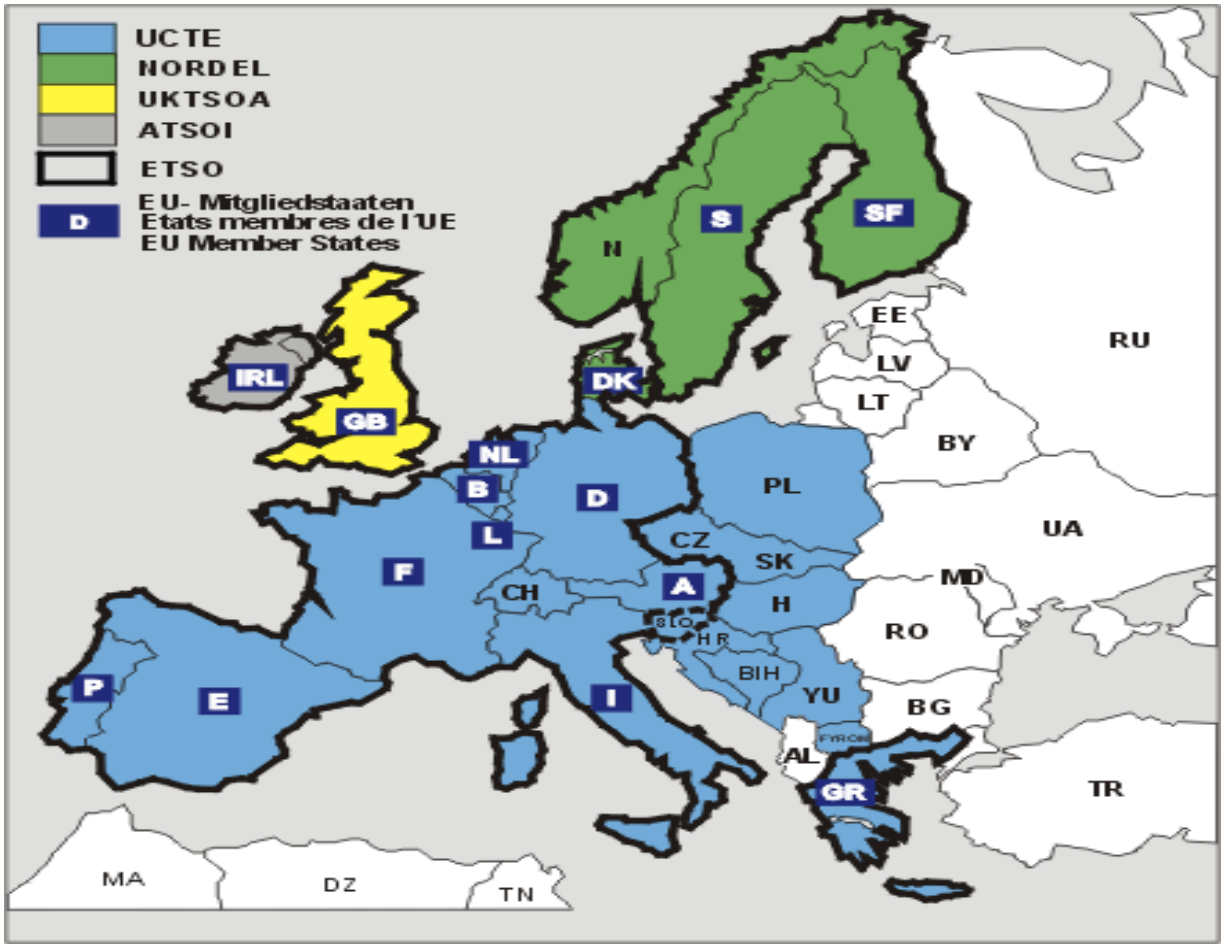
First put the power system again in normal condition. On bus-bar SS1 in Süd-WEST 400 a short circuit appears. We will see the effects. Ask your supervisor to insert a temporary three phase short circuit. Follow the sequence of events on the screen and discuss it using also the protocol.

For your report:
- Hardcopies of load flow picture before and after the events.
- Discussion of the events seen and noted in the protocol.
- Qualify the protection reactions (Which make sense, which seem to be wrong?)

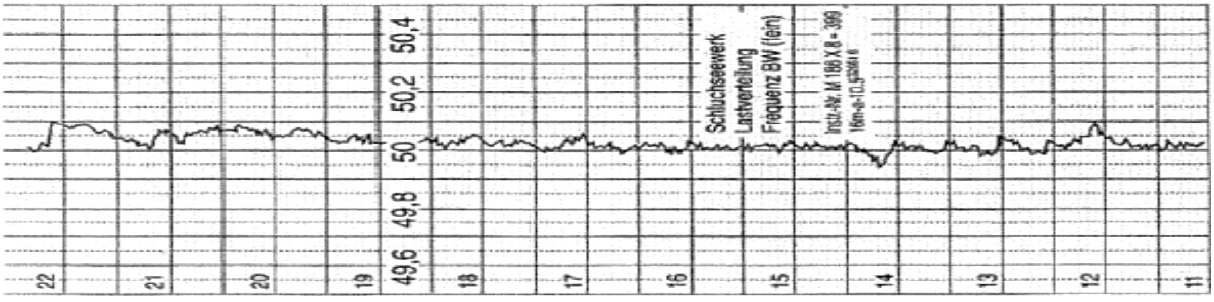
24. Grid separation and frequency changes with load impulses

24.1 Introduction to frequency regulation

The purpose of this exercise is to see and to discuss frequency control behaviour. By doing the experiment please note that this experiment can not be done in a real network because the customers' power supply can be interrupted. We will investigate the reaction of AGC (automatic generation control) of the network on load impulses. A network may be the huge interconnected UCTE-power system but also a small network island.



UCTE grid and other networks in Europe

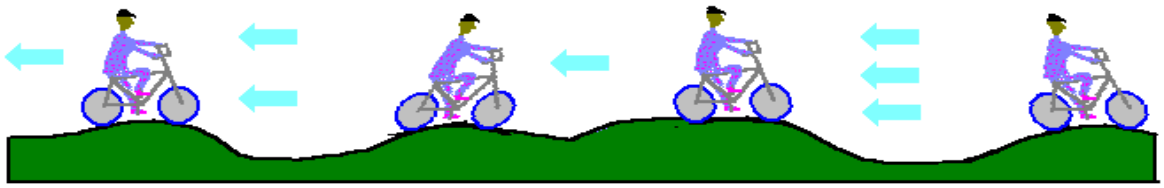


Rather stable frequency in the UCTE grid

In the picture the frequency fluctuations are represented in the European integrated grid system UCTE within several hours. The curve was recorded at a German 400 kV control center (RWE-net). Every location of the UCTE net shows at the same time just the same frequency. All deviations of the 50 Hertz norm are lower than 0.1 Hertz because the regulations immediately step in.

The frequency of 50 Hertz isn't a nature constant it has arisen historically. Other countries like the USA and parts of Japan use 60 Hz. By traditional train technology uses 16 2/3 Hertz. In modern airplanes 400 Hz is used (The higher the frequency, the lighter are the electrical drives at the same power).

Regulations have the task to provide an almost constant frequency. It is a rather difficult task because of some disturbances as uncertain customer consumption. Comparison: Ride a bike **with constant speed** in uneven area and with a changeable headwind.

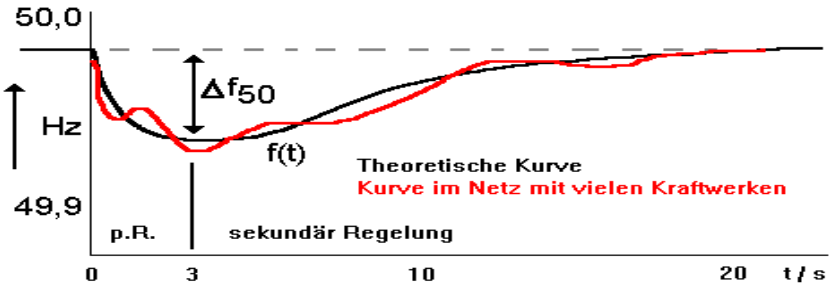


The customer determines about the load: $P_L = f(t)$. There are only a few possibilities of influencing the loads.

- Switch on and off
- Changing voltage

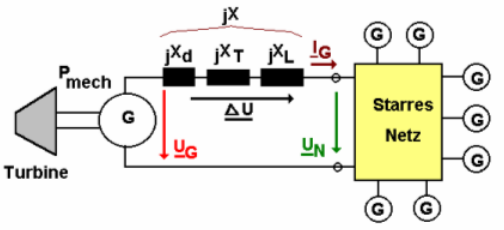
The generation has he task of balancing the load. This is done by steam pressure or by hydro water storage plants. The rotating masses help to compensate load jumps. Missing power generation results in a drop of frequency, overproduction in an increase of frequency. Therefore sensitive frequency sensors are needed. In case of a load impulse or an outage of a big generator the frequency drops and two regulations act:

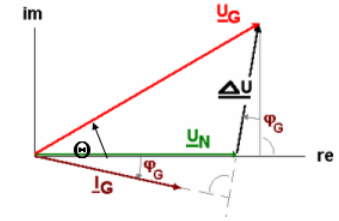
- Primary regulation to stabilize the frequency $df/dt \rightarrow 0$
- Secondary regulation to achieve 50 Hz again $df_{50} \rightarrow 0$



The primary regulation recognizes Δf and regulates Δf against zero. The turbine power is increased for 1… 3 s time constants T at sinking frequency with one. It steps in from

deviations of about 3 mHz and has a purely proportional behaviour. After ending regulation a stationary frequency deviation remains to 50 Hz: Δf_{50} which is recognized by the secondary regulation which regulates the deviation to 50 Hz slowly against zero. Please note: Primary regulation is very fast (1…3 s) mainly using steam bypass. Secondary regulation has time constants of 20 to 30 seconds. How can the power injection of a generator be increased? See next figure:





$$P_G = \frac{3\ U_N U_G \sin\Theta}{X}$$

To increase active the power generation P_G, the turbine has to increase the mechanical power. This results in a grater angle Θ which increase $\Delta\underline{U}$ and $\underline{I}_G$. The injected power is

$$P_G = U_N\, I_G \cos\varphi_G$$

Power network factor V_N:
The factor V_N describes the size of stationary frequency deviation Δf_{stat} resulting from primary regulation affected by a load jump ΔP_{50}. Secondary regulation is OFF.

$$V_N = \Delta P_{50} / \Delta f_{stat}$$

Network static s
Network static stability s describes the size s which arises from the inversion of the figure V_N.

$$s = \frac{\Delta f_{stat}}{\Delta P_{50}} = 1/V_N$$

ΔP_{50} Load jump in 50 Hz
Δf_{stat} stationary deviation of the primary regulator

s describes the expected stationary deviation to 50 Hz in case of a generation outage. Example: In the UCTE net a big power station generation block has an outage. The power deficit is 1000 MW. This causes typically a frequency drop of approximately 0.08 Hz. The network statics s and the figure of merit V of the UCTE net are with that

$$s = 0.08\ \text{Hz} / 1000\ \text{MW}$$

$$V_{N,\,UCTE} = 12.5\ \text{GW/Hz}$$

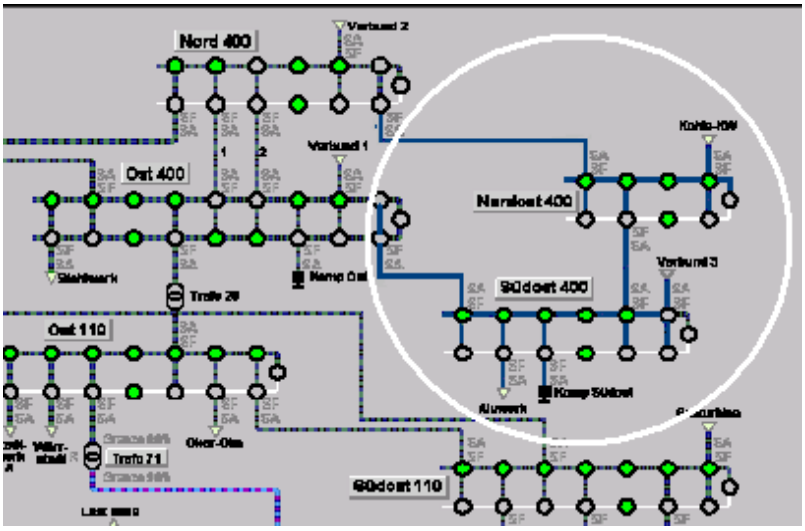
24.2 Exercises

24.2.2 Interconnected UCTE system

In the situation of interconnected UCTE-system we will switch the load of “Aluwerk” in station Südost 400 which means a heavy load impulse of some hundred MVA. Watch the frequency by a measuring instrument (open a frequency window). Frequency is checked approximately every second and displayed by a curve. Repeat ON and OFF switching of the load for a few times, always after stabilization of the frequency.

Make a hardcopy.

24.2.3 Load impulse in islands



Next step will be a separation of the electrical network into two parts, see figure.

Separation can only be done if the separated part is able to survive with it's loads.

Before separation make sure that the small island is being separated with a exchange flow of nearly zero (active and reactive).

Repeat the consideration in the small network by changing Aluwerk load by +/- 10 MW. Watch the frequency by the measuring instrument. Off course...make hardcopies.

For your report:

What does the network's static stability s describes?
Estimate the s and VN for both experiments.

Continue exercise 25 in his power system configuration (island).

25. Frequency dropping and frequency protection relay

The purpose of this exercise is to see and to discuss frequency and protection behaviour. By doing the experiment please note that this experiment can not be done in a real network because the customer's power supply will be interrupted. We will see the overload of a small island and the dropping of the frequency until a self protection (frequency protection) of the power system operates by tripping a breaker of a load to balance the power in the system.

Your supervisor will overload the island by changing the load slightly step by step. Do hardcopies of the load and of the frequency.

For your report:

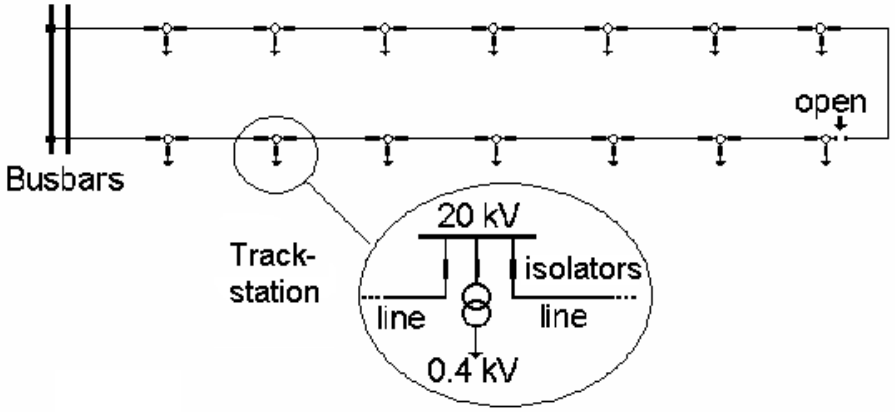
- Describe and discuss the hardcopies.
- Which time constancy (approx.) shows the primary controller (AGC)?
- How is the electrical island reacting on a high load impulses ?
- What can be done to let an overloaded power system survive?
- Give basic rules for reconstructing the power grid after a black out.

26. Short-circuits in a track line

The purpose of this exercise is to locate a permanent short circuit in a 20 kV track line, to isolate the damaged line part and to re-supply all customers again.

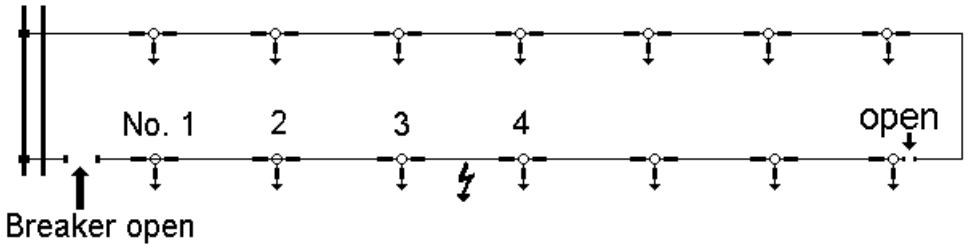
Ask to your instructor to have an introduction into strategy of short-circuit location using short circuit indicators and/or experimental switching. Note that the 20-kV network RTU only control the substations.

Please note: RTU just control the main transformer stations. The full amount of switch gear along the track lines is not controlled by RTU. Switching can only be done manually in the track station. Changing status is not updated automatically – it has to be done manually on the screen by tracking ("Nachführen"). See the grid:



Basic strategies

The basic strategy to locate a short circuit depends from the relay installations as follows. The power grid is often constructed by track lines of loop type which are driven open as it can be seen in the figure. If a short circuit appears, the protection system will switch off the effected track line as it is shown in next figure.



All the customers of track station No. 1 to 7 are off supply. Usually we do not know the location (part of the line) where the short circuit happened. The strategies to locate the fault and to supply all customers again (as fast as possible) differ by the relay equipment installed in the power system.

Strategy if short circuit indicators are installed in the track stations

Then the strategy to locate is to send a team in the grid to see the short circuit indicators installed in the track stations. The search is usually started with track station 1, continued by 2 and so on. E.g. the failure is somewhere between track station 3 and 4 we will have short circuit indication in track station 1, 2 and 3 but not in 4.

Knowing this, the failure is somewhere between track station 3 and 4. As the staff is now in track station 4, the isolator in direction to track station 3 is ordered to open. Then the staff is sent to track station 3 to open the isolator in direction to station 4. By this two open isolators the effected part of the line is disconnected from both sides. To supply customers the breaker can be switched on. This results in the return of supply of track station 1, 2 and 3. Last step is to close the isolator in station 7 so that the track stations 4 to 7 are fedded from the other side.

25.2 Exercise

Your instructor will execute a three phase short-circuit somewhere in the 20 kV network. Your job is to send staff members to the grid stations. Ask for indicator status. By this find the location where the short-circuit happened, isolate the damaged part by orders to the staff and feed all customers.

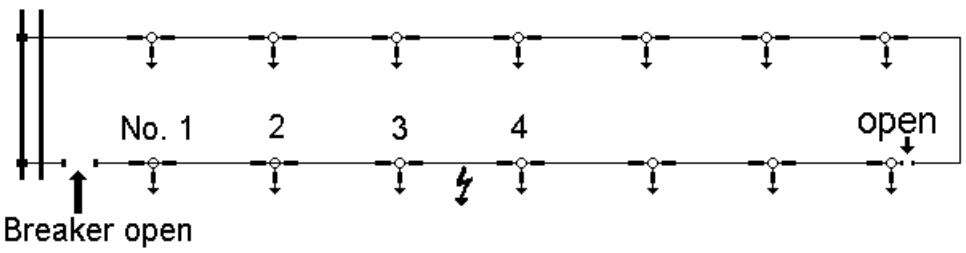
- Watch the protection system report.
- Send a team in the grid reporting the indicator status.
- You are responsible to give the orders for switching operations.
  The personnel communication and switching execution are done by the trainer.
- Think about a strategy to minimize the search time for fault location.
- After a switch has been done in the grid immediately update the SCADA database.

For your report

- Describe the events and the strategy to locate the fault and to re-supply the customers.
- Do some hardcopies for a clear documentation of all actions.

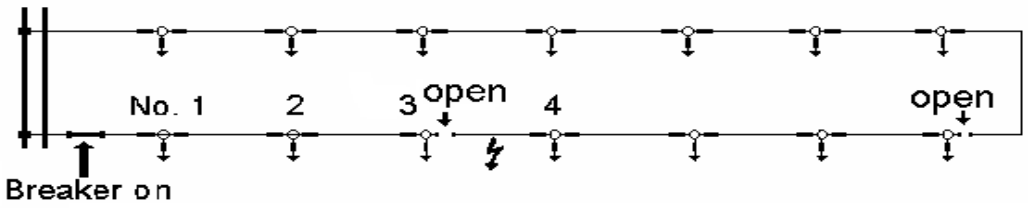
Strategy if no short circuit indicators are installed

In that case the strategy to locate the failure is sending a team in the grid for separating network parts. Separation is coordinated in the control centre. An experimental breaker ON is executed and the results are checked. If the breaker trips again the connected part is effected by the failure. The other part can be connected from the other side. And so on. The search is started with a 50% strategy as it can be seen in the next figures. Imagine a short cut occurs. The relay executes the breaker opening.

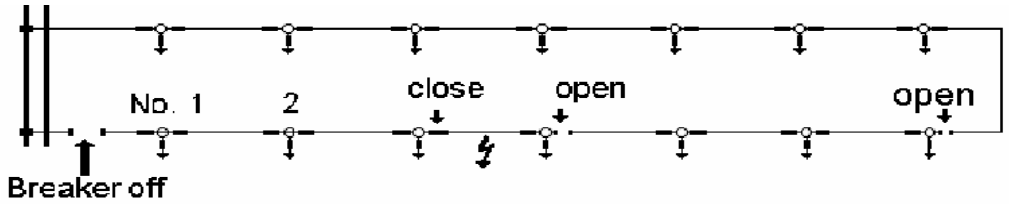


Then staff is sent to track station 3 because it is the middle of the effected track line. Two steps are done: First the isolator in direction to track station 4 is ordered to be open. After this

is done, immediately the screen is updated with the new state. Second step is the breaker switch ON, see next figure:



The result will be no breaker tripping. Knowing this, the failure is somewhere behind track station 3. Next steps are:
The breaker is opened again. The staff is ordered to close the isolator in track station 3 to direction 4 again. Note: The breaker is still open. If not, the staff would have switched with an isolator to a short cut. This is extremely dangerous and might kill workers.



Next steps are: The staff is sent in track station 4. In 4 the isolator in direction to track station 5 is ordered to open. After execution immediately the SCADA data base has to be updated. Then the breaker is closed. The breaker will trip again if the short cut in the connected line. In our case it happened so and we know that the failure is somewhere between track station 3 and 4. As the staff is still in track station 4, the isolator in direction to track station 5 is ordered to close again and the isolator in direction to track station 3 is ordered to open. Then the staff is sent to track station 3 to open isolator in direction to track station 4.

After these switching operations the damaged part of the line is disconnected to both sides. To supply customers the breaker can be switched on. This results in the return of supply of track station 1, 2 and 3. Last step is to give order to close the isolator in track station 7 to the grid staff so that the track stations 4 to 7 are connected again.

25.3 Exercise
Your instructor will execute a 3-phase short-circuit somewhere in the 20 kV network. As in the exercise above your job is to locate it, to isolate the damaged part of the line by switching operations and to re-supply all customers. There are no indicators. Locate by test switching.

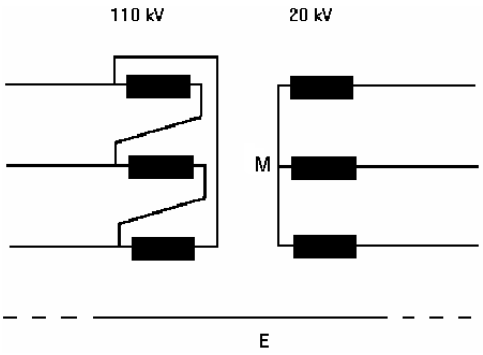
- See the protection system reports and send a team in the network for switching operations. You are responsible to give the orders to the personnel. The staff's operations are performed by the trainer.
- Minimize the amount of switching orders because to send the personnel to the track stations is very time consuming.
- After the switch is executed by the grid staff immediately update the SCADA database.

For your report
- Describe the events and the strategy to locate the fault and to re-supply the customers. Make some hardcopies for a clear documentation of all actions.

26. Petersen Coil for single ground fault compensation

26.1 Treatment of neutral transformer point

In the 3-phase power grids the handling (electrical connection) of the transformer neutral point M is very important especially in the case of a single ground fault which is the main fault in electrical systems. The question is how to connect M and E, see figure.



If we assume a symmetrical grid (symmetrical line to ground capacitors) the potential E is in the centre of the three voltages and therefore on same potential as M . If there is no potential difference between M and E, the connection can be arbitrary. Please note: In the normal power system conditions the star point treatment has no influence.

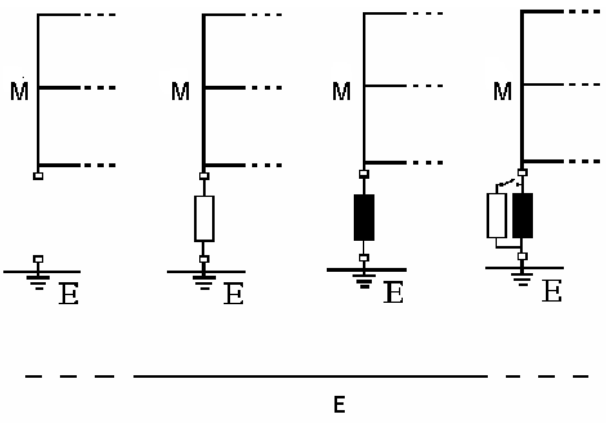
It changes in the case of a single ground fault. E jumps to the potential of the effected line potential. Then the connection between M and E is important. What kind of connections are of technical interest? See figure:

M can be isolated.

M can be connected by a resistor.

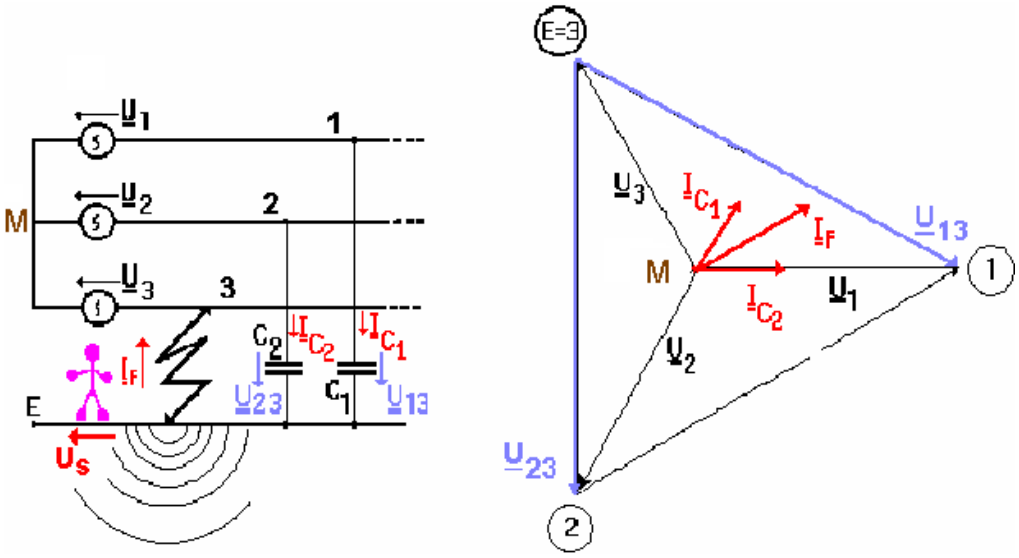
M can be grounded by a coil,

M can be connected a short time by a resistor and then switched to a coil.



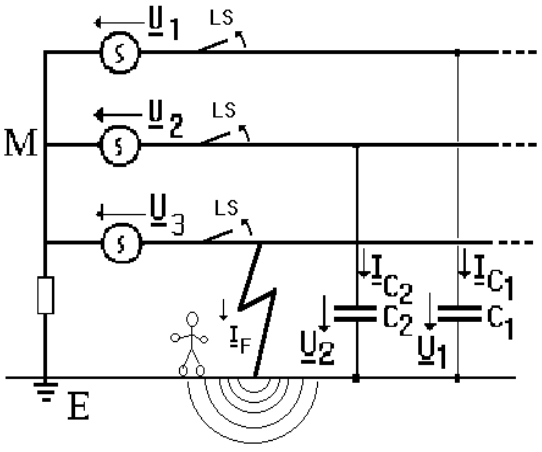
Due to the connection the power system reactions in case of a single ground fault will be different. We will discuss the different types of handling the neutral transformer point and the according effects.

26.2 Isolated star point



We assume a single-phase ground fault in L3. The potential of the point E jumps to L3 potential because of the short-circuit.
The insulations of the lines L1 and L2 are stressed now by the line to line voltage. The loads aren't influenced, full power supply.
Depending from the net size (line to ground capacities) the fault current I_F is high. See the position and angle of IF which is he sum of both capacitive current from Phase 1 and phase 2.
A surface voltage gradient arises which influence step voltage to critical values an put danger to any person or animal near to the failure.
This type of handling the neutral transformer point is used in limited grids of small size or in industry areas that are fenced.

26.3 Resistor grounded star point



Low-impedance grounding causes a single-phase short circuit ground fault in phase L3.

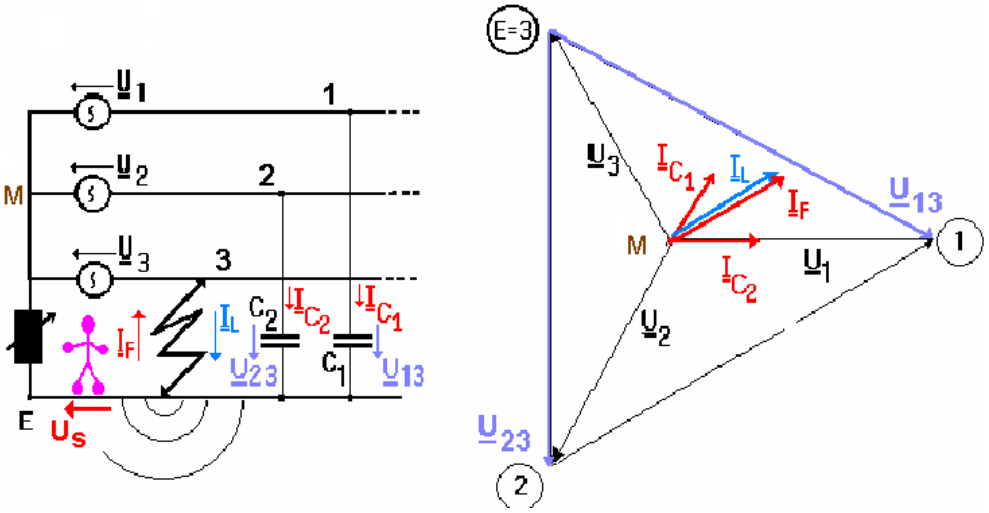
A high short circuit current appears.

For a short time a high surface voltage gradient appears.

Protection system will switch the line typically within 100 ms.

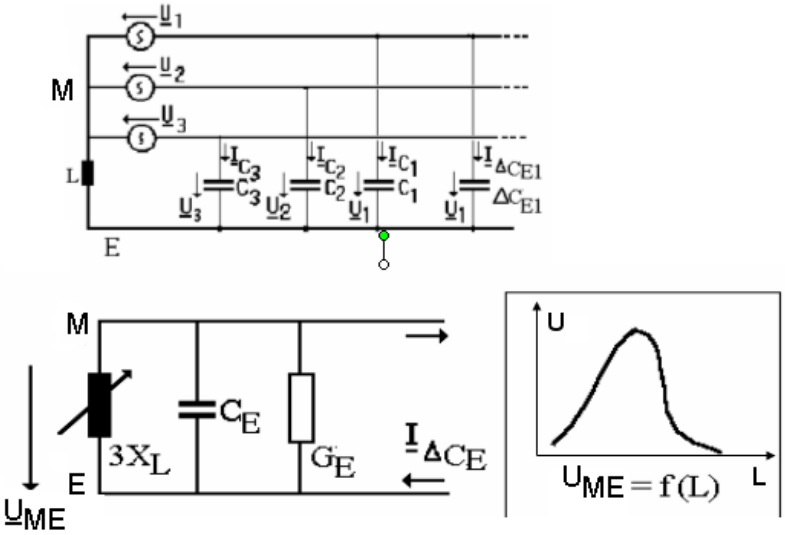
Consequences:
Failure is cut of immediately.
However customers are also switched off.

26.4 Peterson Coil grounded star point



Assuming a single ground fault (in L3) the inductive grounding of the neutral transformer point shows the following consequences:
Potential E goes to potential L3. The fault current I_F appears as the sum of the two capacitive currents I_{C1} and I_{C2} of the not effected phases.

See the angle of I_F with respect to the potential of L_3 (U_3). There is a second electric circle by source U_3, failure and inductivity. It results in an inductive current I_L with respect U_3. This current compensates the failure current I_F **so that an arc may disappear**. It is somehow like a **self-repairing effect of the power system**. Please note that all customers are continuously supplied all the time. In overhead lines the arc usually disappears, in cable systems the arc usually douse not disappear – but the damage is less.



Because of slight line to ground capacitors' differences the coil can be adjusted to the connected grid. The picture shows the network equivalent in case of no failure. If $C_1 \ldots C_3$ are different, the Zero-System presents a resonance network for 50 Hz to adjust L to the grid capacity. After a change of the topology the coil has to be re-adjusted again.

Please note: If there is an extreme low asymmetry concerning line to ground capacities the resonance curve will be extremely flat. In that cases, an artificial 1-phase capacitor will create the needed asymmetry.

26.5 Exercise: Adjustment of the coil as preparation for a single ground fault

For your report: Adjust the coil by fixing at the resonance point. Do hardcopies and discuss the results.

Introduction to control displays
First get an introduction to the displays for controlling the Petersen Coil e.g. in station WEST 20. Be familiar with:

- The transformer display “Erdschluss”: See the bus-bar to switch ON/OFF the coils, the automatic regulation and to control the adjustment drive.
- Control of the asymmetrical 1-phase capacitance to enlarge the displacement voltage if necessary.
- Curve display to see the resonance curve voltage measurement (displacement voltage).

Now do the exercises
1. Switch the automatic coil control to OFF position. Then vary the inductivity and see the resonance curve. Adjust to compensation. Make a hardcopy.
2. Change the grid topology (e.g. switch over the open end line to next substation). Then vary the coil again and follow the resonance voltage curve. Adjust again to compensation. Make a hardcopy.
3. See the effect of asymmetric capacitance by switching to ON and OFF position.

27. Single ground fault location in a compensated grid.

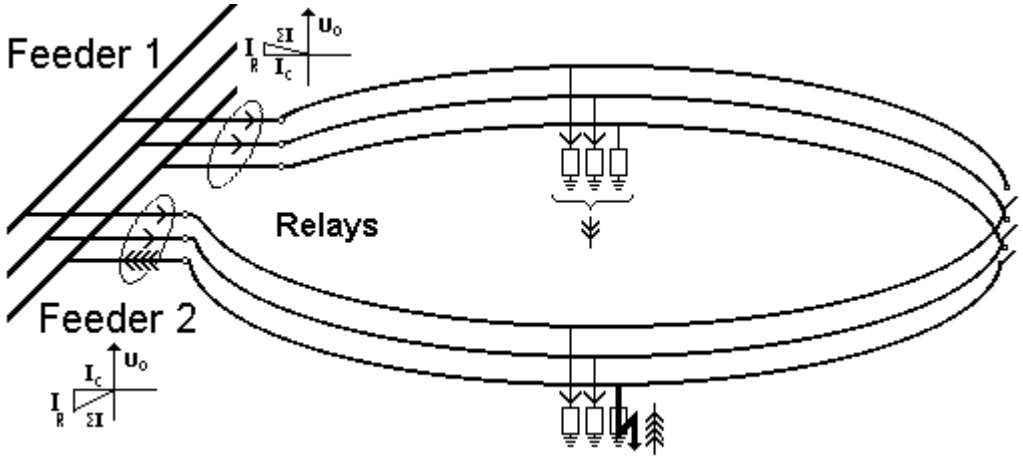
27.1 Objective

The objective of this exercise is locating a permanent 1-phase ground fault in a 20-kV track line. Your instructor will simulate a ground fault and your job is to locate the fault and to isolate the effected part of the track line. Contact your instructor to get a short introduction into the ground fault location strategy and especially in interpreting the ground fault relays.

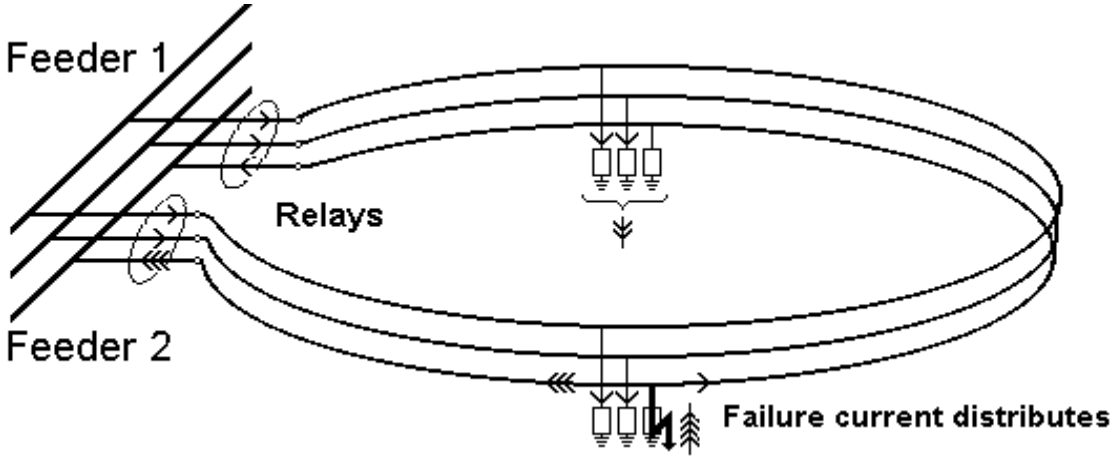
27.2 Ground fault relays

We distinguish two types of relays – a **transient relay** which measures and indicates the voltage change during a ground fault appearance. Please remember: If ground is connected to one line, the two other lines' voltages jump to the square root of three times of former line voltage. This relay type uses the transient change of charge in a capacitor during appearance. The second relay type – a **static relay** uses a balance of all three line currents which normally is zero. If a single ground fault happened, the balance is disturbed as long as the failure is active. See the next pictures, demonstrating the functioning. The boxes are capacitors with a small resistance. Ground fault indication is possible if the line currents' sum is unbalanced.

1-phase ground fault in L3, die loop is open. The static relay indication will work.

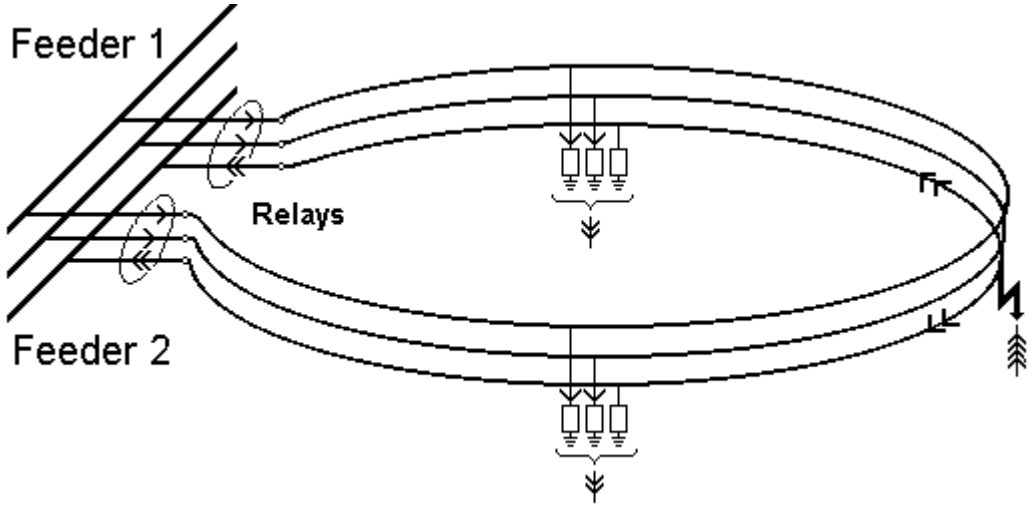


1-phase ground fault in L3, die loop is closed. The static relay indication will work:



1-phase ground fault in L3, die loop is closed. The static relay indication does not work.

In this case the failure is assumed to be in the electrical middle of the track line.



27.3 Exercise: Ground fault location in a track line

See the relays and control system reaction after inserting a single ground fault.

- Check the symbols of the ground fault relays.
- Supervise the grid personnel doing the switches in the track lines.
- Avoid cutting off any load. Close a loop and open the loop for further relay indications. Avoid connecting the 110/20-kV substations and carry off the failure.
- Minimize the amount of switches because you have to send personnel to the transformer stations.
- After the switch has been made immediately update the new switch status on the screen.
- Locate the ground fault and isolate the effected track line element and finally ground it.

For your report:

- Describe what happened and the strategy to locate the fault.
- Hardcopy of grid showing the isolated track element.
- Count the amount of switches. How to minimize?

28. Economic load dispatch and economic indicators

The object of this exercise is to understand the economic side of the power supply taking contracts, regulation supervision and economic load dispatch into account.

Contact your instructor to get a short introduction into the actual state of regulation in Germany. See the displays indication the economic state of the power grid.

Explore the displays indicating economic indicators:

- Electrical losses and cost of losses
- Financial losses by Energy not supplied
- EBT by power transfer charges
- Penalty for customer cuts (CENS)
-

Introduction

Your first task is within half an hour to increase the profit [€/s] as much as possible. In easy words:

Sell as much as possible by transferring power.

Restriction: You are not allowed to switch off the time tables of the loads and to change the tariff contracts. Keep all technical restrictions in mind.

You are allowed to influence voltage level, topology, active and reactive load flow and the plant generation schedules. Some help:

- Lower the losses, but keep short circuit restrictions in mind.
- Sell more energy, but keep voltage level within the range.
- Buy the cheapest energy, but do not violate the limits.
- Do not transfer much reactive power through the long distance lines, but keep the voltages. Use local compensators.

For your report

- Make a hardcopy of the cost display at beginning.
- Make a hardcopy at the end of the exercise.
- Describe your experiences through the exercise – what is the impact of active and reactive production, load flow, compensation etc
- What creates your profit running the power system?

What is the impact of your decisions to the economic indicators? What are the risks?

Appendix

Example of correct switch order communication:

Due to serious accidents which result from misunderstanding during the communication for switching orders it has to be exercised very intensively how to communicate between the control centre and the staff members in the grid.

During fault location, loops have to be opened and closed. Follow the communication between Mr. Smith, engineer in the control centre and Mr. Miller, technican in the grid. Never an isolator can be used to initialize or interrupt a short circuit. This is extremely dangerous to staff members and installations. Now follow the communication:

Smith: "Control centre, Smith speaking. Mr. Miller we have to locate a ground fault and to do some switches in the grid. Please move to the transformer station ERDWEG and report from there."

Miller: "OK, I'm on my way to ERDWEG. It may take 15 minutes. "

...After a while...

Miller: "Miller, reporting from grid station ERDWEG."

Smith: "Yes, go to the feeder CEASARRING and open the isolator in direction to CEASARRING"
Miller: "I’m going to open the isolator of feeder CAESARRING"
Smith: "Yes, execution"

... Now the isolator is opened...

Miller: „Executed"

Smith: "Ok, Mr. Miller now please go to the ...."